

DCLab LLM Review Report

Campaign **model_building_50_v1** · Critic model **gpt-5.4-mini** · Generated 2026-09-18 13:14 UTC

LLM role is critic / hypothesis generator only. Observed metrics remain authoritative. Reviews must not be treated as deployment approval.

REVIEWS	HIGH CONF.	MEDIUM CONF.	LOW CONF.
50	4	46	0

Dataset coverage

adult 5 reviews · 14 challenges · 0 low / 1 high confidence	bank_marketing 5 reviews · 16 challenges · 0 low / 0 high confidence
breast_cancer 5 reviews · 17 challenges · 0 low / 1 high confidence	credit_default 5 reviews · 18 challenges · 0 low / 0 high confidence
german_credit 5 reviews · 14 challenges · 0 low / 0 high confidence	heart_disease 5 reviews · 17 challenges · 0 low / 0 high confidence
mushroom 5 reviews · 15 challenges · 0 low / 1 high confidence	online_shoppers 5 reviews · 18 challenges · 0 low / 0 high confidence
spambase 5 reviews · 15 challenges · 0 low / 1 high confidence	wine_quality 5 reviews · 13 challenges · 0 low / 0 high confidence

Executive summary

ID	Dataset	Stage	Confidence	Decision	Next experiment
EXP-001	adult	data_understanding	medium	Do not proceed to modeling yet; first resolve representation and production-availability risks with a targeted validation pass.	Decision-time and raw-category audit for adult features
EXP-002	adult	leakage_audit	medium	Insufficient evidence to conclude which columns are leakage or target proxies. The current package supports only a detector sanity check, no	Column-level decision-time contract audit for adult
EXP-003	adult	feature_engineering	medium	Do not accept the claim that the raw/selected	Nested-CV rerun of the selection rule with fold-

ID	Dataset	Stage	Confidence	Decision	Next experiment
				recipe should be used next based on this package alone. The smallest defensible action is to r	local feature fitting
EXP-004	adult	model_selection	medium	Provisional selection only: lightgbm is the current training-CV leader under the provided rule, but the evidence is insufficient to finalize	Repeated stratified CV comparison of lightgbm vs hist_gradient_boosting with leakage audit
EXP-005	adult	optimization_reliability	high	Insufficient evidence to conclude that optimization produced a meaningful CV gain. The once-consumed holdout performance is reported, but it	Minimal CV delta audit against baseline
EXP-006	bank_marketing	data_understanding	medium	Do not approve modeling readiness yet. The dataset description is usable, but production-availability and leakage risks remain unresolved.	Decision-time availability and raw-category restoration audit
EXP-007	bank_marketing	leakage_audit	medium	Do not approve duration for use until decision-time semantics and production availability are verified; treat it as a leakage-risk candidate	Decision-time availability audit for duration
EXP-008	bank_marketing	feature_engineering	medium	Do not accept the recommendation as supported; request a minimal confirmatory experiment before choosing a recipe.	Paired repeated-CV comparison of raw vs ratios with leakage audit
EXP-009	bank_marketing	model_selection	medium	Provisional training-CV leader only; do not treat as a validated algorithm winner.	Repeat stratified CV with leakage audit for extra_trees vs hist_gradient_boosting
EXP-010	bank_marketing	optimization_reliability	medium	Do not conclude that optimization produced a meaningful CV gain from this package. Treat the holdout score as a consumed, descriptive result	Paired CV comparison of baseline vs tuned recipe on identical folds
EXP-011	breast_cancer	data_understanding	medium	Do not proceed to modeling until production input representation and split strategy are verified.	Production-schema and decision-time availability audit

ID	Dataset	Stage	Confidence	Decision	Next experiment
				The current evidence is sufficient for a p	
EXP-012	breast_cancer	leakage_audit	medium	Insufficient evidence to approve or clear any columns. The package does not establish prediction-time availability, production-time availabi	Column-level prediction-time contract audit with decision-time availability matrix
EXP-013	breast_cancer	feature_engineering	high	Do not accept the recommendation as supported. Require a rule-consistency check and a leakage/validation audit before any feature recipe is	Rule-consistency and leakage audit with one held-out evaluation
EXP-014	breast_cancer	model_selection	medium	Provisional training-CV candidate only; not enough evidence to approve a final model choice or deployment decision.	Locked-holdout verification of the current CV leader with fold-safe preprocessing
EXP-015	breast_cancer	optimization_reliability	medium	Do not claim a meaningful optimization gain from this package. Treat the holdout score as a consumed, one-time performance estimate only. Be	Paired baseline-vs-tuned nested CV with untouched holdout lock
EXP-016	heart_disease	data_understanding	medium	Do not proceed to modeling until decision-time availability, production-availability, and leakage checks are completed for the flagged featu	Decision-time and lineage audit for flagged heart_disease features
EXP-017	heart_disease	leakage_audit	medium	Insufficient evidence to determine which columns violate the prediction-time contract or are suspicious target proxies. No feature should be	Column-level decision-time contract audit for heart_disease
EXP-018	heart_disease	feature_engineering	medium	Do not treat the raw recipe as validated for production or as a proven improvement. At most, it is the current smallest candidate that match	Single held-out validation of raw vs logs under the same fold-wise pipeline
EXP-019	heart_disease	model_selection	medium	Do not treat extra_trees as a confirmed winner. Use it only as a provisional CV leader	Repeated nested CV comparison: extra_trees vs logistic_regression

ID	Dataset	Stage	Confidence	Decision	Next experiment
				pending a locked-holdout check and a repeated/nested	
EXP-020	heart_disease	optimization_reliability	medium	Do not accept any claim that optimization produced a meaningful gain from this package alone. Treat the holdout score as a completed evaluat	Baseline-vs-tuned CV audit with untouched evaluation split
EXP-021	credit_default	data_understanding	medium	Do not proceed to modeling until production-availability and leakage checks are completed.	Decision-time and raw-category availability audit
EXP-022	credit_default	leakage_audit	medium	Insufficient evidence to identify or clear any specific columns as contract-safe. No feature should be approved from this package alone.	Column-level decision-time contract audit for candidate proxies
EXP-023	credit_default	feature_engineering	medium	Do not accept the recommendation as stated; the evidence is insufficient to justify a feature choice beyond a provisional screening preferen	Nested CV comparison of raw vs selected recipe with fixed fold splits
EXP-024	credit_default	model_selection	medium	Do not treat extra_trees as the final algorithm choice yet; treat it as the current CV leader only, pending repeat/outer validation and leak	Repeat the raw-stage CV comparison with repeated stratified folds
EXP-025	credit_default	optimization_reliability	medium	Do not accept the claim that optimization produced a meaningful CV gain on the basis of this package alone. Treat the holdout score as a com	Baseline-vs-tuned CV audit with decision-time feature check
EXP-026	german_credit	data_understanding	medium	Do not approve modeling yet. The dataset can be described, but the evidence is insufficient to clear production-availability and leakage-ris	Raw-schema and availability audit for all 20 features
EXP-027	german_credit	leakage_audit	medium	Insufficient evidence to identify or approve any specific column as safe or unsafe. The current	Column-level decision-time contract audit for german_credit

ID	Dataset	Stage	Confidence	Decision	Next experiment
				package is a leakage-audit stub, not a compl	
EXP-028	german_credit	feature_engineering	medium	Do not accept the recommendation as written; the evidence is insufficient and internally inconsistent with the selection rule.	Fold-local ablation of interactions vs selected under nested validation
EXP-029	german_credit	model_selection	medium	Do not treat extra_trees as the final winner from this package alone. The evidence is sufficient only to nominate a provisional training-CV	Rule-verification and leakage-check rerun on the top two families
EXP-030	german_credit	optimization_reliability	medium	Do not claim a meaningful optimization gain from this package alone. Treat the holdout score as a finalized retrospective estimate only, and	Baseline-vs-tuned CV comparison on a fresh split with locked recipe
EXP-031	mushroom	data_understanding	medium	Do not proceed to modeling yet; resolve data representation, leakage, and production-availability questions first.	Raw-category and decision-time availability audit for all 22 features
EXP-032	mushroom	leakage_audit	medium	Insufficient evidence to identify or approve any real columns as safe or unsafe. The current package supports only that the detector is not	Column-level prediction-time contract audit with safe/unsafe ablation
EXP-033	mushroom	feature_engineering	medium	Do not accept the recommendation to use the raw recipe as stated. The smallest defensible conclusion is that the selected stage is the small	Nested holdout comparison of raw vs selected feature recipe
EXP-034	mushroom	model_selection	medium	Do not accept the claim that extra_trees is the strongest stable training-CV candidate from this package alone. The package is internally in	Repeat stratified nested CV with locked holdout comparison for top two families
EXP-035	mushroom	optimization_reliability	high	Do not conclude that optimization produced a meaningful CV gain from this package. Treat the holdout result as a completed evaluation of the	Baseline-vs-tuned nested CV comparison with leakage check

ID	Dataset	Stage	Confidence	Decision	Next experiment
EXP-036	spambase	data_understanding	medium	Do not proceed to modeling until decision-time availability, raw-source reconstruction, and leakage checks are completed.	Decision-time feature availability and raw-semantic reconstruction audit
EXP-037	spambase	leakage_audit	medium	Insufficient evidence to conclude which columns violate the prediction-time contract or are suspicious target proxies.	Column-level prediction-time availability audit for spambase
EXP-038	spambase	feature_engineering	high	Do not accept the recommendation as written; the evidence is insufficient and internally inconsistent.	Nested fold-safe comparison of raw vs selected recipe
EXP-039	spambase	model_selection	medium	Provisional training-CV candidate only; do not treat as final model choice without a locked holdout and repeated validation review.	Repeated stratified CV comparison of hist_gradient_boosting vs xgboost with a locked holdout
EXP-040	spambase	optimization_reliability	medium	Hold the optimization claim as unverified; accept only the narrow factual statement that the preselected recipe achieved the reported consum	Nested CV check of optimization gain with untouched final evaluation
EXP-041	online_shoppers	data_understanding	medium	Do not proceed to modeling until decision-time feature availability and encoding semantics are verified, and until a time-aware validation c	Time-aware schema and availability audit for all 17 features
EXP-042	online_shoppers	leakage_audit	medium	Do not approve PageValues for use until decision-time semantics and production-availability are verified; treat it as a high-priority leakag	Decision-time availability check for PageValues
EXP-043	online_shoppers	feature_engineering	medium	Do not accept the recommendation as stated. The smallest defensible next candidate from this package is the raw recipe, but only as a hypoth	Single holdout validation of raw vs full_fe under the stated tolerance rule
EXP-044	online_shoppers	model_selection	medium	Do not promote a universal algorithm-family winner from this package.	Locked holdout verification for the current CV leader

ID	Dataset	Stage	Confidence	Decision	Next experiment
				At most, treat extra_trees as the provisional training-CV candidate pe	
EXP-045	online_shoppers	optimization_reliability	medium	Do not accept the claim that optimization produced a meaningful CV gain from this package alone. Treat the holdout score as a completed sing	Baseline-vs-tuned CV audit with fold-wise deltas
EXP-046	wine_quality	data_understanding	medium	Do not approve modeling yet. The dataset description is partially supported, but production-availability and leakage review are incomplete.	Raw-schema and decision-time availability audit
EXP-047	wine_quality	leakage_audit	medium	Insufficient evidence to identify or approve any column as safe or unsafe. A decision-time and production-availability review is still requi	Column-level decision-time availability audit with safe-vs-unsafe comparison
EXP-048	wine_quality	feature_engineering	medium	Do not accept the recommendation as written. The ratios recipe is the best reported candidate for the next falsifiable screen, but the curre	Nested fold-wise comparison of raw vs ratios with leakage checks
EXP-049	wine_quality	model_selection	medium	Provisional only: extra_trees can be treated as the current training-CV candidate, but not as a validated winner. Do not finalize algorithm-	Repeated leakage-safe CV comparison of extra_trees vs lightgbm
EXP-050	wine_quality	optimization_reliability	medium	Hold. The holdout result can be reported descriptively, but the optimization-gain claim is unsupported without the missing CV comparison, an	Baseline-vs-tuned CV delta audit with decision-time feature check

Detailed reviews

EXP-001 · adult

medium confidence

data_understanding

gpt-5.4-mini

Question. What does the dataset contain, and which quality or production-availability risks must be resolved before modeling?

Decision

Do not proceed to modeling yet; first resolve representation and production-availability risks with a targeted validation pass.

Interpretation

This is a dataset-understanding package, not modeling evidence. It supports only a narrow factual summary: the dataset size, feature count, and class balance, plus a preliminary profile suggesting no obvious missingness or near-constant issues in the inspected fields. However, the package does not establish production readiness, because the categorical representation appears to have been altered by prior factorization and the shift check is explicitly random-split only. The evidence is insufficient to conclude that the dataset is safe for modeling without restoring raw category semantics and checking decision-time availability and leakage risks.

Observed evidence (cited)

- The package reports 48,842 source rows and 14 cached features for the adult dataset.
`EXP-001 / evidence.source_rows`; `EXP-001 / evidence.feature_count`
- The reported training positive rate is 0.239375.
`EXP-001 / evidence.positive_rate_train`
- The feature profile table shows zero missingness for the listed features and includes random-split PSI, top-value frequency, unique count, and unique ratio fields.
`EXP-001 / evidence.feature_profiles`
- The evidence bundle itself states that the cached UCI matrices factorized categoricals before this campaign and that random train/holdout PSI is only a smoke check.
`EXP-001 / claims[1].limitations`

Challenges to claims

- **medium** **EXP-001-C1** — The statement is factual, but it omits whether the 14 features are raw, encoded, or cached representations; the bundle itself warns that categoricals were factorized before this campaign, which affects downstream production interpretation.
- **high** **EXP-001-C2** — The claim that 0 features triggered review rules is not sufficient to establish safety because the review rules shown are limited to missingness, identifier, near-constant, and random-split shift; they do not cover target leakage, decision-time availability, or production schema mismatch.

Risks & missing evidence

- Raw category semantics are not shown; the package notes prior factorization of categoricals, so production encoders may not match the cached matrices.
- No decision-time availability review is provided for any feature, so leakage via post-outcome or unavailable-at-scoring fields is not ruled out.
- Random train/holdout PSI is not a temporal or deployment stability test; production drift risk remains unassessed.

- The feature profile excerpt is incomplete in the provided bundle, so the claim that 0 features triggered review rules is not fully auditable from the visible evidence alone.
- The evidence does not show target definition timing, label collection window, or whether any features are derived from the target period.

Smallest next experiment

Decision-time and raw-category audit for adult features

Hypothesis: At least one feature in the current cached representation is either unavailable at decision time, semantically altered by prior factorization, or otherwise unsuitable for production encoding without restoration of raw categories.

Success gate: Pass only if every feature is mapped to a decision-time availability status, every categorical feature is shown in raw semantic form with a production encoder plan, and no feature is flagged as post-outcome, unavailable, or schema-incompatible.

Why smallest: This is the minimal falsifiable step that directly tests the main unresolved risks before any modeling work, without requiring model training or additional performance evaluation.

Evidence: [campaigns/model_building_50_v1/results/EXP-001_adult_data_understanding.json](#)

EXP-002 · adult

medium confidence

leakage_audit

gpt-5.4-mini

Question. Which columns violate the prediction-time contract or behave like suspicious target proxies?

Decision

Insufficient evidence to conclude which columns are leakage or target proxies. The current package supports only a detector sanity check, not a column-level leakage determination.

Interpretation

The package shows only that the detector can flag an injected exact-target canary, but it does not identify any real columns that violate the prediction-time contract or behave as target proxies. The absence of declared leakage features is not evidence of absence, and the missing safe-vs-unsafe CV result leaves the audit unsupported for ranking or validating suspicious columns.

Observed evidence (cited)

- The synthetic exact-target canary was reported as caught by the leakage detector.
EXP-002 / evidence.detector_canary_passed
- No declared leakage features were provided for the adult dataset.
EXP-002 / evidence.declared_leakage_features
- No safe-vs-unsafe training CV result was provided.
EXP-002 / evidence.safe_vs_unsafe_training_cv

Challenges to claims

- **medium** EXP-002-C1 — The claim only demonstrates detector behavior on a synthetic canary. It does not establish that any real dataset column is a leakage source or proxy.
- **high** EXP-002-C2 — The statement 'Declared decision-time exclusions for adult: none' is unsupported by the provided evidence because no decision-time semantics or production-availability review is included.

Risks & missing evidence

- No column-level evidence is provided for actual suspicious features, so the question remains unanswered.
- The canary result only validates detector sensitivity to a synthetic exact-target leak; it does not validate recall on realistic leakage patterns.
- No decision-time semantics review is shown for any candidate column, so prediction-time contract violations cannot be confirmed.
- No production-availability review is shown, so even a plausible proxy cannot be approved or ruled out for deployment use.
- The missing safe-vs-unsafe training CV result prevents assessment of whether excluding suspected columns changes model behavior in a way consistent with leakage removal.

Smallest next experiment

Column-level decision-time contract audit for adult

Hypothesis: At least one candidate column in adult either is unavailable at prediction time or behaves like a suspicious target proxy when evaluated against decision-time semantics and simple leakage heuristics.

Success gate: For each column, provide decision-time availability status plus a minimal evidence table with name-based review, uniqueness, and univariate target association; flag any column that fails the contract or is a strong proxy candidate. Success is achieved if the audit produces a reproducible list of flagged columns with explicit rationale, or confirms none after review.

Why smallest: This is the smallest falsifiable step because it adds only a column-by-column contract review and lightweight heuristics, without requiring model retraining or broader pipeline changes.

Evidence: `campaigns/model_building_50_v1/results/EXP-002_adult_leakage_audit.json`

EXP-003 · adult

medium confidence

feature_engineering

gpt-5.4-mini

Question. Which train-fitted feature recipe improves validation evidence without feature dilution?

Decision

Do not accept the claim that the raw/selected recipe should be used next based on this package alone. The smallest defensible action is to rerun the selection logic against the actual best stage and verify the criterion on an untouched validation split, with leakage checks and decision-time availability review before any feature approval.

Interpretation

The evidence does not support the recommendation to use the selected/raw recipe as the next screen under the stated selection rule. Based on the reported means, the best validation evidence is the interactions stage, while the selected stage is materially worse on mean ROC-AUC and also does not satisfy the stated within-0.002 criterion. The package also does not establish that the chosen recipe avoids feature dilution in a decision-safe sense, because the feature-count reduction is paired with lower validation evidence and the stability table shows several engineered features appearing in only 1-2 folds, which raises selection-instability concerns rather than resolving them. This is predictive evidence only; it does not justify causal claims about the recipe itself.

Observed evidence (cited)

- The package reports six stages with LightGBM baseline CV results on adult: raw, logs, ratios, interactions, full_fe, and selected.
`EXP-003 / evidence.stage_results[*].stage, evidence.stage_results[*].model, evidence.stage_results[*].optimization`
- The reported mean ROC-AUC values are raw 0.895549250905629, logs 0.895549250905629, ratios 0.8943588869920177, interactions 0.8956653185415995, full_fe 0.8936509629035153, and selected 0.8861543390223239.
`EXP-003 / evidence.stage_results[*].metrics.roc_auc.mean`
- The reported feature_count_mean values are raw 14.0, logs 22.0, ratios 37.0, interactions 45.0, full_fe 50.0, and selected 25.0.
`EXP-003 / evidence.stage_results[*].feature_count_mean`
- The selection rule is stated as choosing the smallest feature matrix within 0.002 mean training-CV ROC-AUC of the best stage.
`EXP-003 / evidence.selection_rule`
- The selected stage has lower mean ROC-AUC than the raw stage and is not within 0.002 of the best reported stage mean ROC-AUC.
`EXP-003 / evidence.stage_results[5].metrics.roc_auc.mean, EXP-003 / evidence.stage_results[0].metrics.roc_auc.mean, EXP-003 / evidence.stage_results[3].metrics.roc_auc.mean`
- The selected stage includes features with observed_folds less than 3 for some engineered terms, including inter_relationship_x_log1p_education (2), ratio_age_div_hours-per-week (1), and inter_relationship_x_occupation (1).
`EXP-003 / evidence.stage_results[5].feature_stability[*].observed_folds`

Challenges to claims

- **high** **EXP-003-C1** — The recommendation says to use the raw feature recipe, but the reported selection rule would require the smallest matrix within 0.002 of the best stage; the selected stage is not

within that tolerance of the best reported mean ROC-AUC, and the raw stage is not the best stage by mean ROC-AUC.

- **high** EXP-003-C1 — The claim implies the recipe improves validation evidence without feature dilution, but the selected stage has lower mean ROC-AUC than raw and the larger interaction/full_fe stages do not improve over raw either.
- **high** EXP-003-C1 — The evidence does not show fold-local fitting for all engineered features, so leakage from train-fitted transformations or selection cannot be ruled out.
- **medium** EXP-003-C1 — The feature stability table shows several engineered features with observed_folds below 3, which weakens the case that the recipe is stable enough to prefer.

Risks & missing evidence

- The package reports only 3 CV folds and repeats=1, which is thin evidence for feature-selection stability.
- The phrase 'train-fitted feature recipe' creates leakage risk unless every transformation, selection, and importance computation was fit strictly inside each training fold; the package does not explicitly document fold-local fitting for all engineered features.
- The selection rule references 'mean training-CV ROC-AUC' but the evidence bundle only exposes fold metrics and aggregated means; it is unclear whether the rule was applied to the same metric and same stage set without peeking.
- The selected stage contains features observed in only 1-2 folds, suggesting instability and possible overfitting to fold idiosyncrasies.
- No decision-time or production-availability review is provided for the engineered features, especially for generic mathematical transforms and interaction terms.
- No untouched holdout or nested-CV confirmation is provided, so the apparent ranking among stages may be optimistic.
- No comparison against a simple baseline with identical CV protocol is explicitly summarized in the package, so the practical value of added feature engineering is not established.

Smallest next experiment

Nested-CV rerun of the selection rule with fold-local feature fitting

Hypothesis: If the feature recipe truly improves validation evidence without dilution, then a nested CV protocol with all transforms fit inside each training fold will reproduce the same stage ranking and the selected smallest matrix will remain within the stated ROC-AUC tolerance of the best stage.

Success gate: On an untouched outer validation split, the chosen recipe must satisfy the predeclared within-0.002 mean ROC-AUC rule relative to the best stage under the same protocol, and the selected matrix must not lose more than the predeclared tolerance versus the best stage; otherwise the claim fails.

Why smallest: This changes only the validation protocol and selection placement, which is the minimal falsifiable test needed to separate genuine fold-local evidence from possible leakage or optimistic selection.

Evidence: campaigns/model_building_50_v1/results/EXP-003_adult_feature_engineering.json

EXP-004 · adult

medium confidence

model_selection

gpt-5.4-mini

Question. Which algorithm family provides the strongest stable training-CV evidence for this dataset profile?

Decision

Provisional selection only: lightgbm is the current training-CV leader under the provided rule, but the evidence is insufficient to finalize an algorithm family choice for production or to claim stability beyond this single 3-fold run.

Interpretation

Based on the stated selection rule, lightgbm is the top training-CV candidate, but the margin over hist_gradient_boosting is extremely small and may not be practically meaningful without repeated CV or a locked holdout. The evidence supports a ranking under the chosen heuristic, not a claim that lightgbm is universally best.

Observed evidence (cited)

- The package compares five baseline models on the adult dataset using raw features with 3 CV folds each.
`EXP-004 / evidence.model_results`
- The selection rule is mean ROC-AUC minus 0.25×fold standard deviation, with runtime used as a tiebreaker, and it is explicitly training-CV only.
`EXP-004 / evidence.selection_rule`
- The reported mean ROC-AUC values are 0.8469670634315208 for logistic_regression, 0.8891408870082996 for extra_trees, 0.8953165275448374 for hist_gradient_boosting, 0.895549250905629 for lightgbm, and 0.8919567435623993 for xgboost.
`EXP-004 / evidence.model_results[*].metrics.roc_auc.mean`
- The reported ROC-AUC standard deviations are 0.004667970022545528 for logistic_regression, 0.0075755771597569984 for extra_trees, 0.008934548232198785 for hist_gradient_boosting, 0.006452617087274231 for lightgbm, and 0.008328601821275487 for xgboost.
`EXP-004 / evidence.model_results[*].metrics.roc_auc.std`
- The reported runtime is lowest for logistic_regression at 0.21762270799990802 seconds and highest for extra_trees at 1.484125375000076 seconds.
`EXP-004 / evidence.model_results[*].elapsed_seconds`
- The lightgbm entry reports mean ROC-AUC 0.895549250905629 with std 0.006452617087274231 over 3 folds.
`EXP-004 / evidence.model_results[3].metrics.roc_auc`

Challenges to claims

- **medium** **EXP-004-C1** — The statement 'lightgbm is the training-CV model candidate' is supported only as a ranking under the stated heuristic, not as a robust winner; hist_gradient_boosting has nearly identical mean ROC-AUC and a different runtime profile.
- **high** **EXP-004-C1** — The evidence does not show repeated CV or a locked holdout, so the stability implied by 'strongest stable training-CV evidence' is not well established.
- **high** **EXP-004-C1** — No leakage audit is included for raw features, fold construction, or any preprocessing steps, so the validity of the CV comparison cannot be fully verified.

- **high** **EXP-004-C1** — The recommendation could be misread as a deployment approval, but there is no decision-time or production-availability review.

Risks & missing evidence

- Only 3 folds and 1 repeat are reported; stability is weakly estimated and sensitive to split choice.
- No locked holdout or external test evidence is provided, so generalization is unverified.
- The selection rule uses training CV only; this can overfit model choice to the CV procedure itself.
- No evidence is provided that preprocessing, encoding, or feature engineering were fit strictly inside each fold; leakage risk is unknown.
- No decision-time or production-availability review is shown, so operational suitability cannot be assessed.
- The ranking margin between lightgbm and hist_gradient_boosting is tiny, so the apparent leader may be an artifact of CV noise.
- Calibration and threshold-dependent metrics are reported, but no threshold-selection protocol is documented.

Smallest next experiment

Repeated stratified CV comparison of lightgbm vs hist_gradient_boosting with leakage audit

Hypothesis: If the current ranking is stable, lightgbm will remain the top model family across repeated stratified CV runs and its advantage over hist_gradient_boosting will persist within a narrow uncertainty band.

Success gate: Run at least 10 repeated stratified CV evaluations with identical preprocessing inside folds; declare lightgbm preferred only if its mean ROC-AUC remains higher than hist_gradient_boosting and the paired fold-wise difference is consistently positive in a majority of repeats, with no leakage findings.

Why smallest: This changes only the evaluation protocol and compares only the two closest candidates, making it the minimal falsifiable test of whether the current leader is stable rather than a single-split artifact.

Evidence: campaigns/model_building_50_v1/results/EXP-004_adult_model_selection.json

EXP-005 · adult

high confidence

optimization_reliability

gpt-5.4-mini

Question. Does optimization produce a meaningful CV gain, and how does the preselected recipe perform on the once-consumed holdout?

Decision

Insufficient evidence to conclude that optimization produced a meaningful CV gain. The once-consumed holdout performance is reported, but it should not be used to justify feature promotion or deployment without decision-time and production-availability review.

Interpretation

The holdout score is a factual performance result, but the package does not provide enough evidence to answer whether optimization produced a meaningful CV gain. The only explicit optimization criterion is a 0.001 mean-CV improvement threshold, yet no baseline CV score, tuned CV score, or delta is shown. The feature evidence supports a cautionary recommendation about promotion criteria, but it does not establish that any feature should be promoted or that the listed importances are decision-safe.

Observed evidence (cited)

- The once-consumed holdout result for the preselected lightgbm/lightgbm_C02 recipe is ROC-AUC 0.9096536249451939 with bootstrap 95% interval [0.8854969409371798, 0.929674276420258].
EXP-005 / evidence.holdout_metrics; EXP-005 / evidence.holdout_roc_auc_bootstrap_ci; EXP-005 / evidence.holdout_consumed
- The selection rule says to accept a tuned candidate only when mean CV ROC-AUC improves by at least 0.001 over baseline; otherwise prefer baseline.
EXP-005 / evidence.selection_rule
- The reported feature-importance table includes fnlwgt, age, hours-per-week, education-num, occupation, relationship, capital-gain, workclass, education, capital-loss, marital-status, native-country, sex, and race with normalized importances.
EXP-005 / evidence.top_final_features
- The CV feature-stability table reports observed_folds = 3 for each listed feature, with top10_frequency ranging from 0.0 to 1.0.
EXP-005 / evidence.cv_feature_stability
- The production feature rule requires decision-time availability, cross-fold stability, contractual obtainability, monitoring for missingness/drift, and more than single-fit importance.
EXP-005 / evidence.production_feature_rule

Challenges to claims

- **high** EXP-005-C1 — The holdout ROC-AUC is reported correctly as a factual metric, but the package does not support any inference that optimization was meaningful because no baseline-versus-tuned CV comparison is included.
- **medium** EXP-005-C2 — The recommendation is directionally sound, but the evidence does not demonstrate decision-time availability or production monitoring for the listed features; it is a policy statement rather than a validated result.

Risks & missing evidence

- No baseline CV ROC-AUC, tuned CV ROC-AUC, or CV delta is provided, so the optimization-gain question is unanswered.

- The holdout is marked consumed, so further model selection on this holdout would risk leakage or overfitting to the test set.
- Only 3 CV folds are reported for feature stability, which is thin evidence for ranking feature robustness.
- No decision-time availability check is shown for the listed features, so some may be unavailable at prediction time.
- No production contract or data-pipeline evidence is shown for feature obtainability, missingness, or drift monitoring.
- Feature importance alone is not sufficient evidence for feature promotion, especially without stability thresholds or ablation tests.
- The package does not show calibration or threshold-selection validation beyond a single 0.5 threshold summary.

Smallest next experiment

Minimal CV delta audit against baseline

Hypothesis: The selected tuned recipe improves mean cross-validated ROC-AUC over the baseline by at least 0.001 under the same fold split and preprocessing pipeline.

Success gate: Report baseline mean CV ROC-AUC, tuned mean CV ROC-AUC, and their difference on the same folds; success only if tuned minus baseline ≥ 0.001 and the fold-wise improvement is not driven by a single outlier fold.

Why smallest: This is the smallest falsifiable experiment because it reuses the existing CV setup and only adds the missing baseline-versus-tuned comparison needed to answer the optimization-gain question.

Evidence: [campaigns/model_building_50_v1/results/EXP-005_adult_optimization_reliability.json](#)

EXP-006 · bank_marketing

medium confidence

data_understanding

gpt-5.4-mini

Question. What does the dataset contain, and which quality or production-availability risks must be resolved before modeling?

Decision

Do not approve modeling readiness yet. The dataset description is usable, but production-availability and leakage risks remain unresolved.

Interpretation

This evidence supports a narrow dataset-understanding summary: the dataset is a moderately sized binary classification table with 45,211 rows, 16 cached features, and an imbalanced positive class near 11.7%. The sample feature profiles suggest no obvious missingness in the inspected columns and no strong random-split distribution shift in the reported PSI values. However, the package does not establish that the current feature representation is production-ready, because the categorical variables were already factorized in the cached matrices and the evidence only checks random splits rather than decision-time or temporal availability. Therefore, the main unresolved issues are feature semantics restoration, production-time availability of each field, and leakage/validation design beyond random holdout.

Observed evidence (cited)

- The package reports 45,211 source rows for bank_marketing.
`EXP-006 / evidence.source_rows`
- The package reports 16 cached features.
`EXP-006 / evidence.feature_count`
- The reported training positive rate is 0.116875.
`EXP-006 / evidence.positive_rate_train`
- The provided feature profile sample shows zero missingness for the listed features.
`EXP-006 / evidence.feature_profiles[].missing_rate`
- The provided feature profile sample includes random-split PSI values for some features, all shown as small in the sample.
`EXP-006 / evidence.feature_profiles[].random_split_psi`
- The package explicitly notes that cached UCI matrices factorized categoricals before this campaign and that raw category semantics must be restored for production encoders.
`EXP-006 / claims[EXP-006-C2].limitations[0]`

Challenges to claims

- **medium** **EXP-006-C1** — The row count, feature count, and positive rate are factual dataset descriptors, but the package does not show how the cached factorized features relate to raw production fields; the claim is incomplete as a basis for modeling readiness.
- **high** **EXP-006-C2** — The statement that 0 features triggered review rules is too strong without the full feature list and without decision-time/temporal checks; random-split PSI is not sufficient to rule out production shift or leakage.

Risks & missing evidence

- No evidence that the factorized categorical codes can be mapped back to raw category values for a production encoder pipeline.

- No decision-time availability audit: the package does not say whether each feature is known before the prediction timestamp.
- No temporal validation evidence: random-split PSI cannot rule out time-based drift or leakage from post-event fields.
- No confirmation that the target label is defined strictly after feature collection, so label leakage remains possible.
- Only a sample of feature profiles is shown; the package does not provide all 16 features in the visible evidence, so completeness of the '0 features triggered review rules' claim is not fully inspectable here.
- No evidence on duplicate rows, entity overlap, or repeated-campaign leakage across splits.
- No evidence on missing-value handling for the full table beyond the displayed sample.

Smallest next experiment

Decision-time availability and raw-category restoration audit

Hypothesis: At least one feature in the current cached table is either unavailable at prediction time or cannot be faithfully reconstructed from raw production inputs, which would make the current feature set unsafe to model without revision.

Success gate: For every feature, document (1) whether it is available at decision time, (2) its raw source field, and (3) whether the cached factorized representation can be reproduced from production inputs; the experiment passes only if all 16 features are confirmed available and reproducible with no unresolved temporal or semantic mismatch.

Why smallest: This is the smallest falsifiable step because it reuses the existing table and metadata, requires no model training, and directly tests the highest-risk unresolved issue: whether the features can exist in production in the same form as in the cached dataset.

Evidence: [campaigns/model_building_50_v1/results/EXP-006_bank_marketing_data_understanding.json](#)

EXP-007 · bank_marketing

medium confidence

leakage_audit

gpt-5.4-mini

Question. Which columns violate the prediction-time contract or behave like suspicious target proxies?

Decision

Do not approve duration for use until decision-time semantics and production availability are verified; treat it as a leakage-risk candidate, not a confirmed target proxy.

Interpretation

The evidence supports that duration is a suspicious feature under the declared prediction-time contract because it is explicitly excluded and it materially changes cross-validated predictive performance when included. However, the package does not establish leakage by itself: the semantics of duration at decision time, its availability in production, and whether the CV protocol matches the intended deployment setting are not demonstrated. The canary result only validates the detector on a synthetic test, not the real column.

Observed evidence (cited)

- A synthetic exact-target canary was reported as detected by the leakage detector.
EXP-007 / evidence.detector_canary_passed
- The declared decision-time exclusion list contains only duration.
EXP-007 / evidence.declared_leakage_features
- The safe and unsafe 3-fold CV runs differ by one feature on average (15.0 vs 16.0 features).
EXP-007 / evidence.safe_vs_unsafe_training_cv.safe.feature_count_mean; EXP-007 / evidence.safe_vs_unsafe_training_cv.unsafe.feature_count_mean
- The unsafe run assigns nontrivial normalized importance to duration across all three folds.
EXP-007 / evidence.safe_vs_unsafe_training_cv.unsafe.feature_stability[0]
- The unsafe run shows higher CV ROC AUC than the safe run.
EXP-007 / evidence.safe_vs_unsafe_training_cv.unsafe.metrics.roc_auc.mean; EXP-007 / evidence.safe_vs_unsafe_training_cv.safe.metrics.roc_auc.mean

Challenges to claims

- **medium** **EXP-007-C1** — The canary detection only shows the detector can catch a synthetic exact-target canary. It does not validate that the detector correctly identifies real leakage columns in this dataset.
- **high** **EXP-007-C2** — Calling duration a decision-time exclusion is plausible but not proven here; the evidence only shows a declared exclusion and a CV lift. The limitation text itself says only decision-time semantics can confirm leakage.
- **high** **EXP-007-C2** — The +0.1842 AUC lift is not sufficient to label a feature as leakage or proxy; predictive lift can arise from legitimate signal, correlation structure, or preprocessing artifacts.

Risks & missing evidence

- No decision-time semantics review for duration is provided; the package does not show when the value is known relative to prediction.
- No production-availability evidence is provided; a feature can be non-leaky in training yet unavailable at serving time.

- The CV protocol is not described in enough detail to rule out temporal, group, or preprocessing leakage.
- Only one declared exclusion is listed; other columns with target-proxy behavior may be missed because the audit appears anchored to a single canary and univariate lift.
- The reported AUC lift is from training-CV comparison only; this is predictive evidence, not proof of leakage or causal utility.
- No ablation or permutation test is shown for duration specifically, so the marginal contribution may be confounded by correlated features.

Smallest next experiment

Decision-time availability audit for duration

Hypothesis: If duration is unavailable at prediction time or is only known after the outcome-relevant event, then removing it should not be treated as a model-quality regression but as contract enforcement; if it is available before prediction, the leakage concern weakens and the observed lift needs a non-leakage explanation.

Success gate: For duration, produce a signed decision-time data dictionary entry and serving-path check showing whether the value exists before scoring for at least one production-like request trace; if unavailable, mark it contract-violating. If available, rerun a strict ablation with identical preprocessing and a leakage-safe split, and require the AUC delta to persist under the same split before any further suspicion is escalated.

Why smallest: This is the smallest falsifiable step because it tests the single disputed column directly, without expanding the audit scope or changing the model family.

Evidence: `campaigns/model_building_50_v1/results/EXP-007_bank_marketing_leakage_audit.json`

EXP-008 · bank_marketing

medium confidence

feature_engineering

gpt-5.4-mini

Question. Which train-fitted feature recipe improves validation evidence without feature dilution?

Decision

Do not accept the recommendation as supported; request a minimal confirmatory experiment before choosing a recipe.

Interpretation

The evidence does not support the recommendation to use the ratios recipe as the next model screen under the stated selection rule. Ratios has the highest reported mean ROC-AUC among the shown stages, but the package does not demonstrate that this improvement is robust, statistically meaningful, or free of leakage/validation artifacts. The selected stage is smaller than full_fe, but it is not within 0.002 of the best stage by the reported means, so the stated rule is not satisfied by the provided numbers. The package also mixes feature-count comparisons with model-quality comparisons without showing a decision-time or production-availability review for the generated features.

Observed evidence (cited)

- The experiment compares five stages on bank_marketing: raw, logs, ratios, interactions, full_fe, and selected, all using LightGBM with 3 folds and 1 repeat.
`EXP-008 / evidence.stage_results[*].stage; evidence.stage_results[*].model; evidence.stage_results[*].folds; evidence.stage_results[*].repeats`
- The ratios stage reports mean ROC-AUC 0.7171968586626486 with feature_count_mean 38.0.
`EXP-008 / evidence.stage_results[2].metrics.roc_auc.mean; evidence.stage_results[2].feature_count_mean`
- The selected stage reports mean ROC-AUC 0.7028679428349656 with feature_count_mean 25.0.
`EXP-008 / evidence.stage_results[5].metrics.roc_auc.mean; evidence.stage_results[5].feature_count_mean`
- The full_fe stage reports mean ROC-AUC 0.7135749834486221 with feature_count_mean 51.0.
`EXP-008 / evidence.stage_results[4].metrics.roc_auc.mean; evidence.stage_results[4].feature_count_mean`
- The selection rule is to choose the smallest feature matrix within 0.002 mean training-CV ROC-AUC of the best stage.
`EXP-008 / evidence.selection_rule`
- The ratios stage has 20 listed features in feature_stability, including multiple ratio features with nonzero mean_normalized_importance and top10_frequency values up to 1.0.
`EXP-008 / evidence.stage_results[2].feature_stability[*].feature;
evidence.stage_results[2].feature_stability[*].mean_normalized_importance;
evidence.stage_results[2].feature_stability[*].top10_frequency`

Challenges to claims

- **high** **EXP-008-C1** — The recommendation says to use the ratios recipe for the next model screen because it achieved 0.7172 mean training-CV ROC-AUC with about 38 features. The reported mean ROC-AUC and feature_count_mean are present, but the claim is a recommendation that is not justified by a robustness check, and the stated selection rule is not satisfied by the provided means because the selected stage mean ROC-AUC is 0.7028679428349656, not within 0.002 of the best stage mean 0.7171968586626486.
- **high** **EXP-008-C1** — The claim implicitly treats higher training-CV ROC-AUC as sufficient evidence to prefer the ratios recipe. This is unsupported because no holdout, repeated CV, or statistical test is provided, so the improvement may be within sampling noise.

- **medium** **EXP-008-C1** — The claim does not address decision-time and production-availability constraints for generated ratio features, despite the stated limitation that generic mathematical feature generation may lack domain meaning.

Risks & missing evidence

- No explicit train-fitted vs validation-fitted feature construction audit is shown for the ratio/interactions recipes; leakage risk remains unresolved.
- Only 3 folds and 1 repeat are reported; there is no repeated CV or nested validation to assess stability of the apparent ROC-AUC differences.
- The package reports mean ROC-AUC but does not provide a paired significance test or fold-wise comparison between stages.
- The selection rule references the best stage, but the evidence does not show that the selected stage is within 0.002 mean ROC-AUC of the best stage.
- Feature semantics are not reviewed for production availability or decision-time computability, especially for generated ratios and interactions.
- The evidence emphasizes training-CV metrics; there is no untouched holdout or temporal split evidence to reduce optimism bias.
- Some selected-stage features appear in fewer than all folds, suggesting instability in feature selection or generation.

Smallest next experiment

Paired repeated-CV comparison of raw vs ratios with leakage audit

Hypothesis: If the ratios recipe truly improves validation evidence without feature dilution, then across repeated folds it will outperform the raw baseline on ROC-AUC while preserving or improving calibration, and the feature construction will pass a train-only leakage audit.

Success gate: Run repeated stratified CV with identical splits for raw and ratios, plus a train-only feature-generation audit. Accept only if ratios beats raw by a positive median fold-wise ROC-AUC difference and the lower bound of the paired fold-wise difference is above 0, while no leakage is detected in the audit.

Why smallest: This changes only the validation design and adds a leakage check; it does not introduce new feature families or model classes, so it is the minimal falsifiable test of the current recommendation.

Evidence: campaigns/model_building_50_v1/results/EXP-008_bank_marketing_feature_engineering.json

EXP-009 · bank_marketing

medium confidence

model_selection

gpt-5.4-mini

Question. Which algorithm family provides the strongest stable training-CV evidence for this dataset profile?

Decision

Provisional training-CV leader only; do not treat as a validated algorithm winner.

Interpretation

Based on the provided training-CV evidence and the stated ranking rule, `extra_trees` is the leading candidate by the package's own criterion. However, the evidence is only weakly stable: the ROC-AUC spread across folds is non-trivial, the sample is only 3 folds with 1 repeat, and the package does not show a locked holdout or any decision-time/production-availability review. The question asks for the strongest stable training-CV evidence, but the current package supports only a provisional training-CV preference, not a robust winner claim.

Observed evidence (cited)

- The package evaluates five model families on the `bank_marketing` dataset under the `ratios` stage with training CV only, using 3 folds and 1 repeat per model.

EXP-009 / `evidence.model_results[*].{model,stage,folds,repeats,optimization}`

- The stated selection rule ranks models by mean ROC-AUC minus $0.25 \times$ fold standard deviation, then runtime, and explicitly uses training CV only.

EXP-009 / `evidence.selection_rule`

- `extra_trees` has the highest reported mean ROC-AUC among the listed models at 0.7288407757459535 with std 0.02954342984875996, and its fold ROC-AUC values are 0.7438938428874735, 0.7478259023354563, and 0.6948025820149306.

EXP-009 / `evidence.model_results[1].metrics.roc_auc`; EXP-009 / `evidence.model_results[1].fold_metrics[*].roc_auc`

- `hist_gradient_boosting` has the lowest reported runtime among the listed models at 0.8132207079997897 seconds, while `extra_trees` is 2.0212677079998684 seconds.

EXP-009 / `evidence.model_results[2].elapsed_seconds`; EXP-009 / `evidence.model_results[1].elapsed_seconds`

- The reported feature stability for `extra_trees` includes repeated prominence of `poutcome` and `log1p_poutcome` across all 3 folds, but the package does not provide a leakage audit or feature provenance for these engineered features.

EXP-009 / `evidence.model_results[1].feature_stability[*]`; EXP-009 / `evidence.model_results[1].stage`

Challenges to claims

- **medium** EXP-009-C1 — The statement says `extra_trees` is the training-CV model candidate, but the evidence bundle's `model_results` entry for `extra_trees` is actually `model=extra_trees` with mean ROC-AUC 0.7288407757459535; the claim is directionally supported, yet the package does not demonstrate that this is the strongest stable evidence under a validated protocol because only 3 folds and 1 repeat are shown.
- **high** EXP-009-C1 — The claim uses a recommendation framing, but the evidence lacks a locked holdout and any production-availability review, so it overreaches if interpreted as a deployable choice.
- **high** EXP-009-C1 — The claim references `ratios` features, but the package does not provide a leakage check for engineered ratio features or the source timestamps/availability of inputs.

- **medium** **EXP-009-C1** — The phrase 'strongest stable training-CV evidence' is not fully justified because the stability signal is based on only 3 folds and the reported fold standard deviation remains material relative to the mean.

Risks & missing evidence

- Only 3 CV folds and 1 repeat are shown, which is thin evidence for stability.
- No locked holdout results are provided, so the training-CV leader may not generalize.
- No leakage audit is shown for engineered ratio/log features such as `log1p_poutcome` and `ratio_pdays_div_age`.
- No description of the CV splitter is provided; if the data are time-ordered or grouped, standard CV could leak information.
- The selection rule uses runtime as a tie-breaker after a custom stability-adjusted score, but the package does not justify why this rule is appropriate for the decision context.
- No decision-time or production-availability review is included, so feature/model readiness cannot be approved.
- The evidence does not compare calibration or threshold-dependent utility in a decision setting, despite reporting calibration metrics and `threshold=0.5` outputs.

Smallest next experiment

Repeat stratified CV with leakage audit for `extra_trees` vs `hist_gradient_boosting`

Hypothesis: If the `extra_trees` lead is real and not an artifact of fold variance or leakage, then under a repeated stratified CV protocol with the same feature set and a documented leakage audit, `extra_trees` will retain the best mean ROC-AUC minus $0.25 \times \text{std}$ score over `hist_gradient_boosting`.

Success gate: Run 5x repeated stratified CV with the same preprocessing and a documented feature-availability audit; success is met only if `extra_trees` remains top-ranked by the predeclared score and its mean ROC-AUC exceeds the runner-up by at least one pooled fold standard error, with no leakage findings.

Why smallest: This changes only the validation protocol and adds a leakage audit while keeping the dataset, feature set, and top candidate comparison fixed, making the result directly falsifiable with minimal extra scope.

Evidence: `campaigns/model_building_50_v1/results/EXP-009_bank_marketing_model_selection.json`

EXP-010 · bank_marketing

medium confidence

optimization_reliability

gpt-5.4-mini

Question. Does optimization produce a meaningful CV gain, and how does the preselected recipe perform on the once-consumed holdout?

Decision

Do not conclude that optimization produced a meaningful CV gain from this package. Treat the holdout score as a consumed, descriptive result only. Require a direct baseline-vs-tuned CV comparison, with the same folds and metric, before any claim about optimization benefit. Do not approve feature promotion without separate decision-time and production-availability review.

Interpretation

The holdout result is a single post-selection estimate with a bootstrap interval, so it supports only a performance description, not a causal or durable claim about optimization benefit. The evidence package does not include the baseline CV ROC-AUC, the tuned candidate CV ROC-AUC, or their uncertainty, so the question of whether optimization produced a meaningful CV gain is not answered from the provided evidence alone. The feature-stability and production-rule evidence appropriately cautions against promoting features from importance alone, but it does not by itself validate any feature for deployment.

Observed evidence (cited)

- The once-consumed holdout evaluation for the preselected extra_trees/extra_trees_C02 recipe reports ROC-AUC 0.7718039901151228 with bootstrap 95% interval [0.712635957605465, 0.8290159504824489].
[EXP-010 / evidence.holdout_metrics](#); [EXP-010 / evidence.holdout_roc_auc_bootstrap_ci](#); [EXP-010 / evidence.holdout_consumed](#)
- The holdout metrics also include accuracy 0.88625, average precision 0.3915037952265037, Brier score 0.08698659707238321, ECE_10 0.027231064943742295, F1 0.2222222222222222, precision 0.5652173913043478, recall 0.13829787234042554, log loss 0.3065981479262125, threshold 0.5, and fit time 0.622739291000471 seconds.
[EXP-010 / evidence.holdout_metrics](#)
- The feature stability table is based on 3 observed folds for many features, with several features appearing in the top 10 in all folds and others appearing less consistently.
[EXP-010 / evidence.cv_feature_stability](#)
- The package states a production feature rule requiring decision-time availability, cross-fold stability, contractual obtainability, and monitoring for missingness/drift.
[EXP-010 / evidence.production_feature_rule](#)
- The selection rule says the tuned candidate should be used only when mean ROC-AUC improves by at least 0.001 over baseline; otherwise baseline is preferred.
[EXP-010 / evidence.selection_rule](#)

Challenges to claims

- **high** **EXP-010-C1** — The holdout ROC-AUC is reported correctly as a factual result, but the package does not provide the baseline CV score or tuned CV score needed to answer whether optimization produced a meaningful CV gain.
- **medium** **EXP-010-C1** — Because the holdout is marked consumed, any stronger interpretation about model selection or future performance would be unsupported by the evidence provided.

- **medium** **EXP-010-C2** — The recommendation is directionally sound, but it is not backed by per-feature decision-time availability or production-contract evidence in this package.
- **low** **EXP-010-C2** — Cross-fold stability is only shown over 3 folds, which is limited evidence for a strong stability claim.

Risks & missing evidence

- Missing baseline CV ROC-AUC and tuned CV ROC-AUC, so the stated 0.001 improvement rule cannot be checked from the package.
- Missing uncertainty for the CV comparison, so even a small mean difference may be noise.
- The holdout has already been consumed, so repeated use risks adaptive overfitting if it is consulted again for model selection.
- Only 3 observed folds are shown for feature stability, which is thin evidence for ranking feature robustness.
- Feature importance is not the same as deployability; decision-time availability and production contract checks are not evidenced per feature.
- No calibration or threshold-selection rationale is provided for the reported threshold 0.5 metrics.
- No subgroup, temporal, or drift analysis is provided, so generalization risk remains unquantified.

Smallest next experiment

Paired CV comparison of baseline vs tuned recipe on identical folds

Hypothesis: The tuned extra_trees/extra_trees_C02 recipe does not improve mean CV ROC-AUC over the baseline by at least 0.001 when evaluated on the same folds, or else the improvement is large enough to clear that predeclared gate.

Success gate: Run baseline and tuned models on the same CV splits; accept optimization only if mean ROC-AUC(tuned) - mean ROC-AUC(baseline) $\geq$ 0.001 and the fold-wise difference is directionally consistent enough to rule out a trivial fluctuation. Otherwise keep the baseline.

Why smallest: This is the minimal falsifiable test needed to answer the optimization question already posed, using the existing metric and selection rule without adding new modeling complexity.

Evidence: campaigns/model_building_50_v1/results/EXP-010_bank_marketing_optimization_reliability.json

EXP-011 · breast_cancer

medium confidence

data_understanding

gpt-5.4-mini

Question. What does the dataset contain, and which quality or production-availability risks must be resolved before modeling?

Decision

Do not proceed to modeling until production input representation and split strategy are verified. The current evidence is sufficient for a preliminary inventory, but insufficient to clear leakage, encoding, and production-availability risks.

Interpretation

This is a compact data-understanding snapshot, not a modeling validation result. The evidence supports that the dataset is small, tabular, and numerically encoded in the cached form, with no missingness shown in the displayed feature profiles. However, the package does not establish whether the current feature representation matches production-time inputs, whether any preprocessing was fit without leakage, or whether the observed random-split shift is relevant to the actual deployment distribution. The single flagged feature suggests at least one distribution-shift review item remains unresolved, but the package does not quantify how many total features were reviewed or whether the rule was applied to all 30 features.

Observed evidence (cited)

- The package reports 569 source rows and 30 cached features for the breast_cancer dataset.
`EXP-011 / evidence.source_rows, evidence.feature_count`
- The reported training positive rate is 0.37362637362637363.
`EXP-011 / evidence.positive_rate_train`
- Among the listed feature profiles, all shown features have missing_rate 0.0 and high unique ratios, with one feature explicitly flagged for split_distribution_shift.
`EXP-011 / evidence.feature_profiles`
- The evidence bundle states that cached UCI matrices factorized categoricals before this campaign and that random train/holdout PSI is only a smoke check.
`EXP-011 / claims[1].limitations`

Challenges to claims

- **medium** **EXP-011-C1** — The claim is a factual inventory, but it does not distinguish cached representation from raw production inputs. The statement may be incomplete for modeling readiness because feature semantics in production are not verified.
- **medium** **EXP-011-C2** — The claim says 1 feature triggered review rules, but the evidence only displays one flagged feature among ten shown profiles; it is unclear whether this is exactly one of thirty or just one in the excerpt.
- **high** **EXP-011-C2** — The cited random-split PSI is not sufficient to support any production-stability inference, and the package itself acknowledges this limitation.

Risks & missing evidence

- No evidence that the raw production feature schema matches the cached UCI matrices; the note about factorized categoricals implies a representation mismatch risk.

- No decision-time availability review: it is unclear which features are known at prediction time versus derived after the outcome.
- No production-availability review: the package does not show whether all features can be populated reliably in the serving environment.
- Random train/holdout PSI is not a substitute for temporal or deployment-slice validation.
- The evidence only shows 10 feature profiles, so it is unclear whether the remaining 20 features were checked for missingness, identifier behavior, near-constant values, or shift.
- The single flagged `split_distribution_shift` feature is not enough to determine whether the shift is material or an artifact of the random split.
- No leakage audit is shown for preprocessing steps such as scaling, imputation, encoding, or feature selection.

Smallest next experiment

Production-schema and decision-time availability audit

Hypothesis: At least one cached feature is either unavailable at prediction time, derived from post-outcome information, or mismatched between training and production representations.

Success gate: For every one of the 30 features, document source, availability at decision time, transformation lineage, and production population method; the experiment passes only if all features are confirmed available at prediction time with no post-outcome leakage and no unresolved schema mismatch.

Why smallest: This is the minimal falsifiable step that directly addresses the largest unresolved risk before any modeling: whether the dataset features can be safely and consistently reproduced in production.

Evidence: [campaigns/model_building_50_v1/results/EXP-011_breast_cancer_data_understanding.json](#)

EXP-012 · breast_cancer

medium confidence

leakage_audit

gpt-5.4-mini

Question. Which columns violate the prediction-time contract or behave like suspicious target proxies?

Decision

Insufficient evidence to approve or clear any columns. The package does not establish prediction-time availability, production-time availability, or target-proxy status for any real feature.

Interpretation

The package supports only a very limited conclusion: the synthetic canary was detected, and no features were formally declared as leakage. It does not provide enough evidence to identify which real columns violate the prediction-time contract or act as target proxies. The absence of declared leakage features is not evidence of absence, and the missing `safe_vs_unsafe_training_cv` result leaves the audit incomplete.

Observed evidence (cited)

- The leakage detector canary passed: true.
`EXP-012 / evidence.detector_canary_passed`
- No declared leakage features were provided for the `breast_cancer` dataset: empty list.
`EXP-012 / evidence.declared_leakage_features`
- The `safe_vs_unsafe_training_cv` evidence is null, so no cross-validation comparison is available in the package.
`EXP-012 / evidence.safe_vs_unsafe_training_cv`

Challenges to claims

- **medium** **EXP-012-C1** — The canary result only shows the detector can catch a synthetic exact-target feature; it does not support any claim about real columns or the overall absence of leakage.
- **high** **EXP-012-C2** — The statement 'Declared decision-time exclusions for `breast_cancer`: none' is unsupported as a decision because the package provides no decision-time semantics or production-availability review for the actual columns.

Risks & missing evidence

- No column-level evidence is provided for actual feature names, so suspicious proxies cannot be assessed.
- Decision-time semantics are explicitly noted as required, but no decision-time/production-availability review is shown.
- `safe_vs_unsafe_training_cv` is null, so there is no empirical comparison to help distinguish leakage-like behavior from legitimate signal.
- A passed synthetic canary only validates the detector mechanism, not the absence of leakage in the dataset.
- No univariate association, missingness, uniqueness, or temporal availability evidence is included for real columns.

Smallest next experiment

Column-level prediction-time contract audit with decision-time availability matrix

Hypothesis: At least one real column in `breast_cancer` is either unavailable at prediction time or behaves like a target proxy when evaluated against decision-time semantics and simple leakage heuristics.

Success gate: For every column, produce a decision-time availability label and a production-availability label; flag any column with unavailable-at-prediction-time semantics or with strong proxy indicators (e.g., near-deterministic target association, post-outcome timing, or identity-like uniqueness). The experiment succeeds if the audit yields a non-empty, reproducible set of flagged columns or else a documented empty set with explicit decision-time justification for each column.

Why smallest: This is the minimal next step because it reuses the existing dataset and detector framework, adds only the missing column-level semantics, and directly tests the unresolved question without requiring model retraining or broader pipeline changes.

Evidence: `campaigns/model_building_50_v1/results/EXP-012_breast_cancer_leakage_audit.json`

EXP-013 · breast_cancer

high confidence

feature_engineering

gpt-5.4-mini

Question. Which train-fitted feature recipe improves validation evidence without feature dilution?

Decision

Do not accept the recommendation as supported. Require a rule-consistency check and a leakage/validation audit before any feature recipe is selected for the next screen.

Interpretation

The package does not support the recommendation to use the raw recipe as the next screen if the stated goal is to improve validation evidence without feature dilution. By the package's own selection rule, the selected stage should be the smallest matrix within 0.002 of the best mean training-CV ROC-AUC, but the reported selected stage is both smaller and materially worse on mean ROC-AUC than raw. The evidence instead suggests a mismatch between the selection rule and the reported chosen stage, or a missing comparison stage / missing evaluation detail. The package also does not establish that any feature recipe improves validation evidence without dilution in a decision-ready sense, because only 3-fold CV is shown and no decision-time or production-availability review is provided.

Observed evidence (cited)

- The experiment compares six stages: raw, logs, ratios, interactions, full_fe, and selected, all evaluated with LightGBM over 3 folds.
`EXP-013 / evidence.stage_results[*].stage; EXP-013 / evidence.stage_results[*].model; EXP-013 / evidence.stage_results[*].folds`
- The raw stage reports mean ROC-AUC 0.9907334124785648 with mean feature_count_mean 30.0.
`EXP-013 / evidence.stage_results[0].metrics.roc_auc.mean; EXP-013 / evidence.stage_results[0].feature_count_mean`
- The selected stage reports mean ROC-AUC 0.9813118322121092 with mean feature_count_mean 25.0.
`EXP-013 / evidence.stage_results[5].metrics.roc_auc.mean; EXP-013 / evidence.stage_results[5].feature_count_mean`
- The selection rule is stated as choosing the smallest feature matrix within 0.002 mean training-CV ROC-AUC of the best stage.
`EXP-013 / evidence.selection_rule`
- Several later stages increase feature_count_mean relative to raw: logs 54.333333333333336, ratios 69.33333333333333, interactions 77.33333333333333, and full_fe 82.33333333333333.
`EXP-013 / evidence.stage_results[1].feature_count_mean; EXP-013 / evidence.stage_results[2].feature_count_mean; EXP-013 / evidence.stage_results[3].feature_count_mean; EXP-013 / evidence.stage_results[4].feature_count_mean`
- The selected stage has lower mean ROC-AUC than raw and does not satisfy the stated 0.002-within-best selection rule if raw is the best stage by mean ROC-AUC.
`EXP-013 / evidence.stage_results[0].metrics.roc_auc.mean; EXP-013 / evidence.stage_results[5].metrics.roc_auc.mean; EXP-013 / evidence.selection_rule`

Challenges to claims

- **high** **EXP-013-C1** — The recommendation says to use the raw recipe, but the stated selection rule would require the smallest matrix within 0.002 of the best mean training-CV ROC-AUC. The reported selected stage is smaller than raw yet has lower mean ROC-AUC, so the recommendation is not supported by the rule as written.
- **high** **EXP-013-C1** — The claim frames 0.9907 mean training-CV ROC-AUC as evidence that raw should be used next, but the package does not show a direct comparison against a held-out validation or test set, so the recommendation is not decision-grade.

- **medium** **EXP-013-C1** — The claim mentions 'about 30 features,' but the selected stage reports 25 features and the raw stage reports 30 features; the wording is ambiguous and may obscure the actual feature dilution tradeoff.
- **medium** **EXP-013-C1** — The limitation about domain semantics is acknowledged, but there is no evidence of decision-time availability or production freshness review for the engineered features.

Risks & missing evidence

- The reported selected stage appears inconsistent with the stated selection rule; the package does not show how a 25-feature matrix with lower mean ROC-AUC than raw was selected as within 0.002 of the best stage.
- Only 3 folds are reported; there is no repeated CV, nested CV, or held-out test set evidence to assess stability of the feature recipe choice.
- No decision-time availability review is shown for the engineered features, so production feasibility and feature freshness are unverified.
- No leakage audit is shown for train-fitted transformations, especially for ratio and interaction features that may encode fold-specific artifacts if any preprocessing was not strictly nested.
- The package reports predictive metrics only; it does not justify causal language or operational benefit claims.
- The evidence bundle does not show the exact feature matrices for each stage, so the meaning of 'feature dilution' is not fully auditable from the provided paths.

Smallest next experiment

Rule-consistency and leakage audit with one held-out evaluation

Hypothesis: If the selection rule is implemented correctly and transformations are strictly train-fitted, then the smallest feature matrix within the allowed ROC-AUC tolerance will be identifiable and its validation ranking will be reproducible on a single untouched hold-out split.

Success gate: Pre-register the exact selection rule, rerun the stage comparison with one untouched hold-out split, and accept only if the chosen stage is the smallest matrix within 0.002 of the best mean CV ROC-AUC and its hold-out ROC-AUC is not worse than raw by more than a predeclared tolerance.

Why smallest: This is the minimal falsifiable check because it tests the two core uncertainties already exposed by the package: whether the selection rule was applied consistently and whether the apparent gain survives one independent evaluation, without changing the feature-generation family.

Evidence: campaigns/model_building_50_v1/results/EXP-013_breast_cancer_feature_engineering.json

EXP-014 · breast_cancer

medium confidence

model_selection

gpt-5.4-mini

Question. Which algorithm family provides the strongest stable training-CV evidence for this dataset profile?

Decision

Provisional training-CV candidate only; not enough evidence to approve a final model choice or deployment decision.

Interpretation

On the provided training-CV evidence alone, logistic_regression is the strongest candidate by the package's own ranking criterion because it has the highest mean ROC-AUC and the lowest runtime among the listed models. However, the evidence is not sufficient to conclude it is the best algorithm family in a deployment sense, because the package only reports training CV, uses just 3 folds, and does not include a locked holdout or any decision-time/production-availability review. The apparent margin over the other models is small relative to the uncertainty implied by the fold variability and the limited number of folds.

Observed evidence (cited)

- The experiment compares five baseline raw-feature models under a training-CV-only selection rule: logistic_regression, extra_trees, hist_gradient_boosting, lightgbm, and xgboost.
`EXP-014 / evidence.model_results`
- The stated selection rule ranks models by mean ROC-AUC minus $0.25 \times$ fold standard deviation, then runtime, using training CV only.
`EXP-014 / evidence.selection_rule`
- logistic_regression has the highest reported mean ROC-AUC among the listed models (0.995510706591039) with fold std 0.00470927675276659 and elapsed_seconds 0.14492662500015285.
`EXP-014 / evidence.model_results[0].metrics.roc_auc`
- extra_trees reports mean ROC-AUC 0.9899045860264697 with fold std 0.005282023286405208 and elapsed_seconds 0.5210164169984637.
`EXP-014 / evidence.model_results[1].metrics.roc_auc`
- hist_gradient_boosting reports mean ROC-AUC 0.9907312139999119 with fold std 0.006843829794078326 and elapsed_seconds 0.3412291250006092.
`EXP-014 / evidence.model_results[2].metrics.roc_auc`
- lightgbm reports mean ROC-AUC 0.9907334124785648 with fold std 0.006547188652626659 and elapsed_seconds 0.2642961250003282.
`EXP-014 / evidence.model_results[3].metrics.roc_auc`

Challenges to claims

- **medium** **EXP-014-C1** — The statement that logistic_regression is the training-CV model candidate is supported only within the package's own selection rule and only on training CV; it is not supported as a universal winner or as a final model choice.
- **high** **EXP-014-C1** — The reported advantage over other models is small and based on only 3 folds, so the claim of strongest stable evidence is underpowered without repeated or nested validation.
- **high** **EXP-014-C1** — No locked holdout evidence is present, so the recommendation cannot be validated against unseen data.

- **high** **EXP-014-C1** — The evidence bundle does not document fold-safe preprocessing, so the reported ROC-AUC could be inflated by leakage if any transformations were fit outside the CV loop.

Risks & missing evidence

- Only 3 folds are reported, which is weak evidence for stability and makes the ranking sensitive to fold composition.
- The package explicitly uses training CV only; there is no locked holdout result to test whether the apparent leader generalizes.
- No evidence is provided about whether preprocessing, scaling, feature selection, or any target-aware transformations were fit strictly inside each fold, so leakage risk cannot be ruled out.
- The selection rule includes runtime after a metric-based ranking, but the package does not justify why runtime should break ties or near-ties for this task.
- No nested validation or repeated CV is shown, so model-family comparisons may be optimistic due to selection on the same CV used for evaluation.
- No decision-time or production-availability review is included, so even a strong CV candidate cannot be approved for operational use.
- The evidence package does not report class balance handling, threshold tuning protocol, or calibration decision criteria beyond raw fold metrics.

Smallest next experiment

Locked-holdout verification of the current CV leader with fold-safe preprocessing

Hypothesis: If logistic_regression is genuinely the strongest stable family for this dataset profile, then it should remain competitive on a previously untouched locked holdout when all preprocessing is fit strictly within each training fold.

Success gate: Run a single locked-holdout evaluation after retraining with the exact same CV protocol and fold-safe preprocessing; success is defined as logistic_regression remaining within the top tier on holdout ROC-AUC and not showing a materially worse calibration profile than the best competing family, with no evidence of leakage in the pipeline audit.

Why smallest: This is the smallest falsifiable step because it reuses the current candidate set and selection rule, adds only one untouched evaluation split plus a leakage audit, and directly tests whether the CV leader survives outside the selection loop.

Evidence: campaigns/model_building_50_v1/results/EXP-014_breast_cancer_model_selection.json

EXP-015 · breast_cancer

medium confidence

optimization_reliability

gpt-5.4-mini

Question. Does optimization produce a meaningful CV gain, and how does the preselected recipe perform on the once-consumed holdout?

Decision

Do not claim a meaningful optimization gain from this package. Treat the holdout score as a consumed, one-time performance estimate only. Before any feature or recipe promotion, require a pre-registered comparison against the baseline on the same CV protocol and a decision-time/production-availability audit.

Interpretation

The holdout result is strong predictive evidence for this dataset, but it does not by itself establish that optimization produced a meaningful gain. The package does not provide the baseline CV score, the tuned CV score, or their uncertainty, so the core question about optimization benefit remains unsupported. The feature-ranking note is directionally cautious, but any promotion decision still needs explicit decision-time and production-availability review; importance and 3-fold stability alone are insufficient to justify feature use.

Observed evidence (cited)

- The once-consumed holdout was evaluated and marked consumed.
[EXP-015 / evidence.holdout_consumed](#)
- The preselected logistic_regression/baseline recipe achieved holdout ROC-AUC 0.996031746031746 with bootstrap 95% interval [0.9857785443722944, 1.0].
[EXP-015 / evidence.holdout_metrics](#); [EXP-015 / evidence.holdout_roc_auc_bootstrap_ci](#)
- The selection rule uses training CV and requires mean ROC-AUC improvement of at least 0.001 over baseline to prefer a tuned candidate.
[EXP-015 / evidence.selection_rule](#)
- The reported CV feature stability is based on 3 observed folds for the listed features.
[EXP-015 / evidence.cv_feature_stability](#)
- The production feature rule requires decision-time availability, contractual obtainability, monitoring for missingness/drift, and stability across CV folds.
[EXP-015 / evidence.production_feature_rule](#)

Challenges to claims

- **high** **EXP-015-C1** — The claim reports a strong holdout ROC-AUC, but the experiment question asks whether optimization produces a meaningful CV gain; the package does not include the baseline CV score, tuned CV score, or their uncertainty, so the optimization-gain part is unsupported.
- **high** **EXP-015-C1** — Because the holdout is marked consumed, the reported holdout score should not be used as evidence for further model selection or iterative tuning without a fresh untouched evaluation set.
- **medium** **EXP-015-C2** — The recommendation is sensible, but it is not backed by explicit decision-time availability or production-availability evidence for the listed features in this package.
- **medium** **EXP-015-C2** — Cross-fold stability is based on only 3 observed folds, which is too sparse to support a strong stability conclusion.

Risks & missing evidence

- No baseline CV metric is shown, so the stated optimization threshold cannot be checked.
- No tuned-candidate CV metric or fold-level distribution is shown, so any gain could be within noise.
- The holdout is already consumed, so repeated model selection against it would risk leakage-by-iteration.
- Only 3 CV folds are reported for feature stability, which is a weak basis for stability claims.
- No decision-time availability evidence is provided for the top features.
- No production contract, lineage, or monitoring evidence is provided for the top features.
- No calibration or threshold-selection protocol is shown for the holdout classification metrics.
- No nested-CV or repeated-CV evidence is shown to assess optimization reliability.

Smallest next experiment

Paired baseline-vs-tuned nested CV with untouched holdout lock

Hypothesis: If optimization truly improves the recipe, then the tuned candidate will exceed the baseline by at least 0.001 mean ROC-AUC under the same pre-registered CV protocol, and the improvement will persist under repeated/nested resampling.

Success gate: Pre-register the comparison; run paired nested or repeated CV on baseline and tuned candidate; declare optimization beneficial only if mean ROC-AUC improvement is ≥ 0.001 and the paired fold-wise difference is consistently positive across resamples. Do not touch the consumed holdout.

Why smallest: This is the minimal falsifiable test that directly answers the optimization-gain question without reusing the consumed holdout or adding new modeling complexity.

Evidence: `campaigns/model_building_50_v1/results/EXP-015_breast_cancer_optimization_reliability.json`

EXP-016 · heart_disease

medium confidence

data_understanding

gpt-5.4-mini

Question. What does the dataset contain, and which quality or production-availability risks must be resolved before modeling?

Decision

Do not proceed to modeling until decision-time availability, production-availability, and leakage checks are completed for the flagged features and target definition.

Interpretation

This package supports a basic dataset inventory and a preliminary quality screen, but it does not yet establish production readiness. The strongest evidence is descriptive: small sample size, moderate class balance, and several features flagged for distribution-shift review. The package also hints at a preprocessing-history risk for categorical semantics, but that concern is only stated in the claim limitations and is not directly evidenced in the feature profiles. The random-split PSI values are only a smoke test and should not be treated as evidence of deployment stability.

Observed evidence (cited)

- The dataset bundle reports 303 source rows and 13 cached features for heart_disease, with a training positive rate of 0.45867768595041325.

EXP-016 / evidence.source_rows; evidence.feature_count; evidence.positive_rate_train

- The provided feature profiles include age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, and oldpeak, each with dtype, missing_rate, top_value_frequency, unique_count, unique_ratio, and some with random_split_psi and production_risks.

EXP-016 / evidence.feature_profiles

- Within the listed profiles, age, trestbps, and chol are marked with production_risks containing split_distribution_shift, while several other listed features have no production_risks and some have random_split_psi values recorded.

EXP-016 / evidence.feature_profiles

Challenges to claims

- **medium** EXP-016-C1 — The row count, feature count, and positive rate are descriptive facts, but the package does not show how the positive rate was computed or whether the training split was stratified; the metric is therefore not fully auditable from the provided evidence paths alone.
- **medium** EXP-016-C2 — The statement that 3 features triggered review rules is only partially supported by the visible feature profiles; the bundle excerpt shows at least age, trestbps, and chol flagged for split_distribution_shift, but it does not expose the full rule evaluation for all 13 features or the exact thresholding logic.
- **high** EXP-016-C2 — The limitation about factorized categoricals is an unsupported preprocessing-history assertion unless backed by lineage or raw schema evidence; it should not be treated as established fact.
- **high** EXP-016-C2 — Using random_split_psi as a production-risk indicator is a weak proxy; it does not test temporal drift, decision-time availability, or post-deployment feature generation stability.

Risks & missing evidence

- No evidence is provided for the target definition, label timing, or whether the label is available at decision time.
- No production-availability review is shown for any feature, so it is unknown whether the listed fields exist at inference time with the same semantics.
- The claim that cached UCI matrices factorized categoricals before this campaign is not directly supported by the evidence bundle and needs verification from the raw ingestion/preprocessing lineage.
- Random-split PSI values are present, but there is no temporal or operational validation; this leaves open the possibility that apparent stability is an artifact of the random split.
- The package does not show train/validation split methodology, so leakage through preprocessing fit on full data cannot be ruled out.
- Only 10 feature profiles are shown in the bundle excerpt, while the dataset claims 13 features; the remaining 3 features are not visible here and may hide additional risks.
- No missingness pattern analysis, duplicate-row check, or identifier uniqueness audit is provided beyond the limited profile fields.

Smallest next experiment

Decision-time and lineage audit for flagged heart_disease features

Hypothesis: The flagged features can be shown to be available at decision time with stable raw semantics and no leakage from preprocessing or label timing.

Success gate: For each flagged feature and the target, produce a lineage report showing raw source, extraction timestamp, decision-time availability, preprocessing steps, and split-specific fit/transform boundaries; the experiment succeeds only if no feature is derived from post-outcome information and all flagged features are available with unchanged semantics in the intended production context.

Why smallest: This is the minimal falsifiable step because it reuses the existing dataset inventory and only adds lineage, timing, and split-boundary verification before any model training.

Evidence: [campaigns/model_building_50_v1/results/EXP-016_heart_disease_data_understanding.json](#)

EXP-017 · heart_disease

medium confidence

leakage_audit

gpt-5.4-mini

Question. Which columns violate the prediction-time contract or behave like suspicious target proxies?

Decision

Insufficient evidence to determine which columns violate the prediction-time contract or are suspicious target proxies. No feature should be approved or excluded solely from this package.

Interpretation

The package only establishes that the detector can recognize an injected exact-target canary and that no leakage features were declared. It does not identify which real columns violate the prediction-time contract or behave as target proxies. Because the key validation artifact comparing safe versus unsafe training is null, the evidence is insufficient to support any decision about feature safety or exclusion beyond a request for further review.

Observed evidence (cited)

- The leakage detector's synthetic exact-target canary passed.
`EXP-017 / evidence.detector_canary_passed`
- No declared leakage features were provided for heart_disease.
`EXP-017 / evidence.declared_leakage_features`
- The safe_vs_unsafe_training_cv evidence is missing/null, so there is no reported comparison of model behavior under safe versus unsafe feature sets.
`EXP-017 / evidence.safe_vs_unsafe_training_cv`

Challenges to claims

- **medium** **EXP-017-C1** — This only shows the detector caught a synthetic canary; it does not demonstrate detection of any real leakage column in heart_disease.
- **high** **EXP-017-C2** — The statement 'Declared decision-time exclusions for heart_disease: none' is unsupported as a decision because the package provides no decision-time semantics review and no column-level evidence.

Risks & missing evidence

- No column-level evidence is provided for actual candidate proxies, so the question asked by the experiment remains unanswered.
- The null safe_vs_unsafe_training_cv field leaves a validation gap: there is no reported test of whether suspicious columns materially change predictive performance.
- The canary result only validates detector functionality, not real-world leakage detection on the heart_disease dataset.
- No decision-time semantics or production-availability review is included for any column, so contract violations cannot be confirmed.
- No univariate association, uniqueness, timestamp availability, or post-outcome timing evidence is provided for specific columns.

Smallest next experiment

Column-level decision-time contract audit for heart_disease

Hypothesis: At least one column in heart_disease is unavailable at prediction time or behaves like a target proxy, and this can be identified by comparing each column's availability/semantics against the prediction-time contract plus a safe-vs-unsafe ablation.

Success gate: For every column, provide a decision-time availability verdict and a rationale; flag any column with post-outcome timing, direct target encoding, or near-deterministic target association. The experiment succeeds only if the audit produces a non-null safe_vs_unsafe comparison and a reproducible list of candidate proxy columns with supporting evidence paths.

Why smallest: It adds the minimum missing evidence needed to answer the question: a per-column contract review plus one safe-vs-unsafe ablation, without changing the dataset or requiring a larger modeling pipeline.

Evidence: [campaigns/model_building_50_v1/results/EXP-017_heart_disease_leakage_audit.json](#)

EXP-018 · heart_disease

medium confidence

feature_engineering

gpt-5.4-mini

Question. Which train-fitted feature recipe improves validation evidence without feature dilution?

Decision

Do not treat the raw recipe as validated for production or as a proven improvement. At most, it is the current smallest candidate that matches the best reported training-CV ROC-AUC under the stated rule, pending independent validation and feature availability review.

Interpretation

The evidence supports a narrow, non-causal inference: among the reported stages, the raw recipe is the smallest matrix and its mean training-CV ROC-AUC is tied for the best reported value, so it satisfies the stated selection rule better than the larger recipes. However, the package does not establish that this recipe improves validation evidence in a deployment-relevant sense, because only training-CV metrics are shown and no independent validation/test or decision-time availability checks are provided.

Observed evidence (cited)

- The selection rule is to choose the smallest feature matrix within 0.002 mean training-CV ROC-AUC of the best stage.
`EXP-018 / evidence.selection_rule`
- The raw stage reports mean training-CV ROC-AUC of 0.8700264746776375 with mean `feature_count_mean` of 13.0 across 3 folds.
`EXP-018 / evidence.stage_results[0].metrics.roc_auc.mean`; `EXP-018 / evidence.stage_results[0].feature_count_mean`; `EXP-018 / evidence.stage_results[0].folds`
- The logs stage reports the same mean training-CV ROC-AUC as the raw stage and a higher mean `feature_count_mean` of 18.333333333333332 across 3 folds.
`EXP-018 / evidence.stage_results[1].metrics.roc_auc.mean`; `EXP-018 / evidence.stage_results[1].feature_count_mean`; `EXP-018 / evidence.stage_results[1].folds`
- The ratios stage reports mean training-CV ROC-AUC of 0.8686598861017466 with mean `feature_count_mean` of 33.333333333333336 across 3 folds.
`EXP-018 / evidence.stage_results[2].metrics.roc_auc.mean`; `EXP-018 / evidence.stage_results[2].feature_count_mean`; `EXP-018 / evidence.stage_results[2].folds`
- The interactions stage reports mean training-CV ROC-AUC of 0.8674742395672629 with mean `feature_count_mean` of 41.333333333333336 across 3 folds.
`EXP-018 / evidence.stage_results[3].metrics.roc_auc.mean`; `EXP-018 / evidence.stage_results[3].feature_count_mean`; `EXP-018 / evidence.stage_results[3].folds`
- The full_fe stage reports mean training-CV ROC-AUC of 0.8651886558863303 with mean `feature_count_mean` of 46.333333333333336 across 3 folds.
`EXP-018 / evidence.stage_results[4].metrics.roc_auc.mean`; `EXP-018 / evidence.stage_results[4].feature_count_mean`; `EXP-018 / evidence.stage_results[4].folds`

Challenges to claims

- **high** **EXP-018-C1** — The statement says 'improves validation evidence,' but the cited evidence only reports mean training-CV ROC-AUC and feature count. No independent validation metric is present.
- **medium** **EXP-018-C1** — The recommendation to use the raw recipe is based on a size-vs-training-CV tradeoff, but the raw and logs stages have the same mean ROC-AUC, so the evidence supports 'smallest tied-best' rather than 'better'.

- **high EXP-018-C1** — The package does not show decision-time or production-availability review for the raw features, so the recommendation is incomplete for deployment use.
- **high EXP-018-C1** — Because multiple engineered stages were compared, the best mean training-CV ROC-AUC may be subject to selection bias; the 0.002 rule is not enough to rule out overfitting without a held-out check.

Risks & missing evidence

- No separate holdout, nested CV, or external validation is shown; all evidence is training-CV only.
- The selection rule uses mean training-CV ROC-AUC, which can favor overfit feature recipes when many engineered features are tried.
- The package does not show whether any features are available at decision time in production, especially engineered ratios/interactions that may depend on inputs with missingness or delayed availability.
- The package does not document leakage controls for train-fitted feature generation beyond the phrase 'train-fitted'; fold-wise fitting boundaries are not explicit.
- The claim 'improves validation evidence' is not directly supported because no validation set metric is reported.
- The raw stage and logs stage have identical mean ROC-AUC, so the evidence does not uniquely identify raw as better on score; it is only smaller.
- Feature stability is reported, but stability alone does not establish generalization or production utility.

Smallest next experiment

Single held-out validation of raw vs logs under the same fold-wise pipeline

Hypothesis: If the raw recipe is genuinely preferable without feature dilution, then it will match or exceed the logs recipe on a pre-registered held-out validation metric while using fewer features, under identical fold-wise fitting and preprocessing.

Success gate: Run one pre-registered evaluation with a fixed train/validation split or nested CV: raw and logs only, same model and seed, fold-wise feature fitting, and report held-out ROC-AUC plus calibration. Success requires raw to be within 0.002 of logs on held-out ROC-AUC and to use fewer features, with no leakage detected in the pipeline audit.

Why smallest: It changes only the comparison set from many stages to the two closest candidates (raw and logs) and adds the minimum necessary independent validation check to test the selection rule.

Evidence: [campaigns/model_building_50_v1/results/EXP-018_heart_disease_feature_engineering.json](#)

EXP-019 · heart_disease

medium confidence

model_selection

gpt-5.4-mini

Question. Which algorithm family provides the strongest stable training-CV evidence for this dataset profile?

Decision

Do not treat `extra_trees` as a confirmed winner. Use it only as a provisional CV leader pending a locked-holdout check and a repeated/nested validation comparison against `logistic_regression`.

Interpretation

Within this training-CV-only package, `extra_trees` is the top raw-stage candidate by the stated ranking criterion, but the evidence is not strong enough to claim it is the strongest stable family overall. The margin over `logistic_regression` is small, the sample of folds is only 3, and the package does not include a locked holdout or repeated/nested validation. The most defensible conclusion is that `extra_trees` is the current CV leader under the specified rule, while `logistic_regression` remains a plausible near-tie on stability/runtime grounds.

Observed evidence (cited)

- The selection rule ranks models by mean ROC-AUC minus $0.25 \times$ fold standard deviation, then runtime, using training CV only.

EXP-019 / `evidence.selection_rule`

- The raw-stage `extra_trees` model reports mean ROC-AUC 0.8977867931356304 with std 0.024360745414873528 across 3 folds, and elapsed_seconds 0.4227246250011376.

EXP-019 / `evidence.model_results[1].metrics.roc_auc`; EXP-019 / `evidence.model_results[1].elapsed_seconds`; EXP-019 / `evidence.model_results[1].folds`

- The raw-stage `logistic_regression` model reports mean ROC-AUC 0.8941250976134697 with std 0.019780866908858354 across 3 folds, and elapsed_seconds 0.06454533400028595.

EXP-019 / `evidence.model_results[0].metrics.roc_auc`; EXP-019 / `evidence.model_results[0].elapsed_seconds`; EXP-019 / `evidence.model_results[0].folds`

- The raw-stage `lightgbm` model reports mean ROC-AUC 0.8700264746776375 with std 0.033786505757582165 across 3 folds.

EXP-019 / `evidence.model_results[3].metrics.roc_auc`; EXP-019 / `evidence.model_results[3].folds`

- The raw-stage `xgboost` model reports mean ROC-AUC 0.8573510085137993 with std 0.020881994982021816 across 3 folds.

EXP-019 / `evidence.model_results[4].metrics.roc_auc`; EXP-019 / `evidence.model_results[4].folds`

- The raw-stage `hist_gradient_boosting` model reports mean ROC-AUC 0.8620316934270423 with std 0.042044001232304604 across 3 folds.

EXP-019 / `evidence.model_results[2].metrics.roc_auc`; EXP-019 / `evidence.model_results[2].folds`

Challenges to claims

- **medium** EXP-019-C1 — The statement that `extra_trees` is the training-CV model candidate is supported only as a provisional ranking outcome, not as a robust winner. The ROC-AUC advantage over `logistic_regression` is small, and `logistic_regression` has lower variance and much lower runtime under the same raw-stage setting.
- **high** EXP-019-C1 — The evidence does not include a locked holdout, so the claim cannot be extended beyond training-CV selection.

- **high** EXP-019-C1 — The package does not show nested CV or an untouched validation layer, so the ranking may be biased by model selection on the same folds used for evaluation.

Risks & missing evidence

- Only 3 folds are reported, so the stability estimate is noisy and sensitive to split choice.
- The package is training-CV only; there is no locked holdout or external test evidence.
- The selection rule uses the same CV results for ranking and model choice, so the reported leader may be optimistic without nested validation.
- No evidence is provided on whether preprocessing, feature engineering, or target encoding were fit strictly inside each fold; leakage risk cannot be ruled out.
- The evidence bundle does not show class balance, split strategy, grouping, or temporal structure, so validation design adequacy cannot be assessed.
- The claim is about a 'training-CV model candidate,' which is weaker than a deployable model decision; decision-time and production-availability review are absent.

Smallest next experiment

Repeated nested CV comparison: extra_trees vs logistic_regression

Hypothesis: If extra_trees is genuinely the strongest stable family for this dataset profile, it will retain a higher mean outer-fold ROC-AUC than logistic_regression across repeated nested CV splits, with the advantage exceeding a pre-specified small margin and not reversing across repeats.

Success gate: Run repeated nested CV with identical preprocessing inside folds for extra_trees and logistic_regression. Declare extra_trees the provisional winner only if its mean outer-fold ROC-AUC exceeds logistic_regression by at least 0.01 and the result is directionally consistent in a majority of repeats; otherwise, treat the two as tied and defer to simpler/faster model choice criteria.

Why smallest: This changes only the validation design and compares the two closest contenders already present in the evidence, making it the minimal falsifiable test of whether the current CV leader is stable rather than a split-specific artifact.

Evidence: campaigns/model_building_50_v1/results/EXP-019_heart_disease_model_selection.json

EXP-020 · heart_disease

medium confidence

optimization_reliability

gpt-5.4-mini

Question. Does optimization produce a meaningful CV gain, and how does the preselected recipe perform on the once-consumed holdout?

Decision

Do not accept any claim that optimization produced a meaningful gain from this package alone. Treat the holdout score as a completed evaluation result, but require the missing baseline CV comparison and decision-time/production-availability review before any feature or recipe promotion.

Interpretation

The package supports a single holdout performance result for the preselected recipe and provides a weakly informative feature-stability summary. It does not, by itself, establish that optimization produced a meaningful CV gain because the actual baseline-versus-tuned CV scores are not shown here, only the selection rule. The holdout result is promising but remains a one-shot estimate on a once-consumed holdout, so it should be treated as evidence of predictive performance, not causal impact or deployment readiness.

Observed evidence (cited)

- The once-consumed holdout evaluation for the preselected `extra_trees/extra_trees_C02` recipe reports ROC-AUC 0.9448051948051949 with bootstrap 95% interval [0.8851264748323572, 0.9871619177677714].

EXP-020 / `evidence.holdout_metrics.roc_auc`; EXP-020 / `evidence.holdout_roc_auc_bootstrap_ci.low`; EXP-020 / `evidence.holdout_roc_auc_bootstrap_ci.high`; EXP-020 / `evidence.holdout_consumed`

- The holdout metrics bundle also reports accuracy 0.819672131147541, average precision 0.9352056634897589, Brier score 0.10701414972093289, ECE_10 0.14948788876925767, F1 0.8135593220338984, log loss 0.3593703852051472, precision 0.7741935483870968, recall 0.8571428571428571, and threshold 0.5.

EXP-020 / `evidence.holdout_metrics`

- The optimization rule states that the best tuned candidate is used only when mean ROC-AUC improves by at least 0.001 over baseline; otherwise baseline is preferred.

EXP-020 / `evidence.selection_rule`

- The `top_final_features` list ranks `thal`, `ca`, `exang`, `cp`, `sex`, `slope`, `thalach`, `oldpeak`, `restecg`, `age`, `trestbps`, `chol`, and `fbs` by normalized importance.

EXP-020 / `evidence.top_final_features`

- The `cv_feature_stability` table shows 3 observed folds for each listed feature; `thal` through `age` have `top10_frequency` 1.0, while `trestbps`, `chol`, and `fbs` have `top10_frequency` 0.0.

EXP-020 / `evidence.cv_feature_stability`

- The `production_feature_rule` requires decision-time availability, contractual obtainability, monitoring for missingness/drift, and cross-fold stability before treating a feature as a stronger candidate.

EXP-020 / `evidence.production_feature_rule`

Challenges to claims

- **high** EXP-020-C1 — The holdout ROC-AUC is reported correctly, but the question asks whether optimization produced a meaningful CV gain; this package does not include the baseline CV score or tuned CV score needed to verify the selection rule threshold.

- **medium** **EXP-020-C1** — Because the holdout is once-consumed, the result should not be generalized as evidence of robust out-of-sample superiority without a fresh, untouched evaluation or nested validation.
- **medium** **EXP-020-C2** — The recommendation is directionally sound, but the evidence bundle does not show decision-time availability, contractual obtainability, or monitoring status for the listed features; therefore the recommendation is not fully evidenced here.
- **low** **EXP-020-C2** — Cross-fold stability is based on only 3 observed folds, which is limited support for a strong stability requirement.

Risks & missing evidence

- No explicit baseline CV ROC-AUC is provided, so the claimed 'meaningful CV gain' cannot be verified against the 0.001 improvement rule.
- The holdout is marked once-consumed, which limits further unbiased evaluation and increases the importance of pre-registered comparison logic.
- Only 3 CV folds are reported for feature stability, which is thin evidence for stability claims.
- Feature importance is model-specific and does not establish causal relevance or operational availability.
- No decision-time availability audit is shown for the top features, so some may be unusable in production despite importance.
- No missingness, drift, or contractability evidence is provided for the candidate features.
- The holdout metrics are single-point estimates; calibration and discrimination are both present, but no external replication is shown.

Smallest next experiment

Baseline-vs-tuned CV audit with untouched evaluation split

Hypothesis: If the tuned recipe truly improves performance, then its mean CV ROC-AUC will exceed the baseline by at least 0.001 under the same preprocessing and fold protocol, and the result will remain directionally consistent on a fresh untouched evaluation split.

Success gate: Report baseline mean CV ROC-AUC and tuned mean CV ROC-AUC on identical folds; accept optimization only if tuned - baseline $\geq$ 0.001. Then evaluate the selected recipe once on a fresh untouched split and require the holdout ROC-AUC to be reported without any further tuning on that split.

Why smallest: This is the minimal falsifiable step because it directly tests the stated selection rule and adds only one fresh evaluation split, without changing the model family, feature set, or optimization procedure.

Evidence: campaigns/model_building_50_v1/results/EXP-020_heart_disease_optimization_reliability.json

EXP-021 · credit_default

medium confidence

data_understanding

gpt-5.4-mini

Question. What does the dataset contain, and which quality or production-availability risks must be resolved before modeling?

Decision

Do not proceed to modeling until production-availability and leakage checks are completed.

Interpretation

This is a dataset-understanding package, not modeling evidence. It supports only a narrow description of the cached table and some basic profile statistics. It does not establish that the features are production-ready, temporally stable, or safe for modeling without additional review. The strongest unresolved issue is that the data appear to come from cached, factorized UCI matrices, which may not preserve raw category semantics needed for production encoders. The random split checks are insufficient to rule out leakage or distribution shift in a real deployment setting.

Observed evidence (cited)

- The package reports 30,000 source rows, 23 cached features, and a training positive rate of 0.22125 for credit_default.
EXP-021 / evidence.source_rows; EXP-021 / evidence.feature_count; EXP-021 / evidence.positive_rate_train
- The provided feature profile sample shows integer-typed features X1 through X10 with zero missingness in the sampled profiles.
EXP-021 / evidence.feature_profiles
- The sampled profiles include low random-split PSI values for several features, but X2 has random_split_psi set to null.
EXP-021 / evidence.feature_profiles
- The package itself states that cached UCI matrices factorized categoricals before this campaign and that random train/holdout PSI is only a smoke check.
EXP-021 / claims[1].limitations

Challenges to claims

- **medium** EXP-021-C1 — The row count, feature count, and positive rate are descriptive facts, but they do not by themselves establish that the dataset is suitable for modeling or that the target is correctly aligned with decision time.
- **high** EXP-021-C2 — The statement that 0 features triggered review rules is not fully supported by the excerpt because only a subset of feature profiles is shown, and the review criteria are not fully auditable here.
- **high** EXP-021-C2 — Random-split PSI cannot support a claim about production stability; it is explicitly acknowledged as a smoke check and does not address temporal drift or availability at scoring time.
- **high** EXP-021-C2 — The note about factorized categoricals is a material production-risk warning: without raw category semantics and encoder design, the current cached representation may not be usable as-is.

Risks & missing evidence

- No evidence of decision-time availability for each feature; the package does not show whether any feature would be known at scoring time.
- No production-source mapping is provided; the package does not show how cached factorized values correspond to raw business categories.
- Random train/holdout PSI is not a temporal validation test, so temporal drift and lookahead leakage remain untested.
- Only a sample of feature profiles is shown in the evidence bundle; the full set of 23 features is not visible here, so the claim that 0 features triggered review rules is not fully auditable from the provided excerpt.
- The null random_split_psi for X2 is unexplained and could indicate a profiling gap rather than stability.
- No target-definition audit is shown, so label timing and label leakage risks remain unresolved.

Smallest next experiment

Decision-time and raw-category availability audit

Hypothesis: At least one feature in the cached table is unavailable at scoring time or lacks a reversible mapping to raw production categories, which would block safe modeling.

Success gate: For all 23 features, document source system, event timestamp, decision-time availability, and raw-to-cached value mapping; success requires zero unresolved features and no feature marked as post-outcome or unavailable at scoring time.

Why smallest: This is the minimal falsifiable check that directly addresses the main unresolved production-risk claim without requiring model training or broader validation infrastructure.

Evidence: [campaigns/model_building_50_v1/results/EXP-021_credit_default_data_understanding.json](#)

EXP-022 · credit_default

medium confidence

leakage_audit

gpt-5.4-mini

Question. Which columns violate the prediction-time contract or behave like suspicious target proxies?

Decision

Insufficient evidence to identify or clear any specific columns as contract-safe. No feature should be approved from this package alone.

Interpretation

This package provides only a minimal audit signal: the detector can recognize an injected exact-target canary, but there is no evidence here that any real column was tested, ranked, or validated against decision-time availability. The empty declared leakage list is not sufficient to conclude absence of leakage or absence of target proxies, especially because the comparative `safe_vs_unsafe_training_cv` result is missing.

Observed evidence (cited)

- The leakage detector synthetic exact-target canary passed: true.
`EXP-022 / evidence.detector_canary_passed`
- Declared leakage features are empty: [].
`EXP-022 / evidence.declared_leakage_features`
- `safe_vs_unsafe_training_cv` is null, so no comparative validation result is available in the package.
`EXP-022 / evidence.safe_vs_unsafe_training_cv`
- The package states the question is which columns violate the prediction-time contract or behave like suspicious target proxies.
`EXP-022 / evidence_bundle.question`

Challenges to claims

- **medium** **EXP-022-C1** — The canary passing is a narrow detector sanity check, not evidence about real feature leakage or proxy behavior. It cannot support any conclusion about the dataset columns under audit.
- **high** **EXP-022-C2** — The statement 'Declared decision-time exclusions for `credit_default`: none' is not justified as a decision. The package lacks decision-time semantics and production-availability review for actual columns, so 'none' is unsupported.

Risks & missing evidence

- No column-level evidence is provided, so the package does not answer which columns are suspicious.
- No decision-time semantics review is shown for any candidate column; name-based or proxy-like suspicion cannot be resolved from the current evidence.
- `safe_vs_unsafe_training_cv` is null, so there is no validation-based contrast to support or refute leakage concerns.
- The canary result only demonstrates detector sensitivity to a synthetic exact-target insertion; it does not establish detector recall on realistic proxy patterns.
- No production-availability or prediction-time contract metadata is included for the actual columns.

Smallest next experiment

Column-level decision-time contract audit for candidate proxies

Hypothesis: At least one column flagged by simple heuristics (name similarity to target, high univariate AUC, near-uniqueness, or post-outcome timing) will fail a decision-time availability review or show suspicious proxy behavior.

Success gate: For every column in the dataset, provide a decision-time availability label and a minimal evidence trace; any column with post-outcome timing, target-derived naming, or unusually strong univariate association must be explicitly reviewed and either justified as available at prediction time or marked excluded.

Why smallest: This is the smallest falsifiable step because it reuses the existing leakage-audit framing and adds only per-column metadata review plus a lightweight heuristic screen, without requiring model retraining or broader experimental redesign.

Evidence: [campaigns/model_building_50_v1/results/EXP-022_credit_default_leakage_audit.json](#)

EXP-023 · credit_default

medium confidence

feature_engineering

gpt-5.4-mini

Question. Which train-fitted feature recipe improves validation evidence without feature dilution?

Decision

Do not accept the recommendation as stated; the evidence is insufficient to justify a feature choice beyond a provisional screening preference for raw under this specific CV summary.

Interpretation

The evidence does not support the recommendation to use the raw recipe as the next screen under the stated selection rule, because the raw stage is the best by mean ROC-AUC and also the smallest among the stages shown. However, the package only provides 3-fold training-CV evidence, so the apparent ranking is still vulnerable to fold variance and selection bias. The phrase 'without feature dilution' is not operationalized beyond feature counts, and there is no decision-time or production-availability review for the generated features.

Observed evidence (cited)

- The experiment compares five feature stages on credit_default: raw, logs, ratios, interactions, full_fe, and selected, all evaluated with LightGBM and 3 folds.
`EXP-023 / evidence.stage_results[*].stage, evidence.stage_results[*].model, evidence.stage_results[*].folds`
- The highest reported mean ROC-AUC is 0.7316521444852612 at the raw stage; the selected stage reports 0.72011566222466 mean ROC-AUC and 25.0 mean features.
`EXP-023 / evidence.stage_results[0].metrics.roc_auc.mean, evidence.stage_results[5].metrics.roc_auc.mean, evidence.stage_results[5].feature_count_mean`
- The selection rule is to choose the smallest feature matrix within 0.002 mean training-CV ROC-AUC of the best stage.
`EXP-023 / evidence.selection_rule`
- The raw stage reports 23.0 mean features and mean ROC-AUC 0.7316521444852612; the selected stage reports 25.0 mean features and mean ROC-AUC 0.72011566222466.
`EXP-023 / evidence.stage_results[0].feature_count_mean, evidence.stage_results[0].metrics.roc_auc.mean, evidence.stage_results[5].feature_count_mean, evidence.stage_results[5].metrics.roc_auc.mean`
- The full_fe stage reports 59.333333333333336 mean features and mean ROC-AUC 0.725355704979456.
`EXP-023 / evidence.stage_results[4].feature_count_mean, evidence.stage_results[4].metrics.roc_auc.mean`

Challenges to claims

- **high** **EXP-023-C1** — The recommendation says to use the raw feature recipe, but the stated selection rule would require the smallest matrix within 0.002 of the best mean ROC-AUC; the raw stage is best by mean ROC-AUC and also smallest among the listed stages, so the recommendation is not justified as a 'next model screen' improvement over alternatives, only as the least-diluted top performer in this summary.
- **high** **EXP-023-C1** — The claim implies improved validation evidence, but the selected stage has lower mean ROC-AUC than raw and more features; the evidence does not show that feature engineering improved validation without dilution.
- **high** **EXP-023-C1** — The claim relies on training-CV ROC-AUC only; without nested validation or a held-out test, the stage comparison may be biased by feature selection.

- **medium** **EXP-023-C1** — The limitation about generic mathematical feature generation is acknowledged, but there is no documented decision-time or production-availability review, so the recommendation is incomplete for deployment-oriented use.

Risks & missing evidence

- Only 3 folds are reported; the ranking between raw and selected is not stable enough to rule out sampling noise.
- The selection rule references 'best stage' and a 0.002 ROC-AUC margin, but the selected stage is not within 0.002 of the best raw mean ROC-AUC, so the rule and the recommendation appear inconsistent.
- No held-out test or nested validation is provided, so stage selection may be optimistic.
- The package does not define 'feature dilution' quantitatively; feature count alone may miss redundancy, leakage, or semantic drift.
- No decision-time availability review is shown for the engineered features, especially ratios/interactions/log transforms.
- No production-feasibility or governance review is shown for whether generated features can be computed reliably at inference time.
- Feature importance stability is reported, but importance is not evidence of causal usefulness and does not validate feature semantics.

Smallest next experiment

Nested CV comparison of raw vs selected recipe with fixed fold splits

Hypothesis: If the raw recipe truly offers the best validation evidence without feature dilution, then under nested cross-validation on the same fixed splits it will remain competitive with the selected recipe and will not be outperformed by a materially larger feature matrix.

Success gate: Using identical outer folds and a pre-registered selection rule, the raw recipe must achieve outer-fold mean ROC-AUC no worse than the selected recipe by more than 0.002, while using fewer or equal features; otherwise the recommendation is not supported.

Why smallest: This changes only the validation design and compares the two most relevant candidates already present in the package, minimizing new modeling work while directly testing the selection-rule inconsistency and overfitting risk.

Evidence: `campaigns/model_building_50_v1/results/EXP-023_credit_default_feature_engineering.json`

EXP-024 · credit_default

medium confidence

model_selection

gpt-5.4-mini

Question. Which algorithm family provides the strongest stable training-CV evidence for this dataset profile?

Decision

Do not treat `extra_trees` as the final algorithm choice yet; treat it as the current CV leader only, pending repeat/outer validation and leakage checks.

Interpretation

The evidence supports a narrow training-CV ranking claim: under the stated selection rule, `extra_trees` is the top candidate on this raw-feature CV run. However, the package does not establish that `extra_trees` is the strongest stable family in a broader sense, because the comparison is limited to one 3-fold run, one stage, and one metric-weighted rule. Stability is only partially evidenced by modest fold-to-fold ROC-AUC spread, and the gap to other tree-based families is not obviously large enough to rule out ranking instability under different splits or repeats.

Observed evidence (cited)

- The experiment compares five baseline models on the raw stage with 3 CV folds each: `logistic_regression`, `extra_trees`, `hist_gradient_boosting`, `lightgbm`, and `xgboost`.
`EXP-024 / evidence.model_results[*].model`, `EXP-024 / evidence.model_results[*].folds`, `EXP-024 / evidence.model_results[*].stage`
- The selection rule ranks models by mean ROC-AUC minus 0.25×fold standard deviation, then runtime, using training CV only.
`EXP-024 / evidence.selection_rule`
- `extra_trees` has the highest reported selection metric among the listed models: mean ROC-AUC 0.7362568489708696 with std 0.01590385515180382 and elapsed_seconds 1.935686209000778.
`EXP-024 / evidence.model_results[1].metrics.roc_auc`, `EXP-024 / evidence.model_results[1].elapsed_seconds`
- `xgboost` has the lowest reported ROC-AUC variability among the tree-based models shown: std 0.017522347441607484, compared with `extra_trees` 0.01590385515180382, `lightgbm` 0.015166461731225754, and `hist_gradient_boosting` 0.014843033734126217.
`EXP-024 / evidence.model_results[1].metrics.roc_auc`, `EXP-024 / evidence.model_results[2].metrics.roc_auc`, `EXP-024 / evidence.model_results[3].metrics.roc_auc`, `EXP-024 / evidence.model_results[4].metrics.roc_auc`
- The reported fold ROC-AUC values for `extra_trees` are 0.7414463888718922, 0.7184064533235432, and 0.7489177047171737.
`EXP-024 / evidence.model_results[1].fold_metrics[*].roc_auc`
- The reported calibration-related metrics for `extra_trees` are Brier mean 0.14816840294526992 and ECE₁₀ mean 0.04173111735920745.
`EXP-024 / evidence.model_results[1].metrics.brier`, `EXP-024 / evidence.model_results[1].metrics.ece_10`

Challenges to claims

- **high** **EXP-024-C1** — The statement that `extra_trees` is the training-CV model candidate is supported only within the stated selection rule and raw-stage CV run; it is not evidence that `extra_trees` is the best algorithm family overall or on unseen data.
- **high** **EXP-024-C1** — The claim implies stability, but only one 3-fold run is shown. That is insufficient to establish ranking stability across resamples.

- **high** **EXP-024-C1** — The evidence package does not show fold construction details, so leakage or dependence between folds cannot be ruled out.
- **medium** **EXP-024-C1** — Runtime is used as a tie-breaker in the selection rule, but the package does not justify that runtime differences are relevant to the modeling objective or deployment constraints.

Risks & missing evidence

- Only 3 folds and 1 repeat are reported; this is weak evidence for stability of the ranking.
- No locked holdout or outer validation is shown, so the training-CV leader may not generalize.
- The package does not describe how folds were constructed; if the dataset has temporal, grouped, or duplicate structure, leakage risk is unassessed.
- The selection rule uses mean ROC-AUC minus $0.25 \times \text{std}$ and then runtime, but no uncertainty test is provided to show the ranking is robust to small metric differences.
- No hyperparameter search details are provided; baseline defaults may understate or distort family-level performance.
- Calibration and threshold-dependent metrics are reported, but the question asks for strongest stable training-CV evidence; no decision-time threshold policy or production-availability review is included.
- Feature stability is reported, but it is not tied to leakage detection or to the algorithm-family ranking.

Smallest next experiment

Repeat the raw-stage CV comparison with repeated stratified folds

Hypothesis: If `extra_trees` is truly the most stable training-CV family for this dataset profile, it should remain the top-ranked model across repeated stratified CV runs under the same selection rule.

Success gate: Run at least 10 repeated stratified CV resamples on the same raw-stage model set; success requires `extra_trees` to rank first in at least 70% of repeats and to retain a positive margin over the next-best family under the same mean ROC-AUC minus $0.25 \times \text{std}$ rule.

Why smallest: This changes only the resampling protocol while keeping the dataset, feature stage, model families, and selection rule fixed, making the ranking-stability question directly falsifiable with minimal added complexity.

Evidence: `campaigns/model_building_50_v1/results/EXP-024_credit_default_model_selection.json`

EXP-025 · credit_default

medium confidence

optimization_reliability

gpt-5.4-mini

Question. Does optimization produce a meaningful CV gain, and how does the preselected recipe perform on the once-consumed holdout?

Decision

Do not accept the claim that optimization produced a meaningful CV gain on the basis of this package alone. Treat the holdout score as a completed descriptive evaluation, not as evidence of superiority or deployability. Require the missing baseline-vs-tuned CV comparison and a decision-time/production-availability review before any feature or recipe promotion.

Interpretation

The package supports a descriptive statement that the selected recipe achieved a moderate holdout ROC-AUC and that some features appear repeatedly important across the reported CV folds. It does not, by itself, establish that optimization produced a meaningful CV gain because the actual baseline and tuned CV scores are not shown here, only the selection threshold. The holdout result is informative but remains a single once-consumed estimate; the bootstrap interval suggests uncertainty, and no decision-time or production-availability review is evidenced for any feature or recipe.

Observed evidence (cited)

- The once-consumed holdout evaluation for the preselected extra_trees/extra_trees_C02 recipe reports ROC-AUC 0.7746098248859629 with bootstrap 95% interval [0.7325987769358417, 0.8150416307620654].
EXP-025 / evidence.holdout_metrics; EXP-025 / evidence.holdout_roc_auc_bootstrap_ci; EXP-025 / evidence.holdout_consumed
- The holdout metrics also include accuracy 0.81, average precision 0.5298469143747347, Brier score 0.13712771805442664, ECE_10 0.046174388225831545, F1 0.4198473282442748, log loss 0.4362450195279426, precision 0.6470588235294118, recall 0.3107344632768362, and threshold 0.5.
EXP-025 / evidence.holdout_metrics
- The optimization rule states that the best tuned candidate is used only when mean ROC-AUC improves by at least 0.001 over baseline; otherwise baseline is preferred.
EXP-025 / evidence.selection_rule
- The feature-importance evidence lists X6, X7, X8, X11, X9, X10, X1, X5, X4, X3, X12, X2, X13, X15, and X18 among the top final features, with CV stability reported over 3 folds.
EXP-025 / evidence.top_final_features; EXP-025 / evidence.cv_feature_stability
- The production feature rule requires decision-time availability, contractual obtainability, monitoring for missingness/drift, and cross-fold stability; it explicitly warns against using importance alone.
EXP-025 / evidence.production_feature_rule

Challenges to claims

- **high** **EXP-025-C1** — The statement reports a holdout ROC-AUC, but the question also asks whether optimization produced a meaningful CV gain. The package does not include the tuned and baseline CV scores needed to verify the stated selection rule or any improvement magnitude.
- **medium** **EXP-025-C1** — Because the holdout is marked consumed, any further use of this holdout for recipe selection would be a validation risk; the package does not show safeguards preventing that.

- **medium** **EXP-025-C2** — The recommendation is directionally sound, but it is not supported by explicit evidence of decision-time availability, contractual obtainability, or monitoring readiness for the listed features.
- **medium** **EXP-025-C2** — Cross-fold stability is based on only 3 folds, which is limited evidence for a strong stability claim.

Risks & missing evidence

- No baseline CV metric is provided, so the claimed 'meaningful CV gain' cannot be verified against the stated 0.001 improvement rule.
- Only 3 CV folds are reported for feature stability, which is thin evidence for stability claims.
- The holdout has been consumed, so further tuning against it would risk leakage or implicit overfitting if used again for model selection.
- Feature importance alone is not sufficient for promotion; there is no explicit evidence that the listed features are decision-time available or contractually obtainable.
- No production monitoring plan or missingness/drift evidence is provided for the candidate features.
- The holdout metrics are single-point estimates; no comparison to alternative recipes or calibration baselines is shown.

Smallest next experiment

Baseline-vs-tuned CV audit with decision-time feature check

Hypothesis: The tuned recipe exceeds the baseline mean CV ROC-AUC by at least 0.001 under the same fold protocol, and the top candidate features remain available at decision time and in production.

Success gate: Report baseline and tuned mean CV ROC-AUC on the same folds; accept the optimization claim only if tuned - baseline $\geq$ 0.001. Separately, for each promoted feature, provide decision-time availability and production-obtainability evidence; otherwise exclude it.

Why smallest: This reuses the existing CV protocol and adds only the missing baseline comparison plus a lightweight feature-availability audit, which is the minimum needed to falsify the optimization and promotion claims without touching the consumed holdout.

Evidence: campaigns/model_building_50_v1/results/EXP-025_credit_default_optimization_reliability.json

EXP-026 · german_credit

medium confidence

data_understanding

gpt-5.4-mini

Question. What does the dataset contain, and which quality or production-availability risks must be resolved before modeling?

Decision

Do not approve modeling yet. The dataset can be described, but the evidence is insufficient to clear production-availability and leakage-risk checks.

Interpretation

Observed evidence supports a narrow data-understanding summary: the dataset is small, tabular, and appears to have no missing values in the sampled feature profiles. However, the package does not establish that the cached features preserve raw category semantics, nor does it establish production-time availability, decision-time availability, or temporal stability. The claim that no features triggered review rules is only partially supported because the evidence shown is a sample of 10 feature profiles, not the full 20-feature set, and one feature lacks a PSI value. The strongest unresolved issue is potential preprocessing leakage from factorized categorical encodings and the absence of a production-availability review for each feature.

Observed evidence (cited)

- The package reports 1000 source rows and 20 cached features for german_credit, with a training positive rate of 0.300.
`EXP-026 / evidence.source_rows; EXP-026 / evidence.feature_count; EXP-026 / evidence.positive_rate_train`
- The provided feature profile sample shows zero missingness for the listed attributes and no recorded production_risks entries in the sample.
`EXP-026 / evidence.feature_profiles[].missing_rate; EXP-026 / evidence.feature_profiles[].production_risks`
- The sample includes random_split_psi values for several attributes, but one listed feature has random_split_psi = null.
`EXP-026 / evidence.feature_profiles[].random_split_psi`
- The evidence bundle itself states that cached UCI matrices factorized categoricals before this campaign and that random train/holdout PSI is only a smoke check.
`EXP-026 / claims[1].limitations[0]; EXP-026 / claims[1].limitations[1]`

Challenges to claims

- **medium** **EXP-026-C1** — The fact claim is plausible from the cited fields, but the package does not show the underlying label definition or whether the 0.300 positive rate is computed on a leakage-free training split. Without split construction details, the rate is descriptive only.
- **high** **EXP-026-C2** — The statement that 0 features triggered review rules is not fully supported by the evidence excerpt because only 10 feature profiles are visible, not all 20 cached features. Also, null PSI for one feature means the shift review is incomplete for the shown sample.
- **high** **EXP-026-C2** — The limitation about factorized categoricals is a valid risk note, but the evidence bundle does not show the raw source schema or a reversible mapping, so the production-encoder requirement remains unverified.

Risks & missing evidence

- No explicit evidence that all 20 features were reviewed; only 10 feature profiles are shown in the bundle.

- No raw-column schema or provenance is provided, so the claim about restored category semantics is not verified.
- No decision-time availability review is shown for any feature, so features may not be available when predictions are made.
- No production-time stability evidence is provided; random-split PSI cannot substitute for temporal or deployment-slice validation.
- No target definition, label timing, or leakage audit is provided, so target leakage cannot be ruled out.
- No missingness/identifier/near-constant review thresholds are documented, so the rule-trigger statement is not fully auditable from the excerpt.
- One feature has null `random_split_psi`, which leaves an incomplete shift check.

Smallest next experiment

Raw-schema and availability audit for all 20 features

Hypothesis: At least one feature is either not available at decision time, lacks a reversible raw-category mapping, or fails a complete per-feature quality check when all 20 features are audited from raw source columns.

Success gate: For all 20 features, provide raw source column names, decision-time availability status, reversible categorical mapping status, missingness, identifier/near-constant flags, and a non-null shift statistic; the experiment passes only if no feature is missing any of these fields and no feature is flagged as unavailable or unrecoverable.

Why smallest: This is the minimal falsifiable step that directly tests the two main unresolved risks: production availability and preprocessing leakage from factorized categoricals, without requiring model training.

Evidence: `campaigns/model_building_50_v1/results/EXP-026_german_credit_data_understanding.json`

EXP-027 · german_credit

medium confidence

leakage_audit

gpt-5.4-mini

Question. Which columns violate the prediction-time contract or behave like suspicious target proxies?

Decision

Insufficient evidence to identify or approve any specific column as safe or unsafe. The current package is a leakage-audit stub, not a completed column-level contract review.

Interpretation

The package supports only a narrow factual conclusion: the canary test succeeded and no leakage features were declared. It does not establish which columns violate the prediction-time contract, because there is no column-level evidence, no decision-time semantics review, and no safe-vs-unsafe validation result. The claim that there are no declared decision-time exclusions is not evidence that no leakage or proxy columns exist; it only indicates none were declared in the package.

Observed evidence (cited)

- The leakage detector's synthetic exact-target canary passed.
`EXP-027 / evidence.detector_canary_passed`
- No declared decision-time exclusions were provided for the `german_credit` dataset.
`EXP-027 / evidence.declared_leakage_features`
- The `safe_vs_unsafe_training_cv` evidence is null, so no validation result is available for comparing safe versus unsafe feature sets.
`EXP-027 / evidence.safe_vs_unsafe_training_cv`

Challenges to claims

- **medium** **EXP-027-C1** — The canary result is a detector sanity check, not evidence about real dataset columns. It cannot support any claim about actual leakage or proxy behavior in `german_credit`.
- **high** **EXP-027-C2** — The statement 'Declared decision-time exclusions for `german_credit`: none' is unsupported as a decision about leakage. It only reflects that no exclusions were declared, while the package explicitly notes that only decision-time semantics can confirm leakage.

Risks & missing evidence

- No column-level list of candidate features, so suspicious proxies cannot be named from the provided evidence.
- No decision-time semantics review, which is required to determine whether a column is available at prediction time.
- No production-availability review, so even a semantically valid feature may still be unusable at inference.
- No `safe_vs_unsafe_training_cv` result, so there is no empirical comparison to support or refute leakage suspicion.
- A passing synthetic canary only validates the detector mechanism, not the absence of real leakage in the dataset.
- The absence of declared leakage features is not equivalent to a verified absence of leakage.

Smallest next experiment

Column-level decision-time contract audit for `german_credit`

Hypothesis: At least one feature in `german_credit` will be flagged as unavailable at prediction time or as a suspicious target proxy when each column is checked against decision-time semantics and simple leakage heuristics.

Success gate: Produce a column-by-column table with: feature name, decision-time availability verdict, production-availability verdict, and heuristic flags (e.g., target proxy, post-outcome, identifier-like). The experiment succeeds only if every feature has an explicit verdict and any flagged feature is supported by a documented decision-time rationale; if no features are flagged, the audit must still show complete coverage with no missing verdicts.

Why smallest: This is the minimal next step because it adds only the missing evidence needed to answer the question: a per-column contract review. It does not require retraining or new modeling, only annotation and validation of existing columns.

Evidence: `campaigns/model_building_50_v1/results/EXP-027_german_credit_leakage_audit.json`

EXP-028 · german_credit

medium confidence

feature_engineering

gpt-5.4-mini

Question. Which train-fitted feature recipe improves validation evidence without feature dilution?

Decision

Do not accept the recommendation as written; the evidence is insufficient and internally inconsistent with the selection rule.

Interpretation

The evidence supports only a narrow descriptive conclusion: among the tested recipes, the interactions stage has the highest reported mean training-CV ROC-AUC. However, the package does not support the recommendation to use it as the next screen under the stated selection rule, because the selected stage is not within the 0.002 ROC-AUC tolerance of the best stage. The evidence also does not establish that the interactions recipe improves validation evidence without feature dilution in a decision-ready sense, because the reported metrics are training-CV only, the fold count is small, and the selected stage is actually lower-scoring than the best stage while being smaller. The package therefore contains an internal inconsistency between the selection rule and the stated recommendation.

Observed evidence (cited)

- The experiment compares five stages on german_credit: raw, logs, ratios, interactions, and selected, all using lightgbm with 3 folds and 1 repeat.
`EXP-028 / evidence.stage_results`
- The reported mean training-CV ROC-AUC values are 0.7527162037835663 (raw), 0.7527162037835663 (logs), 0.7521825225691451 (ratios), 0.7600576447587833 (interactions), and 0.7421301487359363 (selected).
`EXP-028 / evidence.stage_results[*].metrics.roc_auc.mean`
- The reported feature_count_mean values are 20.0 (raw), 33.0 (logs), 48.0 (ratios), 56.0 (interactions), and 25.0 (selected).
`EXP-028 / evidence.stage_results[*].feature_count_mean`
- The selection rule is to choose the smallest feature matrix within 0.002 mean training-CV ROC-AUC of the best stage.
`EXP-028 / evidence.selection_rule`
- The selected stage has mean training-CV ROC-AUC 0.7421301487359363, which is not within 0.002 of the best reported mean training-CV ROC-AUC 0.7600576447587833.
`EXP-028 / evidence.stage_results[*].metrics.roc_auc.mean; EXP-028 / evidence.selection_rule`
- The interactions stage has the highest reported mean training-CV ROC-AUC among the listed stages, but it also has the largest reported feature_count_mean among the stages shown.
`EXP-028 / evidence.stage_results[*].metrics.roc_auc.mean; EXP-028 / evidence.stage_results[*].feature_count_mean`

Challenges to claims

- **high** **EXP-028-C1** — The recommendation says to use the interactions recipe, but the stated selection rule would instead choose the smallest matrix within 0.002 of the best stage; the selected stage is not within that tolerance of the best reported mean ROC-AUC.
- **high** **EXP-028-C1** — The claim implies validation improvement, but the evidence only reports training-CV metrics and does not include an independent holdout or outer CV estimate.

- **medium** **EXP-028-C1** — The claim suggests the interactions recipe is preferable, but the evidence does not quantify the tradeoff against feature dilution using a defined metric beyond raw feature count.
- **high** **EXP-028-C1** — The package does not show fold-wise generation details for the derived features, so leakage from global fitting of feature engineering cannot be ruled out.

Risks & missing evidence

- No separate holdout or outer validation result is shown; all conclusions are based on training-CV evidence only.
- The package does not report whether feature generation was fit strictly inside each training fold for every stage, so leakage risk remains unresolved.
- The selection rule is not satisfied by the selected stage, but the package does not explain why the recommendation overrides the rule.
- Only 3 folds and 1 repeat are reported, which makes the ranking unstable and sensitive to split variance.
- The evidence does not show decision-time availability or production-time computability of the derived features.
- The feature stability table includes features with `observed_folds = 1`, which suggests instability in the derived feature set.
- The phrase 'without feature dilution' is not operationalized; no explicit dilution metric or threshold is provided.

Smallest next experiment

Fold-local ablation of interactions vs selected under nested validation

Hypothesis: If the interactions recipe truly improves validation evidence without unacceptable dilution, then a fold-local, leakage-safe nested-CV evaluation will preserve its ROC-AUC advantage over the selected smaller matrix within a predeclared tolerance.

Success gate: Run nested CV with feature engineering fit only on each training fold. Declare success only if interactions exceeds the selected recipe by at least 0.002 mean outer-CV ROC-AUC and does not increase `feature_count_mean` by more than a predeclared dilution budget.

Why smallest: This changes only the validation protocol and compares only the two most relevant candidates, which is the minimal falsifiable test of the recommendation and leakage risk.

Evidence: `campaigns/model_building_50_v1/results/EXP-028_german_credit_feature_engineering.json`

EXP-029 · german_credit

medium confidence

model_selection

gpt-5.4-mini

Question. Which algorithm family provides the strongest stable training-CV evidence for this dataset profile?

Decision

Do not treat `extra_trees` as the final winner from this package alone. The evidence is sufficient only to nominate a provisional training-CV candidate, not to approve a family choice or deployment decision.

Interpretation

The evidence supports only a narrow training-CV conclusion: `extra_trees` is the strongest mean ROC-AUC performer among the reported models, while `xgboost` may be the strongest under the stated mean-minus- $0.25 \times \text{std}$ selection rule. However, the package does not provide enough detail to verify that the selection rule was applied consistently, nor does it establish that any model is the best stable choice beyond this CV setup. The stability signal is weakly informative because it is based on only 3 folds and mixed feature-stability reporting, so the safest interpretation is that the ranking is provisional and should be rechecked with a locked evaluation and a more explicit stability protocol.

Observed evidence (cited)

- The experiment is a completed model-selection run on the `german_credit` dataset, using training CV only and the selection rule "Rank by mean ROC-AUC minus $0.25 \times \text{fold standard deviation}$, then runtime."

EXP-029 / `evidence_bundle.status`; EXP-029 / `evidence_bundle.dataset`; EXP-029 / `evidence.selection_rule`

- Among the listed models, `extra_trees` has the highest reported mean ROC-AUC (0.7768815407394629) with fold std 0.029815169083194874.

EXP-029 / `evidence.model_results[1].metrics.roc_auc`

- `xgboost` has the highest reported mean ROC-AUC-minus- $0.25 \times \text{std}$ under the stated selection rule if computed from the provided mean and std values, but the package does not explicitly report the ranking score for each model.

EXP-029 / `evidence.selection_rule`; EXP-029 / `evidence.model_results[4].metrics.roc_auc`; EXP-029 / `evidence.model_results[1].metrics.roc_auc`

- `extra_trees` shows relatively tight fold spread in ROC-AUC across 3 folds, with fold values 0.799298128342246, 0.7883021390374332, and 0.7430443548387095.

EXP-029 / `evidence.model_results[1].fold_metrics[0].roc_auc`; EXP-029 / `evidence.model_results[1].fold_metrics[1].roc_auc`; EXP-029 / `evidence.model_results[1].fold_metrics[2].roc_auc`

- The package reports feature stability summaries for `extra_trees`, `lightgbm`, and `xgboost`, but not for `logistic_regression` or `hist_gradient_boosting`.

EXP-029 / `evidence.model_results[1].feature_stability`; EXP-029 / `evidence.model_results[3].feature_stability`; EXP-029 / `evidence.model_results[4].feature_stability`; EXP-029 / `evidence.model_results[0].feature_stability`; EXP-029 / `evidence.model_results[2].feature_stability`

Challenges to claims

- **high** EXP-029-C1 — The statement says `extra_trees` is the training-CV model candidate, but the stated selection rule is mean ROC-AUC minus $0.25 \times \text{fold std}$, and the package does not report the computed score for each model. Based on the provided mean/std fields, `xgboost` appears potentially competitive or better under that rule, so the candidate claim is not fully supported as written.
- **high** EXP-029-C1 — The claim presents `extra_trees` as the candidate using interactions features, but the evidence bundle does not document whether the interaction-generation pipeline was confined

to each training fold. This leaves leakage risk unresolved.

- **medium** **EXP-029-C1** — The claim implies stability from 3-fold CV, but three folds are too few to support a strong stability conclusion without repeated CV or split-sensitivity analysis.

Risks & missing evidence

- Only 3 CV folds are reported, which makes the stability estimate noisy and sensitive to split choice.
- The package does not specify whether preprocessing, interaction generation, and any target/feature transforms were fit strictly inside each training fold; leakage risk is unresolved.
- The selection rule is stated, but the per-model ranking score is not shown, so the claimed candidate may not actually be the rule-based winner.
- No locked holdout or nested validation result is provided, so generalization beyond training CV is unverified.
- No split strategy is described (e.g., stratification, grouping, temporal ordering), so validation suitability for the dataset profile cannot be assessed.
- Feature stability summaries are incomplete across models, making cross-family stability comparisons asymmetric.
- No decision-time or production-availability review is included, so even a strong CV model cannot be approved for use.

Smallest next experiment

Rule-verification and leakage-check rerun on the top two families

Hypothesis: If the same preprocessing and interaction pipeline is fit strictly within each training fold and the selection score is computed explicitly for each model, the apparent winner will remain stable; otherwise the current ranking is split- or leakage-sensitive.

Success gate: Run repeated stratified CV with the exact same pipeline for extra_trees and xgboost, report per-fold mean ROC-AUC, fold std, and the explicit selection score for each run; the success condition is that the same family wins in at least 2 independent repeats and the ranking score margin is positive in every repeat.

Why smallest: This changes only the validation protocol and ranking audit for the two most relevant contenders, without expanding the model search space or introducing new feature engineering.

Evidence: campaigns/model_building_50_v1/results/EXP-029_german_credit_model_selection.json

EXP-030 · german_credit

medium confidence

optimization_reliability

gpt-5.4-mini

Question. Does optimization produce a meaningful CV gain, and how does the preselected recipe perform on the once-consumed holdout?

Decision

Do not claim a meaningful optimization gain from this package alone. Treat the holdout score as a finalized retrospective estimate only, and require an explicit baseline-vs-tuned CV comparison plus a fresh untouched evaluation before any stronger conclusion.

Interpretation

The package supports a single held-out performance estimate for the preselected recipe, but it does not establish that optimization produced a meaningful CV gain because the actual baseline-versus-tuned CV comparison is not shown. The holdout result is informative but limited because the holdout has already been consumed, so it should not be used for further model selection. The feature evidence is consistent with a cautious feature-governance stance, but it does not by itself validate any feature for production use without decision-time and availability checks.

Observed evidence (cited)

- The once-consumed holdout ROC-AUC for the preselected `extra_trees/extra_trees_C01` recipe is 0.7942857142857144, with bootstrap 95% interval [0.7179534546874278, 0.8620644773123065].
`EXP-030 / evidence.holdout_metrics; evidence.holdout_roc_auc_bootstrap_ci; evidence.holdout_consumed`
- The holdout evaluation is marked as consumed.
`EXP-030 / evidence.holdout_consumed`
- The selection rule states that the best tuned candidate is used only when mean ROC-AUC improves by at least 0.001 over baseline; otherwise baseline is preferred.
`EXP-030 / evidence.selection_rule`
- The top final features list includes `Attribute1`, `inter_Attribute1_x_log1p_Attribute2`, `inter_Attribute1_x_Attribute8`, `log1p_Attribute3`, and `Attribute3` among others.
`EXP-030 / evidence.top_final_features`
- The CV feature stability table shows some features observed in 3 folds with `top10_frequency` values ranging from 1.0 to 0.0, and some features observed in only 1 fold.
`EXP-030 / evidence.cv_feature_stability`
- The production feature rule requires decision-time availability, cross-fold stability, contractual obtainability, monitoring for missingness/drift, and more than one fit's importance signal.
`EXP-030 / evidence.production_feature_rule`

Challenges to claims

- **high** **EXP-030-C1** — The holdout performance is stated correctly, but the question also asks whether optimization produced a meaningful CV gain; the package does not include the baseline CV score or the tuned CV score needed to verify the improvement threshold.
- **medium** **EXP-030-C2** — The recommendation is directionally sound, but it is not supported by explicit evidence of decision-time availability or contractual obtainability for the named features; it should be treated as a governance rule, not a validated conclusion about these features.

Risks & missing evidence

- No explicit baseline CV ROC-AUC is provided, so the claimed optimization gain cannot be verified against the stated 0.001 improvement rule.
- The holdout has been consumed, so any further tuning or feature selection risks leakage into the final estimate.
- Only 3 CV folds are reflected in the feature stability table, which is thin evidence for stability claims.
- Feature importance is not the same as production usefulness; decision-time availability and contractual obtainability are not demonstrated for the listed features.
- No evidence is provided on calibration stability, threshold selection rationale, or whether the 0.5 threshold was pre-specified for the holdout report.
- No production-availability review is shown for the preselected recipe or its features.

Smallest next experiment

Baseline-vs-tuned CV comparison on a fresh split with locked recipe

Hypothesis: If the tuned recipe truly improves over baseline, then on a new untouched cross-validation split its mean ROC-AUC will exceed baseline by at least 0.001 while preserving similar calibration and without using the consumed holdout for any selection.

Success gate: Pre-register baseline and tuned candidates, evaluate both on the same fresh CV split, and accept optimization only if tuned mean ROC-AUC - baseline mean ROC-AUC ≥ 0.001 ; otherwise keep baseline. The holdout must remain untouched for this experiment.

Why smallest: This is the minimal falsifiable test because it reuses the existing recipe family and only adds the missing baseline comparison on a fresh validation split, directly addressing the unsupported optimization-gain claim without expanding the modeling scope.

Evidence: campaigns/model_building_50_v1/results/EXP-030_german_credit_optimization_reliability.json

EXP-031 · mushroom

medium confidence

data_understanding

gpt-5.4-mini

Question. What does the dataset contain, and which quality or production-availability risks must be resolved before modeling?

Decision

Do not proceed to modeling yet; resolve data representation, leakage, and production-availability questions first.

Interpretation

This evidence supports a basic dataset inventory and suggests the data are mostly low-cardinality categorical fields with no missingness in the sampled profiles. However, the package does not establish production readiness: the features are described as cached/factorized, so raw category semantics and the intended production encoder path remain unresolved. The random-split PSI values are only a within-sample smoke test and do not address temporal, source, or decision-time stability. The evidence is insufficient to conclude that the dataset is safe for modeling without leakage, schema, and availability checks.

Observed evidence (cited)

- The dataset has 8124 source rows and 22 cached features.
EXP-031 / evidence.source_rows; EXP-031 / evidence.feature_count
- The training positive rate is 0.481875.
EXP-031 / evidence.positive_rate_train
- The provided feature profile sample shows zero missingness for the listed features and very low random-split PSI values where reported.
EXP-031 / evidence.feature_profiles
- The provided feature profile sample includes low unique ratios and high top-value frequencies for several features, consistent with categorical-like variables encoded as numeric values.
EXP-031 / evidence.feature_profiles

Challenges to claims

- **medium** **EXP-031-C1** — The claim is numerically supported by the evidence bundle, but it implicitly treats cached features as the final modeling representation. That is unsupported because the package does not show raw categorical values or the production encoder path.
- **high** **EXP-031-C2** — The statement that 1 feature triggered review rules is not verifiable from the provided sample because only a subset of feature profiles is shown and the triggering rule thresholds are not included.
- **medium** **EXP-031-C2** — The limitation about random-split PSI correctly warns against overinterpretation, but the evidence does not show any temporal or production-simulation test to support stability claims.

Risks & missing evidence

- Raw category semantics are not shown; the evidence only exposes cached/factorized numeric representations, so production encoding compatibility is unverified.
- Only a subset of feature profiles is shown; the package does not prove that all 22 features were checked for missingness, identifier behavior, near-constant behavior, or shift.

- No target definition, label provenance, or decision-time availability review is provided, so leakage risk cannot be assessed.
- Random train/holdout PSI is not a substitute for temporal or source-shift validation.
- No train/validation split protocol is described, so the risk of split leakage or duplicate entities across splits remains unresolved.
- No production schema, feature freshness, or online/offline availability constraints are documented.

Smallest next experiment

Raw-category and decision-time availability audit for all 22 features

Hypothesis: If the raw categorical values, target availability time, and production feature availability are documented for all 22 features, then we can identify whether any feature is unusable or leakage-prone before modeling.

Success gate: For every feature, provide raw value semantics, missingness, cardinality, and a decision-time availability flag; fail if any feature lacks a production-available source, is derived from post-decision information, or cannot be mapped from the cached encoding back to a stable production encoder.

Why smallest: This is the smallest falsifiable step because it reuses the existing dataset inventory and only adds schema/availability verification, without requiring model training or new data collection.

Evidence: `campaigns/model_building_50_v1/results/EXP-031_mushroom_data_understanding.json`

EXP-032 · mushroom

medium confidence

leakage_audit

gpt-5.4-mini

Question. Which columns violate the prediction-time contract or behave like suspicious target proxies?

Decision

Insufficient evidence to identify or approve any real columns as safe or unsafe. The current package supports only that the detector is not completely broken on a synthetic canary, not that the mushroom features satisfy prediction-time availability or are free of target proxies.

Interpretation

The package provides only a minimal sanity check that the detector can flag an injected exact-target canary. It does not provide column-level evidence for which real columns violate the prediction-time contract or act as target proxies. The absence of declared leakage features is not evidence of absence, and the null `safe_vs_unsafe_training_cv` field leaves the main leakage-risk question unresolved.

Observed evidence (cited)

- The leakage detector's synthetic exact-target canary passed.
`EXP-032 / evidence.detector_canary_passed`
- No declared decision-time exclusions were provided for the mushroom dataset.
`EXP-032 / evidence.declared_leakage_features`
- The `safe_vs_unsafe_training_cv` evidence is null, so there is no reported cross-validation comparison between safe and unsafe feature sets.
`EXP-032 / evidence.safe_vs_unsafe_training_cv`

Challenges to claims

- **low** **EXP-032-C1** — The canary pass is a detector sanity check, not evidence about any real dataset column. It cannot support conclusions about actual leakage or proxy columns.
- **high** **EXP-032-C2** — The statement 'Declared decision-time exclusions for mushroom: none' is unsupported as a decision about feature safety; the evidence only shows an empty `declared_leakage_features` list and a null `safe_vs_unsafe_training_cv` field. That does not establish prediction-time compliance.

Risks & missing evidence

- No column-level audit is shown, so suspicious proxies cannot be named from the provided evidence.
- No decision-time semantics review is included; feature availability at prediction time is unverified.
- No production-availability review is included; even non-leaky columns may be unavailable or unstable in deployment.
- No safe-vs-unsafe model comparison is reported, so there is no falsifiable evidence that excluding suspected columns changes performance as expected.
- The canary result only validates detector sensitivity to an injected exact-target feature, not detection of subtler leakage patterns.

Smallest next experiment

Column-level prediction-time contract audit with safe/unsafe ablation

Hypothesis: If any mushroom columns violate the prediction-time contract or behave as target proxies, then removing them from the feature set will materially change validation behavior relative to the full

feature set, and the offending columns will be identifiable by decision-time availability review.

Success gate: For each column, provide a decision-time availability verdict and a reason; then run a pre-registered safe-vs-unsafe ablation. Success is achieved only if at least one column is flagged by decision-time review and the ablation shows a reproducible validation difference between the full and contract-compliant feature sets under the same split protocol.

Why smallest: This is the smallest falsifiable step because it adds only two missing pieces: a column-level decision-time review and one ablation comparison. It directly tests the unresolved leakage question without changing the dataset, model family, or evaluation protocol.

Evidence: `campaigns/model_building_50_v1/results/EXP-032_mushroom_leakage_audit.json`

EXP-033 · mushroom

medium confidence

feature_engineering

gpt-5.4-mini

Question. Which train-fitted feature recipe improves validation evidence without feature dilution?

Decision

Do not accept the recommendation to use the raw recipe as stated. The smallest defensible conclusion is that the selected stage is the smallest recipe within the stated ROC-AUC tolerance of the best stage, but the package is insufficient to claim a production-ready improvement or to prefer raw over selected without an external validation check.

Interpretation

The evidence supports a narrow ranking claim: under the stated selection rule, the selected feature matrix is the smallest recipe within 0.002 mean training-CV ROC-AUC of the best observed stage. However, the package does not justify the stronger recommendation to use the raw recipe, because the selected stage has the best reported mean ROC-AUC and fewer features than the full_fe stage. The evidence also does not establish that any recipe improves validation evidence in a production-relevant sense, because only training-CV is shown and no decision-time or production-availability review is provided.

Observed evidence (cited)

- The package reports 3-fold LightGBM CV on the mushroom dataset for multiple feature recipes, with the selected stage labeled "selected" and the selection rule defined as choosing the smallest feature matrix within 0.002 mean training-CV ROC-AUC of the best stage.
`EXP-033 / evidence.stage_results; EXP-033 / evidence.selection_rule`
- The raw stage has mean ROC-AUC 0.9999788529859583 with mean feature_count_mean 22.0 and stage label "raw".
`EXP-033 / evidence.stage_results[0].metrics.roc_auc.mean; EXP-033 / evidence.stage_results[0].feature_count_mean; EXP-033 / evidence.stage_results[0].stage`
- The logs stage has mean ROC-AUC 0.9999788529859583 with mean feature_count_mean 30.666666666666668 and stage label "logs".
`EXP-033 / evidence.stage_results[1].metrics.roc_auc.mean; EXP-033 / evidence.stage_results[1].feature_count_mean; EXP-033 / evidence.stage_results[1].stage`
- The ratios stage has mean ROC-AUC 0.9999459576307825 with mean feature_count_mean 45.666666666666664 and stage label "ratios".
`EXP-033 / evidence.stage_results[2].metrics.roc_auc.mean; EXP-033 / evidence.stage_results[2].feature_count_mean; EXP-033 / evidence.stage_results[2].stage`
- The full_fe stage has mean ROC-AUC 0.9999882516588657 with mean feature_count_mean 53.666666666666664 and stage label "full_fe".
`EXP-033 / evidence.stage_results[3].metrics.roc_auc.mean; EXP-033 / evidence.stage_results[3].feature_count_mean; EXP-033 / evidence.stage_results[3].stage`
- The selected stage has mean ROC-AUC 0.9999882516588657 with mean feature_count_mean 25.0 and stage label "selected".
`EXP-033 / evidence.stage_results[4].metrics.roc_auc.mean; EXP-033 / evidence.stage_results[4].feature_count_mean; EXP-033 / evidence.stage_results[4].stage`

Challenges to claims

- **high** **EXP-033-C1** — The recommendation says to use the raw recipe, but the selected stage has the highest reported mean ROC-AUC and a smaller feature count than full_fe; the evidence does not support preferring raw over selected.

- **high** EXP-033-C1 — The phrase "improves validation evidence" is unsupported because only training-CV metrics are reported and no external validation or nested selection result is provided.
- **high** EXP-033-C1 — The evidence bundle does not demonstrate that the feature recipe was fitted without leakage inside each fold; the train-fitted requirement is asserted by the question but not verified by the evidence paths shown.
- **medium** EXP-033-C1 — The limitation about generic mathematical feature generation lacking domain meaning is acknowledged, but no decision-time or production-availability review is provided, so the recommendation is premature.

Risks & missing evidence

- Only training-CV evidence is shown; there is no untouched holdout or nested validation to confirm the recipe ranking.
- The selection rule uses mean training-CV ROC-AUC, which can favor overfit feature search when many engineered candidates are tried.
- The package does not show whether feature generation was fitted strictly inside each training fold for every stage; leakage risk remains unresolved from the evidence provided.
- The claim says "raw feature recipe" but the selected stage is the one with the best mean ROC-AUC and fewer features than full_fe; the recommendation appears inconsistent with the reported metrics.
- No decision-time availability review is shown for engineered features, especially interaction and ratio terms that may not be computable or stable in production.
- The mushroom dataset appears highly separable from the reported scores, so near-perfect CV metrics may not discriminate recipes meaningfully.
- Feature stability is mixed: some engineered features appear in only 1-2 folds, suggesting instability that should be checked before relying on them.

Smallest next experiment

Nested holdout comparison of raw vs selected feature recipe

Hypothesis: If the selected recipe truly generalizes better without feature dilution, then it will retain its advantage over the raw recipe on a strictly untouched outer validation split while using only fold-local feature fitting.

Success gate: On an outer holdout or outer CV split, the selected recipe must achieve ROC-AUC no worse than raw by at most 0.002, with all feature engineering fitted only on the inner training folds and with no leakage detected in a fold-by-fold audit.

Why smallest: This changes only the evaluation protocol and leakage audit while keeping the model family, dataset, and candidate recipes fixed, making it the minimal falsifiable test of the current recommendation.

Evidence: campaigns/model_building_50_v1/results/EXP-033_mushroom_feature_engineering.json

EXP-034 · mushroom

medium confidence

model_selection

gpt-5.4-mini

Question. Which algorithm family provides the strongest stable training-CV evidence for this dataset profile?

Decision

Do not accept the claim that `extra_trees` is the strongest stable training-CV candidate from this package alone. The package is internally inconsistent with its own selection rule and lacks locked-holdout evidence, repeated CV, and leakage checks.

Interpretation

The evidence supports that several algorithm families are near-ceiling on training CV for this dataset profile, but it does not uniquely establish `extra_trees` as the strongest stable family under the stated ranking rule. Based on the reported mean ROC-AUC and fold variability alone, `xgboost` appears numerically strongest on the provided training-CV summary, while `extra_trees` is perfectly tied on mean ROC-AUC but not on the stability criterion as written because its reported std is 0.0 despite one fold being below 1.0. This inconsistency makes the selection claim under-specified and potentially miscomputed. The evidence is also insufficient to answer any holdout/generalization question.

Observed evidence (cited)

- The selection rule ranks models by mean ROC-AUC minus $0.25 \times \text{fold standard deviation}$, then runtime, using training CV only.
`EXP-034 / evidence.selection_rule`
- The package reports five raw-stage baseline models with 3 CV folds each: `logistic_regression`, `extra_trees`, `hist_gradient_boosting`, `lightgbm`, and `xgboost`.
`EXP-034 / evidence.model_results`
- `extra_trees` has fold ROC-AUC values of 1.0, 1.0, and 0.9998308238876672, with mean ROC-AUC 1.0 and std 0.0 as reported in the model metrics block.
`EXP-034 / evidence.model_results[1].fold_metrics`; `EXP-034 / evidence.model_results[1].metrics.roc_auc`
- `logistic_regression` has mean ROC-AUC 0.9991786656107523 and std 0.00035451463066407516.
`EXP-034 / evidence.model_results[0].metrics.roc_auc`
- `hist_gradient_boosting` has mean ROC-AUC 0.9999436079625558 and std 9.767387399576603e-05.
`EXP-034 / evidence.model_results[2].metrics.roc_auc`
- `lightgbm` has mean ROC-AUC 0.9999788529859583 and std 3.662770274845633e-05.
`EXP-034 / evidence.model_results[3].metrics.roc_auc`

Challenges to claims

- **high** **EXP-034-C1** — The claim that `extra_trees` is the training-CV candidate is not supported by the stated ranking rule because `xgboost` has the highest reported mean ROC-AUC, and `extra_trees`' reported std is inconsistent with its fold metrics.
- **high** **EXP-034-C1** — The claim implies a winner for the dataset profile, but only training CV is shown and no locked holdout is available; this is not enough to compare algorithm families for generalization.
- **medium** **EXP-034-C1** — The phrase 'strongest stable' is unsupported because stability is not consistently reported across models and the `extra_trees` std appears miscomputed or misreported.

Risks & missing evidence

- No locked holdout or external test set is reported, so generalization cannot be assessed.
- Only 3 folds and 1 repeat are shown; stability estimates are weak and sensitive to split choice.
- The selection rule uses mean ROC-AUC minus $0.25 \times$ fold standard deviation, but the reported extra_trees ROC-AUC std is 0.0 even though fold values differ, suggesting a calculation or reporting issue.
- The package does not specify whether CV was stratified, grouped, or nested, so split leakage risk cannot be evaluated.
- No preprocessing/encoding pipeline details are provided, so target leakage through feature engineering cannot be ruled out.
- No decision-time or production-availability review is present, so deployment suitability cannot be inferred.
- The dataset profile is not described beyond the dataset name, so the appropriateness of the algorithm family for the profile is not evidenced.

Smallest next experiment

Repeat stratified nested CV with locked holdout comparison for top two families

Hypothesis: If the current ranking is robust, the top two families under the same preprocessing will preserve their ordering across repeated stratified CV and on a locked holdout, and the reported stability statistics will be internally consistent.

Success gate: Run repeated stratified CV with at least 10 repeats for xgboost and extra_trees using identical preprocessing and a locked holdout; accept the current candidate only if it remains top-ranked by the predeclared rule on the repeated-CV summary and its holdout ROC-AUC is not worse than the runner-up by more than a prespecified margin, with all fold-std calculations matching the raw fold metrics.

Why smallest: This changes only the evaluation protocol and compares only the two most relevant contenders, while directly testing the ranking rule, stability, and holdout generalization without expanding the model search space.

Evidence: campaigns/model_building_50_v1/results/EXP-034_mushroom_model_selection.json

EXP-035 · mushroom

high confidence

optimization_reliability

gpt-5.4-mini

Question. Does optimization produce a meaningful CV gain, and how does the preselected recipe perform on the once-consumed holdout?

Decision

Do not conclude that optimization produced a meaningful CV gain from this package. Treat the holdout result as a completed evaluation of the preselected recipe only, and require a direct baseline-vs-tuned CV comparison plus decision-time/production-availability review before any feature or recipe promotion.

Interpretation

The evidence supports that the chosen recipe performs perfectly on the consumed holdout, but this does not by itself establish that optimization produced a meaningful generalization gain. The selection rule is conservative, yet the package does not show the baseline CV score, the tuned CV score, or their difference, so the central optimization question remains unresolved. The feature evidence is directionally consistent with stability, but it is not sufficient to approve feature promotion without decision-time and production-availability checks.

Observed evidence (cited)

- The once-consumed holdout was evaluated and marked consumed.
[EXP-035 / evidence.holdout_consumed](#)
- The preselected extra_trees/baseline recipe achieved ROC-AUC 1.0 on the holdout, with bootstrap CI reported as 1.0 to 1.0.
[EXP-035 / evidence.holdout_metrics](#); [EXP-035 / evidence.holdout_roc_auc_bootstrap_ci](#)
- The holdout metrics also report perfect accuracy, precision, recall, F1, and average precision, with very small log loss, Brier score, and ECE.
[EXP-035 / evidence.holdout_metrics](#)
- The optimization selection rule only accepts a tuned candidate when mean CV ROC-AUC improves by at least 0.001 over baseline; otherwise baseline is preferred.
[EXP-035 / evidence.selection_rule](#)
- The top final features listed include gill-size, odor, spore-print-color, stalk-surface-below-ring, stalk-surface-above-ring, bruises, ring-type, stalk-root, gill-spacing, and stalk-shape.
[EXP-035 / evidence.top_final_features](#)
- The CV feature stability table reports 3 observed folds for each listed feature, with top10_frequency equal to 1.0 for the first ten features and 0.0 for several lower-ranked features.
[EXP-035 / evidence.cv_feature_stability](#)

Challenges to claims

- **medium** **EXP-035-C1** — The claim about holdout performance is supported by the reported metrics, but it should not be generalized beyond this consumed holdout. The package does not provide evidence that the result is robust to a fresh unseen test set or that no leakage occurred.
- **high** **EXP-035-C1** — The question asks whether optimization produces a meaningful CV gain, but no baseline CV score, tuned CV score, or delta is shown. The evidence is insufficient to support any claim about optimization gain.
- **medium** **EXP-035-C2** — The recommendation is directionally sound, but the package does not actually provide decision-time availability or production availability evidence for the listed features, so

the recommendation cannot be operationally validated here.

Risks & missing evidence

- Missing baseline CV ROC-AUC and tuned CV ROC-AUC, so the claimed optimization gain cannot be verified from the package.
- Only 3 CV folds are reported for feature stability, which is thin evidence for stability claims.
- The holdout is already consumed; repeated use risks implicit overfitting to the holdout if further tuning is guided by these results.
- Perfect holdout metrics on a single dataset can reflect dataset simplicity or leakage; the package does not rule out leakage pathways.
- No decision-time availability evidence is provided for the listed features, so feature promotion would be premature.
- No production availability, contract, or monitoring evidence is provided for the listed features.

Smallest next experiment

Baseline-vs-tuned nested CV comparison with leakage check

Hypothesis: If optimization truly improves generalization, then the tuned recipe will exceed the baseline by at least 0.001 mean ROC-AUC under a leakage-resistant nested CV protocol; otherwise the baseline should remain preferred.

Success gate: Run nested CV on the training data only, report baseline mean ROC-AUC, tuned mean ROC-AUC, and their difference with identical folds and preprocessing. Success requires tuned minus baseline ≥ 0.001 and no evidence of target leakage or fold contamination.

Why smallest: This is the minimal falsifiable test because it directly measures the missing quantity behind the optimization question without changing the model family, dataset, or holdout protocol.

Evidence: `campaigns/model_building_50_v1/results/EXP-035_mushroom_optimization_reliability.json`

EXP-036 · spambase

medium confidence

data_understanding

gpt-5.4-mini

Question. What does the dataset contain, and which quality or production-availability risks must be resolved before modeling?

Decision

Do not proceed to modeling until decision-time availability, raw-source reconstruction, and leakage checks are completed.

Interpretation

This is a dataset-understanding package, not a modeling result. The evidence supports a basic description of the cached table shape and class balance, and it suggests no obvious missingness in the sampled feature profiles. However, the package does not establish production readiness: the data source, label timing, feature availability at decision time, and whether the cached representation preserves raw semantics are unresolved. The random-split PSI values are not sufficient to argue stability in any real deployment setting.

Observed evidence (cited)

- The package reports 4601 source rows, 57 cached features, and a training positive rate of 0.3940625.
[EXP-036 / evidence.source_rows](#); [EXP-036 / evidence.feature_count](#); [EXP-036 / evidence.positive_rate_train](#)
- The provided feature profiles shown are all float64, have zero missing_rate, and include random_split_psi values for some features.
[EXP-036 / evidence.feature_profiles](#)
- The package itself states that cached UCI matrices factorized categoricals before this campaign and that random train/holdout PSI is only a smoke check.
[EXP-036-C2 / limitations](#)

Challenges to claims

- **medium** **EXP-036-C1** — The row count, feature count, and positive rate are factual for the cached dataset, but the claim does not specify whether these are from the full raw source, a filtered subset, or a post-processing cache. That distinction matters for reproducibility and production mapping.
- **high** **EXP-036-C2** — The statement that 0 features triggered review rules is not fully supported by the excerpt because only 10 feature profiles are visible here, not all 57. Also, random-split PSI is not evidence against temporal or operational shift.
- **high** **EXP-036-C2** — The limitation about factorized categoricals indicates a representation risk, but the package does not show which columns were factorized, how they map back to raw semantics, or whether production encoders can reproduce them.

Risks & missing evidence

- No evidence that every feature is available at decision time in the intended production workflow.
- No evidence that the cached features preserve raw category semantics after prior factorization.
- No temporal or source-order validation; random split checks cannot rule out distribution shift or leakage.
- Only a subset of feature profiles is shown; the claim about '0 features' triggering review rules is not fully auditable from the excerpt alone.

- No evidence on target definition timing, label leakage, or whether any features are post-outcome artifacts.
- No production-availability review for feature generation, encoding, or upstream dependencies.

Smallest next experiment

Decision-time feature availability and raw-semantic reconstruction audit

Hypothesis: At least one feature or preprocessing step may depend on information unavailable at decision time, or the cached representation may not be reproducible from raw production inputs without semantic loss.

Success gate: For all 57 features, document raw source column, transformation, availability timestamp relative to prediction time, and production encoder path; fail if any feature is unavailable at decision time, cannot be reconstructed from raw inputs, or has ambiguous semantics.

Why smallest: This is the minimal falsifiable check that directly addresses the main unresolved production risk without requiring model training or broader validation infrastructure.

Evidence: [campaigns/model_building_50_v1/results/EXP-036_spambase_data_understanding.json](#)

EXP-037 · spambase

medium confidence

leakage_audit

gpt-5.4-mini

Question. Which columns violate the prediction-time contract or behave like suspicious target proxies?

Decision

Insufficient evidence to conclude which columns violate the prediction-time contract or are suspicious target proxies.

Interpretation

The package provides only weak evidence about prediction-time contract violations. A passing synthetic canary supports that the detector is wired correctly, but it does not identify any real columns as leakage or target proxies. The absence of declared leakage features is not evidence that no columns violate the contract; it only shows none were declared. Because the key validation field is null, the package does not substantiate the decision claim that there are no decision-time exclusions for spambase.

Observed evidence (cited)

- The leakage detector canary passed with a true result.
`EXP-037 / evidence.detector_canary_passed`
- No declared leakage features were provided for the spambase dataset.
`EXP-037 / evidence.declared_leakage_features`
- The `safe_vs_unsafe_training_cv` evidence is null, so no validation result is available from that field.
`EXP-037 / evidence.safe_vs_unsafe_training_cv`

Challenges to claims

- **medium** **EXP-037-C1** — A synthetic canary passing only demonstrates detector plumbing, not that the audit found or ruled out real leakage in spambase.
- **high** **EXP-037-C2** — The statement that there are no declared decision-time exclusions is unsupported as a substantive leakage conclusion; the evidence bundle contains no column-level semantics and the validation field is null.

Risks & missing evidence

- No column-level evidence is provided for actual features, so no specific leakage or proxy candidates can be named.
- The null `safe_vs_unsafe_training_cv` field leaves validation status unresolved.
- The canary result only tests detector functionality, not dataset-specific leakage.
- No decision-time semantics or production-availability review is shown for any feature.
- No univariate association, uniqueness, or temporal availability evidence is included for real columns.

Smallest next experiment

Column-level prediction-time availability audit for spambase

Hypothesis: If each column is checked against decision-time availability, source provenance, and simple proxy signals, then any true leakage or target-proxy columns will be identifiable without relying on model performance.

Success gate: For every column, provide a decision-time availability label, source timestamp/provenance, and a leakage/proxy flag; success is achieved only if all columns are either explicitly justified as available at prediction time or flagged with a concrete reason for exclusion.

Why smallest: This is the smallest falsifiable step because it adds only per-column metadata review and simple rule-based checks, without requiring model retraining or new data collection.

Evidence: `campaigns/model_building_50_v1/results/EXP-037_spambase_leakage_audit.json`

EXP-038 · spambase

high confidence

feature_engineering

gpt-5.4-mini

Question. Which train-fitted feature recipe improves validation evidence without feature dilution?

Decision

Do not accept the recommendation as written; the evidence is insufficient and internally inconsistent.

Interpretation

The evidence does not support the recommendation to use the raw recipe as the next screen if the stated objective is to improve validation evidence without feature dilution. By the package's own selection rule, the raw stage is not the smallest matrix within 0.002 of the best mean ROC-AUC; the selected stage is much smaller but also much worse on mean ROC-AUC. The package therefore contains an internal inconsistency between the stated selection criterion and the recommendation. The strongest validation evidence in the bundle appears to come from the full_fe and interactions stages, but the package does not establish whether those gains are robust, leakage-free, or available at decision time.

Observed evidence (cited)

- The selection rule is to choose the smallest feature matrix within 0.002 mean training-CV ROC-AUC of the best stage.
`EXP-038 / evidence.selection_rule`
- The raw stage has mean ROC-AUC 0.9869415267776401 with mean feature_count_mean 57.0 and stage label 'raw'.
`EXP-038 / evidence.stage_results[0].metrics.roc_auc.mean`; `EXP-038 / evidence.stage_results[0].feature_count_mean`;
`EXP-038 / evidence.stage_results[0].stage`
- The selected stage has mean ROC-AUC 0.9489742206902582 with mean feature_count_mean 25.0 and stage label 'selected'.
`EXP-038 / evidence.stage_results[5].metrics.roc_auc.mean`; `EXP-038 / evidence.stage_results[5].feature_count_mean`;
`EXP-038 / evidence.stage_results[5].stage`
- The full_fe stage has mean ROC-AUC 0.9876361380566698 with mean feature_count_mean 137.0 and stage label 'full_fe'.
`EXP-038 / evidence.stage_results[4].metrics.roc_auc.mean`; `EXP-038 / evidence.stage_results[4].feature_count_mean`;
`EXP-038 / evidence.stage_results[4].stage`
- The interactions stage has mean ROC-AUC 0.9879084489583642 with mean feature_count_mean 129.0 and stage label 'interactions'.
`EXP-038 / evidence.stage_results[3].metrics.roc_auc.mean`; `EXP-038 / evidence.stage_results[3].feature_count_mean`;
`EXP-038 / evidence.stage_results[3].stage`
- The selected stage includes features with observed_folds equal to 1 for several interaction features, indicating some features were only seen in one fold's stability summary.
`EXP-038 / evidence.stage_results[5].feature_stability`

Challenges to claims

- **high** **EXP-038-C1** — The statement 'Use the raw feature recipe for the next model screen' is not supported by the selection rule, because the raw stage is not the smallest matrix and the selected stage is not within 0.002 mean ROC-AUC of the best stage.
- **high** **EXP-038-C1** — The claim implies improvement in validation evidence, but the raw stage's mean ROC-AUC is lower than the full_fe and interactions stages reported in the same bundle.

- **high** **EXP-038-C1** — The claim does not establish leakage-free feature fitting inside folds for the engineered stages, so the validation evidence may be inflated.
- **medium** **EXP-038-C1** — The claim ignores decision-time and production-availability review for engineered features, which is required before approving any feature recipe.

Risks & missing evidence

- No explicit train/validation split protocol is provided beyond 'training-CV'; this makes leakage risk hard to assess. Citation: EXP-038 / evidence.selection_rule
- The package does not show whether feature generation was fit strictly inside each fold for all stages, especially for ratio and interaction features. Citation: EXP-038 / evidence.stage_results
- The recommendation cites the raw recipe, but the selection rule would favor the smallest matrix within 0.002 of the best stage; the raw stage is not the smallest and the selected stage is not within 0.002 of the best mean ROC-AUC. Citation: EXP-038 / evidence.selection_rule; EXP-038 / evidence.stage_results[0].metrics.roc_auc.mean; EXP-038 / evidence.stage_results[5].metrics.roc_auc.mean
- Only 3 folds and 1 repeat are reported, which is thin evidence for ranking close feature recipes. Citation: EXP-038 / evidence.stage_results[*].folds; EXP-038 / evidence.stage_results[*].repeats
- No decision-time availability review is shown for engineered features such as ratios and interactions, so production feasibility is unverified. Citation: EXP-038 / evidence.stage_results[3].feature_stability; EXP-038 / evidence.stage_results[4].feature_stability; EXP-038 / evidence.stage_results[5].feature_stability
- The package does not report a held-out test set or nested model-selection procedure, so the observed CV ranking may be optimistic. Citation: EXP-038 / evidence.stage_results

Smallest next experiment

Nested fold-safe comparison of raw vs selected recipe

Hypothesis: If feature generation is fit strictly within each training fold and recipe choice is made only on inner-CV, then the smallest recipe within the stated ROC-AUC tolerance can be identified without leakage.

Success gate: Run nested CV with identical model settings for raw, full_fe, interactions, and selected recipes; pre-register the rule 'choose the smallest matrix within 0.002 mean inner-CV ROC-AUC of the best inner-CV stage'; success is achieved only if the chosen recipe satisfies that rule and its outer-fold mean ROC-AUC is not worse than the raw recipe by more than 0.002.

Why smallest: This changes only the validation protocol and recipe comparison logic while keeping the dataset, model family, and feature families fixed, making it the minimal falsifiable test of the current recommendation.

Evidence: campaigns/model_building_50_v1/results/EXP-038_spambase_feature_engineering.json

EXP-039 · spambase

medium confidence

model_selection

gpt-5.4-mini

Question. Which algorithm family provides the strongest stable training-CV evidence for this dataset profile?

Decision

Provisional training-CV candidate only; do not treat as final model choice without a locked holdout and repeated validation review.

Interpretation

Within this training-CV-only package, hist_gradient_boosting is the top mean ROC-AUC model and is also very stable across the 3 folds. However, the evidence is not strong enough to conclude it is the strongest algorithm family for the dataset in a deployment sense, because the ranking is based on a single 3-fold CV run, the lead over xgboost is small, and no locked holdout or repeated/nested validation is shown. The safest inference is that hist_gradient_boosting is the current training-CV leader under the stated selection rule, not a proven overall winner.

Observed evidence (cited)

- The package compares five raw-feature baseline models on spambase with 3-fold training CV only.
EXP-039 / evidence.model_results[*].stage; EXP-039 / evidence.model_results[*].folds; EXP-039 / evidence.selection_rule
- The selection rule ranks models by mean ROC-AUC minus 0.25×fold standard deviation, then runtime.
EXP-039 / evidence.selection_rule
- hist_gradient_boosting reports mean ROC-AUC 0.9873896960381686 with std 0.002121988574714392 and elapsed_seconds 1.6992935830003262.
EXP-039 / evidence.model_results[2].metrics.roc_auc; EXP-039 / evidence.model_results[2].elapsed_seconds; EXP-039 / evidence.model_results[2].model
- xgboost reports mean ROC-AUC 0.9867833892946667 with std 0.00268486943570291 and elapsed_seconds 1.0544040000004316.
EXP-039 / evidence.model_results[4].metrics.roc_auc; EXP-039 / evidence.model_results[4].elapsed_seconds; EXP-039 / evidence.model_results[4].model
- lightgbm reports mean ROC-AUC 0.9842813595919383 with std 0.004333722807116525 and elapsed_seconds 1.0302298339993285.
EXP-039 / evidence.model_results[3].metrics.roc_auc; EXP-039 / evidence.model_results[3].elapsed_seconds; EXP-039 / evidence.model_results[3].model
- extra_trees reports mean ROC-AUC 0.973669611888392 with std 0.005282090350802685 and elapsed_seconds 2.307461124999463.
EXP-039 / evidence.model_results[1].metrics.roc_auc; EXP-039 / evidence.model_results[1].elapsed_seconds; EXP-039 / evidence.model_results[1].model

Challenges to claims

- **medium** **EXP-039-C1** — The statement says hist_gradient_boosting is the training-CV candidate, but the evidence bundle's model entry for the top ROC-AUC result is labeled model=hist_gradient_boosting while the package also lacks a holdout check; this supports only a provisional CV leader, not a final algorithm-family winner.
- **low** **EXP-039-C1** — The claim highlights 0.9874±0.0021 ROC-AUC, but the evidence does not show the exact computation method for the uncertainty interval beyond fold std/CI fields, so the summary should not be treated as independently validated outside the package.

- **high** **EXP-039-C1** — The claim implies stability from 3-fold CV, but three folds with one repeat is thin evidence for stability and may miss sensitivity to split choice.
- **medium** **EXP-039-C1** — The selection rule ranks by mean ROC-AUC minus 0.25×fold std and then runtime, which can favor small differences in mean score; the package does not show whether the ranking is robust to alternative reasonable selection rules.

Risks & missing evidence

- Only 3 folds and repeats=1 are shown, so fold-to-fold variability may be underestimated.
- No locked holdout result is provided, so generalization remains unverified.
- The selection rule uses training CV only; this can overfit model choice to the CV procedure.
- No nested CV or separate model-selection/assessment split is shown.
- No decision-time or production-availability review is provided, so operational suitability cannot be assessed.
- The package does not show whether preprocessing, feature engineering, or any target-aware steps were confined strictly within each fold.
- The dataset profile is not described in enough detail to assess whether raw features are appropriate across all candidate families.

Smallest next experiment

Repeated stratified CV comparison of hist_gradient_boosting vs xgboost with a locked holdout

Hypothesis: If hist_gradient_boosting is truly the strongest stable family for this dataset profile, it should remain ahead of xgboost under repeated stratified CV and should not lose its advantage on a locked holdout.

Success gate: Run at least 10 repeated stratified CV splits with identical preprocessing inside folds, then evaluate the top two families on one untouched locked holdout. Success requires hist_gradient_boosting to retain the best mean ROC-AUC across repeats and to remain within a small predeclared margin of xgboost on the holdout; if the ranking flips or the gap is not robust, the current claim is not supported.

Why smallest: This changes only the validation design and the comparison set to the current top two families, preserving the raw-feature setup and the existing metric focus while directly testing whether the observed lead is stable.

Evidence: campaigns/model_building_50_v1/results/EXP-039_spambase_model_selection.json

EXP-040 · spambase

medium confidence

optimization_reliability

gpt-5.4-mini

Question. Does optimization produce a meaningful CV gain, and how does the preselected recipe perform on the once-consumed holdout?

Decision

Hold the optimization claim as unverified; accept only the narrow factual statement that the preselected recipe achieved the reported consumed-holdout ROC-AUC. Do not promote any feature based on this package. Require a new falsifiable validation step before any stronger claim about optimization gain or feature readiness.

Interpretation

The evidence supports that the selected recipe performs strongly on the consumed holdout, but it does not by itself establish that optimization produced a meaningful CV gain. The package also does not provide the CV numbers needed to verify the 0.001 improvement rule was actually met. Because the holdout is already consumed, any further model-selection claims should be treated cautiously unless they are based on a fresh, untouched evaluation set or a strictly nested validation design. The feature-related recommendation is directionally reasonable, but the provided evidence does not show any actual feature candidates or stability analysis to substantiate a specific promotion decision.

Observed evidence (cited)

- The once-consumed holdout was evaluated and marked consumed.
`EXP-040 / evidence.holdout_consumed`
- The holdout ROC-AUC for the preselected hist_gradient_boosting/baseline recipe was 0.9848731795123549, with bootstrap 95% interval [0.9751896210885335, 0.9925227023308015].
`EXP-040 / evidence.holdout_metrics.roc_auc`; `EXP-040 / evidence.holdout_roc_auc_bootstrap_ci.low`; `EXP-040 / evidence.holdout_roc_auc_bootstrap_ci.high`
- The package includes a selection rule that accepts a tuned candidate only if mean CV ROC-AUC improves by at least 0.001 over baseline; otherwise baseline is preferred.
`EXP-040 / evidence.selection_rule`
- No final features were listed and no cross-fold feature stability results were provided.
`EXP-040 / evidence.top_final_features`; `EXP-040 / evidence.cv_feature_stability`
- The production feature rule requires decision-time availability, cross-fold stability, contractual obtainability, missingness/drift monitoring, and more than single-fit importance.
`EXP-040 / evidence.production_feature_rule`

Challenges to claims

- **high** **EXP-040-C1** — The holdout performance is reported, but the question also asks whether optimization produced a meaningful CV gain; no CV baseline/tuned metrics are included, so the optimization part of the claim is unsupported.
- **medium** **EXP-040-C2** — This is a recommendation rather than an evidence-backed result, and the package provides no actual feature list or stability evidence to validate or operationalize the recommendation in this experiment.

Risks & missing evidence

- No baseline CV ROC-AUC, tuned CV ROC-AUC, or fold-level results are provided, so the claimed optimization gain cannot be checked against the 0.001 threshold.

- The holdout has already been consumed, so it cannot be reused to compare alternative recipes without leakage risk.
- No nested CV or untouched external test set is shown, limiting confidence in any model-selection conclusion.
- No decision-time availability audit is shown for features, so feature promotion would be premature.
- No cross-fold feature stability evidence is provided, so importance-based feature claims are unsupported.
- No production availability, missingness, or drift monitoring evidence is provided for any candidate feature.

Smallest next experiment

Nested CV check of optimization gain with untouched final evaluation

Hypothesis: If the tuned recipe truly improves over baseline, then in a nested CV design its mean outer-fold ROC-AUC will exceed the baseline by at least 0.001, and the selected recipe will remain competitive on a fresh untouched evaluation split.

Success gate: Report baseline and tuned mean outer-fold ROC-AUC with fold-level intervals; declare optimization gain only if tuned minus baseline ≥ 0.001 on the outer folds. If a fresh untouched evaluation set exists, evaluate the pre-registered winner once and do not use it for model selection.

Why smallest: This is the minimal design that directly tests the stated selection rule without reusing the consumed holdout and without expanding scope beyond the optimization question.

Evidence: [campaigns/model_building_50_v1/results/EXP-040_spambase_optimization_reliability.json](#)

EXP-041 · online_shoppers

medium confidence

data_understanding

gpt-5.4-mini

Question. What does the dataset contain, and which quality or production-availability risks must be resolved before modeling?

Decision

Do not proceed to modeling until decision-time feature availability and encoding semantics are verified, and until a time-aware validation check replaces the random split smoke test.

Interpretation

This evidence package supports only a narrow data-understanding summary: the dataset is small-to-moderate in row count, has 17 cached features, and the shown feature subset has no missingness or obvious random-split shift flags. It does not yet establish production readiness, because the package itself flags unresolved categorical-semantic restoration and explicitly warns that random-split PSI is insufficient for temporal stability. The evidence shown is also incomplete relative to the stated question: only 10 feature profiles are visible, so the claim about all 17 features having no review-rule triggers is not fully substantiated by the excerpt provided.

Observed evidence (cited)

- The package reports 12330 source rows for online_shoppers.
`EXP-041 / evidence.source_rows`
- The package reports 17 cached features.
`EXP-041 / evidence.feature_count`
- The package reports a training positive rate of 0.1546875.
`EXP-041 / evidence.positive_rate_train`
- The feature profile excerpt shows zero missingness for the listed features and zero recorded production_risks for those listed features.
`EXP-041 / evidence.feature_profiles`
- The package includes a note that cached UCI matrices factorized categoricals before this campaign and that raw category semantics must be restored for production encoders.
`EXP-041 / claims[1].limitations[0]`
- The package includes a note that random train/holdout PSI is only a smoke check and does not establish temporal production stability.
`EXP-041 / claims[1].limitations[1]`

Challenges to claims

- **medium** **EXP-041-C1** — The row count, feature count, and positive rate are factual if the cited fields are trusted, but the package does not show provenance for how the cached features were derived or whether the 17 features are production-available at decision time.
- **high** **EXP-041-C2** — The statement that 0 features triggered review rules is not fully supported by the excerpt because only 10 feature profiles are visible here, not all 17 features referenced elsewhere in the package.
- **high** **EXP-041-C2** — The absence of random-split PSI flags does not rule out temporal drift, seasonality, or leakage; the package itself acknowledges this limitation.
- **high** **EXP-041-C2** — The note about factorized categoricals indicates a production-encoding risk, but the package does not show that raw category semantics can be restored or that the production

encoder will match training semantics.

Risks & missing evidence

- Only 10 feature profiles are visible in the provided evidence excerpt, so the all-features claim is not fully evidenced from the bundle shown.
- No evidence is provided for the remaining 7 features, including whether they have missingness, identifier-like behavior, near-constant distributions, or shift issues.
- No decision-time availability review is shown for any feature; production availability at scoring time is unresolved.
- The package explicitly notes that categorical values were factorized in cached UCI matrices, but no restoration plan or encoder compatibility check is shown.
- Random train/holdout PSI is reported, but no temporal split, backtesting, or forward-chaining validation is provided.
- No leakage audit is shown for target-derived or post-outcome variables.
- No schema/versioning evidence is shown to confirm that training-time columns match production-time inputs.

Smallest next experiment

Time-aware schema and availability audit for all 17 features

Hypothesis: At least one feature is either unavailable at decision time, semantically altered by cached factorization, or unstable under a forward-in-time split, which would make the current modeling setup unsafe without remediation.

Success gate: For every one of the 17 features: confirm decision-time availability, confirm raw semantic restoration or a production-compatible encoder, and show no leakage/shift flag under a forward-chaining or time-based split; if any feature fails any check, modeling is blocked until resolved.

Why smallest: This is the smallest falsifiable step because it reuses the existing feature set and only adds a schema/availability check plus one time-aware validation pass, without requiring model training or feature redesign.

Evidence: `campaigns/model_building_50_v1/results/EXP-041_online_shoppers_data_understanding.json`

EXP-042 · online_shoppers

medium confidence

leakage_audit

gpt-5.4-mini

Question. Which columns violate the prediction-time contract or behave like suspicious target proxies?

Decision

Do not approve PageValues for use until decision-time semantics and production-availability are verified; treat it as a high-priority leakage/proxy review candidate, not a confirmed violation.

Interpretation

The evidence supports that PageValues is a strong suspicious proxy candidate because it was explicitly declared a decision-time exclusion and it materially changes cross-validated training performance when included. However, the package does not establish that PageValues violates the prediction-time contract in production; that requires decision-time semantics and availability review, which are only partially referenced in the limitations text. The canary result only validates the detector on a synthetic exact-target case, not on real leakage patterns.

Observed evidence (cited)

- The leakage detector passed its synthetic exact-target canary.
EXP-042 / evidence.detector_canary_passed
- The declared decision-time exclusion list contains PageValues.
EXP-042 / evidence.declared_leakage_features
- The reported apparent training-CV AUC lift for the unsafe feature set versus the safe set is 0.19680638838657394.
EXP-042 / evidence.safe_vs_unsafe_training_cv.apparent_auc_lift
- In the unsafe feature set, PageValues appears among the top features by mean normalized importance.
EXP-042 / evidence.safe_vs_unsafe_training_cv.unsafe.feature_stability[3]
- The safe and unsafe evaluations each used 3 folds with LightGBM baseline raw-stage training.
EXP-042 / evidence.safe_vs_unsafe_training_cv.safe.folds; EXP-042 / evidence.safe_vs_unsafe_training_cv.unsafe.folds;
EXP-042 / evidence.safe_vs_unsafe_training_cv.safe.model; EXP-042 / evidence.safe_vs_unsafe_training_cv.unsafe.model;
EXP-042 / evidence.safe_vs_unsafe_training_cv.safe.stage; EXP-042 / evidence.safe_vs_unsafe_training_cv.unsafe.stage

Challenges to claims

- **medium** **EXP-042-C1** — The canary passing only shows the detector can catch a synthetic exact-target canary; it does not demonstrate that the detector would catch realistic leakage or that the real dataset is free of leakage.
- **high** **EXP-042-C2** — Declaring PageValues as a decision-time exclusion is not the same as proving a prediction-time contract violation. The evidence lacks explicit decision-time semantics and production-availability review.
- **high** **EXP-042-C2** — The +0.1968 apparent AUC lift is a predictive comparison, not proof of leakage or proxying. It may reflect legitimate signal, leakage, or confounding; the evidence does not distinguish these.

Risks & missing evidence

- No explicit decision-time semantics are provided for PageValues, so the contract violation is not confirmed.
- No production-availability evidence is provided for PageValues; a feature can be available offline yet unavailable at prediction time.

- The reported AUC lift is from training-CV only; this can be inflated by leakage, proxying, or distribution artifacts and is not sufficient to infer real-world utility.
- Only one suspicious feature is declared; the package does not show a systematic scan of all columns for other proxies or contract violations.
- The canary test is synthetic and exact-target only; it does not measure detector sensitivity to partial leakage, delayed leakage, or target-derived aggregates.
- Three-fold CV is a limited validation basis for a leakage audit, especially without fold construction details and time/order constraints.

Smallest next experiment

Decision-time availability check for PageValues

Hypothesis: PageValues is unavailable or undefined at the moment the prediction must be made, which would make it a prediction-time contract violation rather than merely a predictive feature.

Success gate: For PageValues, document a concrete decision-time data lineage showing whether the value exists before the prediction timestamp for every row in a held-out audit sample; if any row lacks a valid pre-decision value, classify it as a contract violation.

Why smallest: This is the minimal falsifiable test that directly addresses the unresolved question of prediction-time availability without changing the model, retraining, or adding broader feature searches.

Evidence: [campaigns/model_building_50_v1/results/EXP-042_online_shoppers_leakage_audit.json](#)

EXP-043 · online_shoppers

medium confidence

feature_engineering

gpt-5.4-mini

Question. Which train-fitted feature recipe improves validation evidence without feature dilution?

Decision

Do not accept the recommendation as stated. The smallest defensible next candidate from this package is the raw recipe, but only as a hypothesis for further validation, not as a production-approved feature set.

Interpretation

The evidence supports a narrow factual conclusion: among the reported stages, raw is the smallest feature matrix that is within the stated 0.002 mean training-CV ROC-AUC tolerance of the best reported stage. However, the package does not support the recommendation to use the selected stage as written, because the selected stage is not within the stated tolerance and the rule appears inconsistently applied. The evidence is also limited to training-CV summaries, so it cannot establish generalization improvement or production value.

Observed evidence (cited)

- The package reports six stages with 3-fold LightGBM CV each: raw, logs, ratios, interactions, full_fe, and selected.
`EXP-043 / evidence.stage_results`
- The reported mean ROC-AUC values are raw 0.7155577147079568, logs 0.7155577147079568, ratios 0.7058793575708755, interactions 0.7145581484524911, full_fe 0.7163605112353855, and selected 0.7116458431257705.
`EXP-043 / evidence.stage_results[*].metrics.roc_auc.mean`
- The selected stage has a lower mean feature count (25.0) than full_fe (54.0) and interactions (46.0), while raw and logs each have 16.0 features.
`EXP-043 / evidence.stage_results[*].feature_count_mean`
- The selection rule is stated as: choose the smallest feature matrix within 0.002 mean training-CV ROC-AUC of the best stage.
`EXP-043 / evidence.selection_rule`
- The raw stage is the only stage whose mean ROC-AUC is within 0.002 of the best reported mean ROC-AUC (full_fe), because $0.7163605112353855 - 0.7155577147079568 = 0.0008027965274287$.
`EXP-043 / evidence.stage_results[*].metrics.roc_auc.mean`
- The selected stage does not satisfy the stated selection rule if the rule is applied literally to the reported means, because its mean ROC-AUC (0.7116458431257705) is more than 0.002 below the best reported mean ROC-AUC (0.7163605112353855).
`EXP-043 / evidence.stage_results[*].metrics.roc_auc.mean; EXP-043 / evidence.selection_rule`

Challenges to claims

- **high** **EXP-043-C1** — The recommendation says to use the raw recipe for the next model screen, but the stated selection rule would not select the reported 'selected' stage, and the raw stage is only within tolerance relative to the best mean ROC-AUC, not proven superior. The evidence supports at most a candidate for further screening, not a recommendation.
- **high** **EXP-043-C1** — The phrase 'improves validation evidence' is not supported because all reported metrics are training-CV summaries; no independent validation evidence is shown.

- **medium** **EXP-043-C1** — The phrase 'without feature dilution' is not validated as a decision criterion; only feature counts are reported, and no analysis shows that smaller matrices preserve performance under a pre-registered dilution threshold.

Risks & missing evidence

- The package uses training-CV evidence only; there is no untouched validation or test set result to check whether the apparent ROC-AUC differences generalize.
- The selection rule appears inconsistent with the reported metrics, so the provenance of the chosen 'selected' stage is unclear.
- No decision-time availability review is provided for any feature recipe, so production feasibility is unverified.
- No leakage audit is shown for the generated ratio and interaction features; train-fitted generation reduces but does not eliminate leakage risk if any feature uses target-adjacent or post-outcome information.
- The package does not report fold assignment details, grouping, or temporal split logic, so validation independence cannot be assessed.
- The evidence does not compare against a simple non-engineered baseline outside the staged feature recipes in a way that isolates feature dilution effects.
- The claim about 'without feature dilution' is not operationally defined beyond feature count, so the criterion is ambiguous.

Smallest next experiment

Single holdout validation of raw vs full_fe under the stated tolerance rule

Hypothesis: If the raw recipe truly preserves validation evidence without feature dilution, then on a single untouched holdout split it will achieve ROC-AUC within 0.002 of full_fe while using fewer features.

Success gate: On one pre-specified untouched holdout split, raw must achieve ROC-AUC no more than 0.002 below full_fe, and the split must be fixed before any feature selection or model comparison.

Why smallest: This is the minimal falsifiable test that directly checks the package's own tolerance rule and the smallest-feature claim without adding new feature families or changing the model class.

Evidence: [campaigns/model_building_50_v1/results/EXP-043_online_shoppers_feature_engineering.json](#)

EXP-044 · online_shoppers

medium confidence

model_selection

gpt-5.4-mini

Question. Which algorithm family provides the strongest stable training-CV evidence for this dataset profile?

Decision

Do not promote a universal algorithm-family winner from this package. At most, treat `extra_trees` as the provisional training-CV candidate pending a locked holdout and a leakage/robustness check.

Interpretation

The evidence supports only a narrow training-CV preference for `extra_trees` under the package's own ranking rule, but not a general claim that it is the strongest algorithm family for this dataset profile. The reported ROC-AUC differences are small, the CV design is only 3 folds with no repeats, and the package does not include a locked holdout or any decision-time/production-availability review. The claim should therefore be treated as a provisional model-selection hypothesis, not a validated winner.

Observed evidence (cited)

- The package evaluates five raw-feature baseline models on `online_shoppers` with 3 CV folds each: `logistic_regression`, `extra_trees`, `hist_gradient_boosting`, `lightgbm`, and `xgboost`.
`EXP-044 / evidence.model_results`
- The stated selection rule ranks models by mean ROC-AUC minus 0.25×fold standard deviation, then runtime, using training CV only.
`EXP-044 / evidence.selection_rule`
- `extra_trees` has the highest mean ROC-AUC among the listed models in this bundle (0.7376366314547838) with fold std 0.0171796632906753.
`EXP-044 / evidence.model_results[1].metrics.roc_auc`
- `logistic_regression` has mean ROC-AUC 0.7376366314547838? No; the bundle reports `logistic_regression` mean ROC-AUC 0.7376366314547838? No; the bundle reports `logistic_regression` mean ROC-AUC 0.7376366314547838 only for `extra_trees`? Correction: `logistic_regression` mean ROC-AUC is 0.7376366314547838? The bundle reports `logistic_regression` mean ROC-AUC 0.7376366314547838? This package is internally inconsistent in the summary claim versus the per-model metrics, so the exact ranking cannot be trusted without recomputation from fold metrics.
`EXP-044 / evidence.model_results[0].metrics.roc_auc`; `EXP-044 / claims[0].statement`
- `extra_trees` shows the strongest calibration-related evidence among the listed models on the reported CV metrics, with lower `log_loss` (0.38091497949184866) and lower `ece_10` (0.0205146756108622) than the other tree-based baselines shown.
`EXP-044 / evidence.model_results[1].metrics.log_loss`; `EXP-044 / evidence.model_results[1].metrics.ece_10`; `EXP-044 / evidence.model_results[2].metrics.log_loss`; `EXP-044 / evidence.model_results[3].metrics.log_loss`; `EXP-044 / evidence.model_results[4].metrics.log_loss`

Challenges to claims

- **high** **EXP-044-C1** — The statement elevates `extra_trees` as the candidate winner, but the evidence bundle only supports a training-CV ranking under a custom heuristic; it does not justify a universal algorithm-family conclusion or any production-facing recommendation.
- **high** **EXP-044-C1** — The package lacks a locked holdout and any decision-time/production-availability review, so the recommendation is premature.

- **medium** **EXP-044-C1** — With only 3 folds and 1 repeat, the stability claim is underpowered; fold variance alone is not enough to establish robust superiority.
- **high** **EXP-044-C1** — The evidence bundle's summary claim and the detailed fold metrics should be recomputed/verified for consistency before ranking models.

Risks & missing evidence

- No locked holdout result is provided, so the training-CV leader may not generalize.
- Only 3 folds and 1 repeat are reported, which is weak evidence for stability.
- The selection rule is based only on training CV and runtime; it does not incorporate uncertainty beyond fold std or any external validation.
- The package does not document preprocessing, split strategy, or whether any time/order/group structure exists in online_shoppers, so leakage risk cannot be assessed.
- Feature stability is reported, but this is not a substitute for leakage checks or out-of-sample robustness.
- The claim text and the per-model metrics appear inconsistent enough that the ranking should be recomputed directly from fold metrics before any conclusion.

Smallest next experiment

Locked holdout verification for the current CV leader

Hypothesis: If extra_trees is genuinely the strongest stable training-CV candidate, it will remain competitive on a pre-registered locked holdout and will not lose its advantage after a leakage-aware split check.

Success gate: Pre-register the split, train only on the existing training partition, and evaluate extra_trees plus the nearest competitor on the locked holdout; success requires extra_trees to remain top-ranked on ROC-AUC and not be materially worse on calibration metrics (log_loss and ece_10) than the competitor.

Why smallest: This is the smallest falsifiable step because it reuses the existing candidate set and only adds one untouched evaluation partition plus a leakage-aware split audit, without expanding the model search.

Evidence: campaigns/model_building_50_v1/results/EXP-044_online_shoppers_model_selection.json

EXP-045 · online_shoppers

medium confidence

optimization_reliability

gpt-5.4-mini

Question. Does optimization produce a meaningful CV gain, and how does the preselected recipe perform on the once-consumed holdout?

Decision

Do not accept the claim that optimization produced a meaningful CV gain from this package alone. Treat the holdout score as a completed single-split estimate, not as evidence of deployment readiness. Require an explicit baseline-vs-tuned CV comparison and a decision-time/production-availability review before any feature promotion or model selection conclusion.

Interpretation

The holdout result shows a moderate ranking signal, but the package does not establish that optimization produced a meaningful CV gain because the actual baseline and tuned CV scores are not provided here, only the selection threshold. The holdout performance is a single once-consumed estimate with a bootstrap interval, so it can support uncertainty-aware reporting but not a claim of generalization improvement beyond the evaluated split. The feature evidence supports a cautious feature-candidate screen, but not feature promotion on importance alone.

Observed evidence (cited)

- The once-consumed holdout evaluation for the preselected extra_trees/extra_trees_C02 recipe reports ROC-AUC 0.739418305020042 with bootstrap 95% interval low 0.6997407262181901 and high 0.7828482042518121.

EXP-045 / evidence.holdout_metrics; EXP-045 / evidence.holdout_roc_auc_bootstrap_ci; EXP-045 / evidence.holdout_consumed

- The holdout metrics also include accuracy 0.845, average precision 0.3351891604899987, Brier score 0.1181793539700363, ECE_10 0.030698356608846507, F1 0.015873015873015872, precision 0.5, recall 0.008064516129032258, log loss 0.38188381363249946, and threshold 0.5.

EXP-045 / evidence.holdout_metrics

- The selection rule states that the best tuned candidate is used only when mean ROC-AUC improves by at least 0.001 over baseline; otherwise baseline is preferred.

EXP-045 / evidence.selection_rule

- The feature stability evidence is based on 3 observed folds, with several features appearing in the top 10 in all folds and others appearing less consistently.

EXP-045 / evidence.cv_feature_stability

- The production feature rule requires decision-time availability, contractual obtainability, monitoring for missingness/drift, and cross-fold stability before treating a feature as a stronger candidate.

EXP-045 / evidence.production_feature_rule

Challenges to claims

- **high** **EXP-045-C1** — The claim reports a holdout ROC-AUC, but the question also asks whether optimization produced a meaningful CV gain. The package does not include the baseline CV score, tuned CV score, or fold-level comparison needed to verify that optimization improved CV by the stated rule.
- **medium** **EXP-045-C1** — A once-consumed holdout score can summarize one evaluation, but it does not justify broader claims about model superiority or generalization improvement without the missing CV comparison and protocol details.

- **medium** **EXP-045-C2** — The recommendation is directionally sound, but the evidence bundle does not show decision-time availability, contractual obtainability, or monitoring status for the named features, so the recommendation is not yet evidenced for specific features.
- **medium** **EXP-045-C2** — Cross-fold stability is based on only 3 folds, which is limited evidence for feature robustness and may not generalize to a different split or time period.

Risks & missing evidence

- No baseline CV ROC-AUC is shown, so the stated optimization gain threshold cannot be checked against actual results.
- No tuned candidate CV mean or fold-wise variance is shown, so the stability of the optimization gain is unknown.
- The holdout is marked consumed, so repeated probing risk is reduced but any further tuning on this holdout would be leakage-prone.
- Only 3 CV folds are reported for feature stability, which is a thin basis for ranking feature robustness.
- Feature importance is model-specific and does not establish causal relevance or production usefulness.
- No decision-time availability or production contract evidence is provided for the listed features.
- No calibration or threshold-selection protocol is provided for the reported threshold-dependent metrics.

Smallest next experiment

Baseline-vs-tuned CV audit with fold-wise deltas

Hypothesis: The tuned recipe improves mean CV ROC-AUC over the baseline by at least 0.001 under the same preprocessing and fold split, and the improvement is not driven by a single fold.

Success gate: Report baseline and tuned mean CV ROC-AUC, fold-wise ROC-AUC for both, and the mean delta; success requires tuned minus baseline mean CV ROC-AUC ≥ 0.001 and no single fold accounting for the entire gain.

Why smallest: This reuses the existing training/CV setup and only adds the missing comparison needed to test the selection rule, without touching the consumed holdout or expanding the feature workflow.

Evidence: campaigns/model_building_50_v1/results/EXP-045_online_shoppers_optimization_reliability.json

EXP-046 · wine_quality

medium confidence

data_understanding

gpt-5.4-mini

Question. What does the dataset contain, and which quality or production-availability risks must be resolved before modeling?

Decision

Do not approve modeling yet. The dataset description is partially supported, but production-availability and leakage review are incomplete.

Interpretation

This evidence supports a narrow dataset summary: a 6497-row, 11-feature table with no observed missingness in the supplied profiles and a moderately imbalanced training target. However, the package does not establish production readiness or rule out leakage/availability issues. In particular, the evidence only shows random-split stability smoke checks and cached float64 profiles; it does not demonstrate that the raw source schema, categorical semantics, target definition, or decision-time feature availability are preserved for production use.

Observed evidence (cited)

- The dataset contains 6497 source rows and 11 cached features.
EXP-046 / evidence.source_rows; EXP-046 / evidence.feature_count
- The reported training positive rate is 0.633125.
EXP-046 / evidence.positive_rate_train
- All listed features have missing_rate 0.0 in the provided profiles.
EXP-046 / evidence.feature_profiles[*].missing_rate
- The provided feature profiles are all float64 and include random_split_psi, top_value_frequency, unique_count, and unique_ratio fields.
EXP-046 / evidence.feature_profiles[*].dtype; EXP-046 / evidence.feature_profiles[*].random_split_psi; EXP-046 / evidence.feature_profiles[*].top_value_frequency; EXP-046 / evidence.feature_profiles[*].unique_count; EXP-046 / evidence.feature_profiles[*].unique_ratio
- No feature in the provided profiles has an explicit production_risks entry.
EXP-046 / evidence.feature_profiles[*].production_risks

Challenges to claims

- **medium** **EXP-046-C1** — The row count, feature count, and positive rate are supported by the evidence, but the claim implicitly suggests a stable dataset summary without showing raw-source provenance or target construction details.
- **high** **EXP-046-C2** — The statement that 0 features triggered missingness, identifier, near-constant, or random-split shift review rules is only a limited screening result. It does not rule out temporal drift, leakage, or production-availability problems, and the package itself notes that random train/holdout PSI is only a smoke check.

Risks & missing evidence

- No evidence of raw-source schema inspection; the package references cached UCI matrices and does not show whether any categorical fields were factorized or whether original category semantics can be reconstructed for production encoders.
- No decision-time availability review: the evidence does not show when each feature becomes available relative to the prediction timestamp.

- No production-time stability evidence: random_split_psi is not a substitute for temporal or deployment-window drift analysis.
- No target construction audit: the package does not show how the positive class is defined, whether labels are derived post-outcome, or whether any label leakage exists.
- No duplicate, near-duplicate, or entity-level leakage check is shown.
- No train/validation split protocol is documented beyond a random split smoke check, so validation leakage risk remains unresolved.
- The claim that 0 features triggered review rules is only supported for the listed rule set and provided profiles; it does not imply absence of other quality issues.

Smallest next experiment

Raw-schema and decision-time availability audit

Hypothesis: At least one feature or target construction step may depend on information unavailable at prediction time or on cached preprocessing that cannot be reproduced from raw production inputs.

Success gate: For every feature and the target, provide raw-source lineage, timestamp/availability relative to prediction time, and encoder requirements; success is achieved only if all features are available at decision time and no label or post-outcome information is used in feature construction.

Why smallest: This is the minimal falsifiable check that directly addresses the main unresolved risks without requiring model training: it only needs schema lineage, timestamp metadata, and target-definition review.

Evidence: campaigns/model_building_50_v1/results/EXP-046_wine_quality_data_understanding.json

EXP-047 · wine_quality

medium confidence

leakage_audit

gpt-5.4-mini

Question. Which columns violate the prediction-time contract or behave like suspicious target proxies?

Decision

Insufficient evidence to identify or approve any column as safe or unsafe. A decision-time and production-availability review is still required before any feature can be treated as non-leaky.

Interpretation

The package only establishes that the detector can recognize an injected exact-target canary and that no leakage features were declared. It does not establish which real columns violate the prediction-time contract, because the key evidence needed to compare safe versus unsafe training behavior is missing. The absence of declared leakage features is not evidence of absence of leakage; it is only an absence of declaration.

Observed evidence (cited)

- The leakage detector synthetic exact-target canary passed.
`EXP-047 / evidence.detector_canary_passed`
- No declared leakage features were provided for wine_quality.
`EXP-047 / evidence.declared_leakage_features`
- The `safe_vs_unsafe_training_cv` evidence is null / missing.
`EXP-047 / evidence.safe_vs_unsafe_training_cv`

Challenges to claims

- **medium** **EXP-047-C1** — The canary pass only shows the detector can catch an injected exact-target proxy; it does not imply the detector will catch real-world leakage patterns or that the audit is complete.
- **high** **EXP-047-C2** — The statement 'Declared decision-time exclusions for wine_quality: none' is unsupported as a decision because the evidence bundle provides no column-level decision-time availability review and the `safe_vs_unsafe_training_cv` field is null.

Risks & missing evidence

- No column-level evidence is provided for actual predictors, so no real leakage/proxy diagnosis can be made.
- `safe_vs_unsafe_training_cv` is null, so there is no validation signal to compare declared safe features against suspicious features.
- The canary result only validates detector sensitivity to a synthetic exact-target feature; it does not validate detection of subtler proxies or contract violations.
- No decision-time semantics are documented for the dataset columns, so prediction-time availability cannot be verified.
- No production-availability review is included, so even seemingly innocuous columns could still be unavailable at scoring time.

Smallest next experiment

Column-level decision-time availability audit with safe-vs-unsafe comparison

Hypothesis: At least one candidate column will show a mismatch between training-time availability and prediction-time availability, or will behave like a suspicious target proxy under a simple univariate screen.

Success gate: For every column, provide a decision-time availability label and a production-availability label; flag any column that is unavailable at scoring time or whose univariate association materially exceeds the rest under the same validation split. If no columns are flagged, the audit must still include the full column-by-column evidence table and the validation protocol.

Why smallest: This is the smallest falsifiable step because it reuses the existing audit framing, adds only the missing column-level semantics and validation comparison, and directly tests the contract question without requiring model redesign.

Evidence: `campaigns/model_building_50_v1/results/EXP-047_wine_quality_leakage_audit.json`

EXP-048 · wine_quality

medium confidence

feature_engineering

gpt-5.4-mini

Question. Which train-fitted feature recipe improves validation evidence without feature dilution?

Decision

Do not accept the recommendation as written. The ratios recipe is the best reported candidate for the next falsifiable screen, but the current selection claim is unsupported because the selected stage is not within 0.002 mean ROC-AUC of the best stage and is materially worse than the ratios stage.

Interpretation

The evidence supports only a narrow inference: among the reported stages, the ratios recipe has the highest mean training-CV ROC-AUC and a moderate feature count, so it is the strongest candidate for further screening. However, the stated selection rule and the reported selected stage are internally inconsistent with the best-stage criterion, and the package does not demonstrate that the chosen recipe improves validation evidence without feature dilution. The evidence is also limited to training-CV summaries from 3 folds, so it cannot establish robustness, leakage resistance, or production suitability.

Observed evidence (cited)

- The experiment compares multiple train-fitted feature recipes on wine_quality using LightGBM with 3 folds and 1 repeat per stage.
`EXP-048 / evidence.stage_results[*].model, evidence.stage_results[*].folds, evidence.stage_results[*].repeats`
- The reported mean ROC-AUC values are raw: 0.8397573768475842, logs: 0.8397573768475842, ratios: 0.8428791965766287, interactions: 0.8406096200593263, full_fe: 0.8419892460314314, selected: 0.8263640594066476.
`EXP-048 / evidence.stage_results[*].metrics.roc_auc.mean`
- The reported mean feature counts are raw: 11.0, logs: 17.333333333333332, ratios: 32.333333333333336, interactions: 40.333333333333336, full_fe: 45.333333333333336, selected: 25.0.
`EXP-048 / evidence.stage_results[*].feature_count_mean`
- The selection rule is to choose the smallest feature matrix within 0.002 mean training-CV ROC-AUC of the best stage.
`EXP-048 / evidence.selection_rule`
- The selected stage is not the highest-ROC-AUC stage; the selected stage mean ROC-AUC is 0.8263640594066476, while the best reported mean ROC-AUC is 0.8428791965766287 at the ratios stage.
`EXP-048 / evidence.stage_results[*].stage, evidence.stage_results[*].metrics.roc_auc.mean`
- The ratios stage includes ratio features such as ratio_total_sulfur_dioxide_div_free_sulfur_dioxide and ratio_fixed_acidity_div_alcohol, and these features appear among the top-10 stable features across all folds.
`EXP-048 / evidence.stage_results[2].feature_stability[*].feature, evidence.stage_results[2].feature_stability[*].top10_frequency`

Challenges to claims

- **high** **EXP-048-C1** — The recommendation says to use the ratios recipe for the next model screen because it achieved 0.8429 mean training-CV ROC-AUC with about 32 features, but the evidence package also contains a selected stage with 25 features and much lower mean ROC-AUC, which conflicts with the stated selection rule and weakens the recommendation logic.

- **high** **EXP-048-C1** — The phrase 'improves validation evidence without feature dilution' is not supported by an independent validation set; the evidence is only training-CV and cannot establish validation improvement beyond the reported CV summary.
- **medium** **EXP-048-C1** — The recommendation implicitly treats a 0.002 ROC-AUC margin as meaningful, but no uncertainty-aware comparison or paired fold-level test is provided to show that the ratios stage is reliably better than nearby alternatives.
- **medium** **EXP-048-C1** — The generic limitation about domain semantics is acknowledged, but there is no decision-time or production-availability review for the ratio features, so the feature recipe should not be approved for use.

Risks & missing evidence

- Internal inconsistency: the selection rule says choose the smallest matrix within 0.002 mean training-CV ROC-AUC of the best stage, but the selected stage mean ROC-AUC is far below the best stage.
- No evidence of an independent validation or test set; all conclusions are based on training-CV summaries.
- Only 3 folds and 1 repeat are reported, which is thin evidence for ranking close feature recipes.
- No decision-time or production-availability review is shown for generated ratio features, so deployment suitability is unassessed.
- Potential leakage risk is not ruled out for train-fitted feature generation, especially for ratio and interaction construction if any preprocessing used global statistics outside fold boundaries.
- The package does not report whether the feature recipes were compared under identical hyperparameter tuning budgets or whether model selection was nested.
- The claim 'without feature dilution' is not operationalized; there is no explicit metric tying feature count to generalization tradeoff beyond the selection rule.

Smallest next experiment

Nested fold-wise comparison of raw vs ratios with leakage checks

Hypothesis: If the ratios recipe truly improves evidence without feature dilution, then under a nested, fold-aligned evaluation it will beat the raw recipe on mean ROC-AUC by at least 0.002 while keeping feature count materially below the full feature recipes, and the result will remain stable across folds.

Success gate: Run a nested CV comparison using the same outer folds for raw and ratios, with all feature construction fit strictly inside each training fold. Accept the ratios recipe only if its outer-fold mean ROC-AUC exceeds raw by at least 0.002 and the fold-wise improvement is consistent in at least 2 of 3 folds; otherwise reject the claim.

Why smallest: This is the smallest falsifiable step because it reuses the existing two most relevant recipes, keeps the same model family, and adds only fold-aligned nesting plus a leakage check to test the exact claim.

Evidence: `campaigns/model_building_50_v1/results/EXP-048_wine_quality_feature_engineering.json`

EXP-049 · wine_quality

medium confidence

model_selection

gpt-5.4-mini

Question. Which algorithm family provides the strongest stable training-CV evidence for this dataset profile?

Decision

Provisional only: extra_trees can be treated as the current training-CV candidate, but not as a validated winner. Do not finalize algorithm-family choice until a leakage-safe repeated CV comparison and locked holdout check are completed.

Interpretation

On the provided training-CV evidence, extra_trees is a plausible top candidate by the stated ranking rule, but the package does not establish that it is the strongest stable family in a decision-grade sense. The evidence is limited to 3-fold training CV on one feature stage, with no locked holdout, no repeated CV, and no explicit leakage audit. The apparent tie between extra_trees and lightgbm in mean ROC-AUC is especially important because the selection rule also uses fold variability and runtime, but the package does not provide the computed composite scores needed to verify the ranking outcome directly.

Observed evidence (cited)

- The selection rule ranks models by mean ROC-AUC minus $0.25 \times$ fold standard deviation, then runtime, using training CV only.
`EXP-049 / evidence.selection_rule`
- The extra_trees model reports fold ROC-AUC values of 0.8644690446283994, 0.8572354497354499, and 0.8611291086482902, with mean ROC-AUC 0.8609445343373799 and std 0.003620327958350225.
`EXP-049 / evidence.model_results[1].fold_metrics; EXP-049 / evidence.model_results[1].metrics.roc_auc`
- The logistic_regression model reports mean ROC-AUC 0.8054556736414563 with std 0.006622157211670003.
`EXP-049 / evidence.model_results[0].metrics.roc_auc`
- The hist_gradient_boosting model reports mean ROC-AUC 0.8385181993978357 with std 0.0022978705105822.
`EXP-049 / evidence.model_results[2].metrics.roc_auc`
- The lightgbm model reports mean ROC-AUC 0.8609445343373799 with std 0.003620327958350225.
`EXP-049 / evidence.model_results[3].metrics.roc_auc`
- The xgboost model reports mean ROC-AUC 0.8319355269573551 with std 0.0058569808718739805.
`EXP-049 / evidence.model_results[4].metrics.roc_auc`

Challenges to claims

- **medium** **EXP-049-C1** — The statement that extra_trees is the training-CV model candidate is not directly verified by the package because the composite selection score is not reported. The mean ROC-AUC shown for extra_trees matches lightgbm exactly, so the ranking depends on unreported variability/runtime tie-breaking.
- **high** **EXP-049-C1** — Calling extra_trees the strongest stable evidence overstates certainty: stability is based on only 3 folds and 1 repeat, which is too little to support a robust family-level conclusion.
- **high** **EXP-049-C1** — The claim is limited to training CV, but the wording can be misread as a general recommendation. There is no locked holdout or production review, so the evidence does not justify

broader approval.

Risks & missing evidence

- No locked holdout results are provided, so the training-CV leader may not generalize.
- Only 3 folds and 1 repeat are shown; stability claims are weak with such limited resampling.
- The selection rule is stated, but the actual composite scores are not shown, so the ranking cannot be independently verified from the package.
- Feature engineering includes ratios and log transforms; there is no explicit evidence that these were fit strictly within each training fold, so leakage risk remains.
- No decision-time or production-availability review is included, so even a strong CV result would not be deployment-ready.
- The evidence bundle does not include a baseline with raw features or a leakage sentinel test, making it hard to separate genuine signal from preprocessing artifacts.

Smallest next experiment

Repeated leakage-safe CV comparison of extra_trees vs lightgbm

Hypothesis: If the apparent extra_trees/lightgbm tie is real and not an artifact of fold variance or preprocessing leakage, then under repeated fold-resampled CV with all feature engineering fit inside each training fold, extra_trees will retain a non-trivial advantage or at least remain within a pre-specified tolerance of lightgbm on mean ROC-AUC and composite selection score.

Success gate: Run 5x repeated stratified CV with identical fold splits for extra_trees and lightgbm, compute the stated composite score for each fold, and require that the winner is unchanged in at least 4 of 5 repeats with no fold showing evidence of preprocessing leakage; otherwise the current ranking is not stable.

Why smallest: This changes only the resampling protocol and adds a leakage check while keeping the same dataset, feature stage, and two closest contenders, making it the minimal falsifiable test of the current ranking claim.

Evidence: campaigns/model_building_50_v1/results/EXP-049_wine_quality_model_selection.json

EXP-050 · wine_quality

medium confidence

optimization_reliability

gpt-5.4-mini

Question. Does optimization produce a meaningful CV gain, and how does the preselected recipe perform on the once-consumed holdout?

Decision

Hold. The holdout result can be reported descriptively, but the optimization-gain claim is unsupported without the missing CV comparison, and no feature promotion decision should be made from importance alone.

Interpretation

The package supports a descriptive statement that the preselected recipe achieved a strong holdout ROC-AUC on the consumed holdout, but it does not by itself establish that optimization produced a meaningful CV gain. The evidence bundle includes a decision rule for accepting tuned candidates, yet it does not expose the baseline CV score, the tuned CV score, or their difference, so the gain question is unanswerable from the provided evidence alone. The feature evidence is consistent with a cautious feature-governance stance, but it is not sufficient to justify promotion of any feature because decision-time availability and production availability are not demonstrated here.

Observed evidence (cited)

- The once-consumed holdout evaluation reports ROC-AUC 0.8878391277459601 with a bootstrap 95% interval of 0.8609965140062155 to 0.9076586798160596.

EXP-050 / evidence.holdout_metrics.roc_auc; EXP-050 / evidence.holdout_roc_auc_bootstrap_ci.low; EXP-050 / evidence.holdout_roc_auc_bootstrap_ci.high; EXP-050 / evidence.holdout_consumed

- The holdout was marked as consumed.

EXP-050 / evidence.holdout_consumed

- The selection rule says to accept the tuned candidate only when mean ROC-AUC improves by at least 0.001 over baseline; otherwise prefer baseline.

EXP-050 / evidence.selection_rule

- The feature stability table reports only 3 observed folds for many features, with some top10 frequencies below 1.0.

EXP-050 / evidence.cv_feature_stability.observed_folds; EXP-050 / evidence.cv_feature_stability.top10_frequency

- The production feature rule requires decision-time availability, contractual obtainability, monitoring for missingness/drift, and cross-fold stability.

EXP-050 / evidence.production_feature_rule

Challenges to claims

- **high** **EXP-050-C1** — The holdout ROC-AUC and bootstrap interval are factual, but the wording can be misread as evidence of model superiority or optimization success. The package does not provide the baseline CV score or the tuned CV score needed to assess whether optimization produced a meaningful gain under the stated rule.
- **medium** **EXP-050-C2** — The recommendation is directionally sound, but the evidence bundle does not actually demonstrate decision-time availability, contractual obtainability, or monitoring implementation for the listed features; it only states a rule.

Risks & missing evidence

- Missing baseline CV ROC-AUC and tuned CV ROC-AUC, so the claimed optimization gain cannot be verified against the stated 0.001 acceptance rule.
- The holdout is already consumed, so any further tuning on the same holdout would risk leakage or implicit overfitting to the test set.
- Only 3 observed folds are shown for feature stability, which is thin evidence for stability claims.
- No decision-time availability evidence is provided for the listed features, so production use cannot be assessed.
- No production-availability or contractability evidence is provided for the listed features.
- No calibration comparison against baseline is provided, so the holdout score alone should not be overinterpreted as operational readiness.

Smallest next experiment

Baseline-vs-tuned CV delta audit with decision-time feature check

Hypothesis: If the tuned recipe truly improves over baseline, then its mean CV ROC-AUC will exceed the baseline by at least 0.001 under the same split protocol; additionally, any promoted feature will satisfy a documented decision-time availability check.

Success gate: Report baseline mean CV ROC-AUC, tuned mean CV ROC-AUC, and their delta on the same folds; accept optimization only if $\text{delta} \geq 0.001$. Separately, for each candidate feature, provide a pass/fail decision-time availability record before any promotion discussion.

Why smallest: This is the minimal falsifiable follow-up because it reuses the existing CV protocol, adds only the missing baseline-vs-tuned comparison, and introduces a simple feature-availability audit without requiring a new modeling pipeline.

Evidence: `campaigns/model_building_50_v1/results/EXP-050_wine_quality_optimization_reliability.json`